

# Summary Tables of MAXCUT Data

## 1 Tables for depth $P = 1$

|              | ER         | HH           | LB           | RR          |
|--------------|------------|--------------|--------------|-------------|
| MPS (Aer)    |            |              |              |             |
| Fixed Angle* | 10 [50]    | -            | 9 [100]      | 7 [100]     |
| Fixed Angle† | 10 [50]    | -            | 9 [100]      | 7 [100]     |
| Fourier*     | 20 [40–50] | 20 [39–144]  | 55 [40–100]  | 17 [40–100] |
| Interp.*     | 20 [40–50] | 20 [39–144]  | 55 [40–100]  | 17 [40–100] |
| Linear Ramp  | 10 [50]    | 10 [144]     | 10 [100]     | 7 [100]     |
| Linear Ramp* | 10 [50]    | 10 [144]     | 10 [100]     | 7 [100]     |
| TQA          | 10 [50]    | 10 [144]     | 10 [100]     | 7 [100]     |
| TQA*         | 10 [50]    | 10 [144]     | 10 [100]     | 7 [100]     |
| MPS (Quimb)  |            |              |              |             |
| Fixed Angle* | 10 [50]    | -            | 9 [100]      | 7 [100]     |
| Fixed Angle† | 10 [50]    | -            | 9 [100]      | 7 [100]     |
| Fourier*     | 20 [40–50] | 31 [39–144]  | 130 [40–100] | 58 [30–100] |
| Interp.*     | 20 [40–50] | 31 [39–144]  | 130 [40–100] | 58 [30–100] |
| Linear Ramp  | 10 [50]    | 10 [144]     | 10 [100]     | 7 [100]     |
| Linear Ramp* | 10 [50]    | 10 [144]     | 10 [100]     | 7 [100]     |
| Recursive TS | 10 [50]    | 11 [106–144] | 10 [100]     | 9 [40–100]  |
| TQA          | 10 [50]    | 10 [144]     | 10 [100]     | 7 [100]     |
| TQA*         | 10 [50]    | 10 [144]     | 10 [100]     | 7 [100]     |
| PP           |            |              |              |             |
| Fixed Angle* | 10 [50]    | -            | 9 [100]      | 7 [100]     |
| Fixed Angle† | 10 [50]    | -            | 9 [100]      | 7 [100]     |
| Fourier*     | 20 [40–50] | 31 [39–144]  | 130 [40–100] | 58 [30–100] |
| Interp.*     | 20 [40–50] | 31 [39–144]  | 130 [40–100] | 58 [30–100] |
| Linear Ramp  | 10 [50]    | 20 [105–144] | 20 [40–100]  | 8 [100]     |
| Linear Ramp* | 10 [50]    | 20 [105–144] | 20 [40–100]  | 8 [100]     |
| Recursive TS | 10 [50]    | 11 [106–144] | 10 [100]     | 9 [40–100]  |
| TQA          | 10 [50]    | 10 [144]     | 10 [100]     | 7 [100]     |
| TQA*         | 10 [50]    | 10 [144]     | 10 [100]     | 7 [100]     |
| SV           |            |              |              |             |
| Fixed Angle* | 64 [10–20] | -            | 19 [20]      | 273 [10–20] |
| Fixed Angle† | 64 [10–20] | -            | 19 [20]      | 289 [10–20] |
| Fourier*     | 92 [10–20] | 20 [12–21]   | 784 [10–20]  | 570 [10–20] |
| Interp.*     | 92 [10–20] | 20 [12–21]   | 828 [10–22]  | 382 [10–20] |
| Linear Ramp  | 10 [20]    | 10 [21]      | 10 [20]      | 7 [20]      |
| Linear Ramp* | -          | -            | -            | -           |
| Recursive TS | 92 [10–20] | 20 [12–21]   | 505 [10–22]  | 381 [10–20] |
| TQA          | 74 [10–20] | 10 [21]      | 611 [10–20]  | 280 [10–20] |
| TQA*         | 72 [10–20] | 10 [21]      | 603 [10–20]  | 66 [10–20]  |

Table 1: **Number of Instances (num. nodes in brackets) for  $P = 1$ .** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Cells are colored based on the number of instances for the given configuration, with darker colors indicating more instances.

|                          | ER             | HH             | LB             | RR             |
|--------------------------|----------------|----------------|----------------|----------------|
| MPS (Aer)                |                |                |                |                |
| Fixed Angle <sup>*</sup> | 76.0 $\pm$ 1.2 | -              | 72.4 $\pm$ 2.3 | 71.3 $\pm$ 1.3 |
| Fixed Angle <sup>†</sup> | 75.4 $\pm$ 1.3 | -              | 72.2 $\pm$ 2.3 | 70.5 $\pm$ 1.6 |
| Fourier <sup>*</sup>     | 81.5 $\pm$ 1.2 | 75.6 $\pm$ 1.2 | 73.1 $\pm$ 2.1 | 77.3 $\pm$ 2.1 |
| Interp. <sup>*</sup>     | 81.5 $\pm$ 1.2 | 75.6 $\pm$ 1.2 | 73.1 $\pm$ 2.1 | 77.3 $\pm$ 2.1 |
| Linear Ramp              | 75.7 $\pm$ 1.3 | 74.6 $\pm$ 0.9 | 71.9 $\pm$ 2.7 | 71.2 $\pm$ 1.3 |
| Linear Ramp <sup>*</sup> | 75.8 $\pm$ 1.3 | 74.6 $\pm$ 0.9 | 71.9 $\pm$ 2.7 | 71.2 $\pm$ 1.3 |
| TQA                      | 75.5 $\pm$ 1.3 | 74.5 $\pm$ 1.0 | 68.9 $\pm$ 4.3 | 71.0 $\pm$ 1.4 |
| TQA <sup>*</sup>         | 75.6 $\pm$ 1.3 | 74.7 $\pm$ 1.0 | 71.8 $\pm$ 2.7 | 71.2 $\pm$ 1.4 |
| MPS (Quimb)              |                |                |                |                |
| Fixed Angle <sup>*</sup> | 75.6 $\pm$ 1.2 | -              | 72.3 $\pm$ 2.2 | 69.3 $\pm$ 2.6 |
| Fixed Angle <sup>†</sup> | 74.9 $\pm$ 1.1 | -              | 72.3 $\pm$ 2.2 | 68.1 $\pm$ 2.9 |
| Fourier <sup>*</sup>     | 81.5 $\pm$ 1.2 | 75.4 $\pm$ 1.2 | 72.7 $\pm$ 1.7 | 77.1 $\pm$ 1.8 |
| Interp. <sup>*</sup>     | 81.5 $\pm$ 1.2 | 75.4 $\pm$ 1.2 | 72.7 $\pm$ 1.7 | 77.1 $\pm$ 1.8 |
| Linear Ramp              | 75.8 $\pm$ 1.1 | 68.2 $\pm$ 1.1 | 71.9 $\pm$ 2.6 | 69.3 $\pm$ 2.6 |
| Linear Ramp <sup>*</sup> | 75.8 $\pm$ 1.1 | 67.4 $\pm$ 1.3 | 71.9 $\pm$ 2.6 | 69.3 $\pm$ 2.5 |
| Recursive TS             | 80.1 $\pm$ 1.1 | 74.6 $\pm$ 0.8 | 71.3 $\pm$ 2.0 | 77.2 $\pm$ 2.1 |
| TQA                      | 75.8 $\pm$ 1.2 | 67.3 $\pm$ 1.2 | 68.6 $\pm$ 4.5 | 68.9 $\pm$ 3.0 |
| TQA <sup>*</sup>         | 75.8 $\pm$ 1.1 | 67.4 $\pm$ 1.3 | 71.9 $\pm$ 2.6 | 69.4 $\pm$ 2.4 |
| PP                       |                |                |                |                |
| Fixed Angle <sup>*</sup> | 81.9 $\pm$ 1.1 | -              | 73.0 $\pm$ 1.4 | 78.8 $\pm$ 2.2 |
| Fixed Angle <sup>†</sup> | 81.6 $\pm$ 1.0 | -              | 73.0 $\pm$ 1.4 | 78.7 $\pm$ 2.2 |
| Fourier <sup>*</sup>     | 81.5 $\pm$ 1.2 | 75.4 $\pm$ 1.2 | 72.7 $\pm$ 1.7 | 77.1 $\pm$ 1.8 |
| Interp. <sup>*</sup>     | 81.5 $\pm$ 1.2 | 75.4 $\pm$ 1.2 | 72.7 $\pm$ 1.7 | 77.1 $\pm$ 1.8 |
| Linear Ramp              | 81.9 $\pm$ 1.1 | 75.3 $\pm$ 1.0 | 72.6 $\pm$ 1.6 | 78.4 $\pm$ 2.3 |
| Linear Ramp <sup>*</sup> | 81.9 $\pm$ 1.1 | 75.3 $\pm$ 1.0 | 72.6 $\pm$ 1.6 | 78.4 $\pm$ 2.3 |
| Recursive TS             | 80.1 $\pm$ 1.1 | 74.6 $\pm$ 0.8 | 71.3 $\pm$ 2.0 | 77.2 $\pm$ 2.1 |
| TQA                      | 77.8 $\pm$ 0.9 | 75.1 $\pm$ 0.9 | 68.9 $\pm$ 4.2 | 76.9 $\pm$ 1.0 |
| TQA <sup>*</sup>         | 81.9 $\pm$ 1.1 | 75.4 $\pm$ 0.9 | 72.9 $\pm$ 1.4 | 78.8 $\pm$ 2.2 |
| SV                       |                |                |                |                |
| Fixed Angle <sup>*</sup> | 79.0 $\pm$ 2.8 | -              | 73.4 $\pm$ 2.6 | 81.7 $\pm$ 4.3 |
| Fixed Angle <sup>†</sup> | 78.4 $\pm$ 2.5 | -              | 73.4 $\pm$ 2.6 | 80.7 $\pm$ 3.1 |
| Fourier <sup>*</sup>     | 78.8 $\pm$ 2.8 | 78.3 $\pm$ 2.6 | 74.7 $\pm$ 4.0 | 82.0 $\pm$ 4.1 |
| Interp. <sup>*</sup>     | 78.8 $\pm$ 2.8 | 78.3 $\pm$ 2.6 | 74.6 $\pm$ 3.9 | 81.9 $\pm$ 4.2 |
| Linear Ramp              | 78.4 $\pm$ 1.9 | 77.4 $\pm$ 1.7 | 74.2 $\pm$ 1.9 | 79.9 $\pm$ 2.8 |
| Linear Ramp <sup>*</sup> | -              | -              | -              | -              |
| Recursive TS             | 78.8 $\pm$ 2.8 | 78.3 $\pm$ 2.6 | 74.9 $\pm$ 4.8 | 81.9 $\pm$ 4.2 |
| TQA                      | 77.6 $\pm$ 2.5 | 75.9 $\pm$ 2.7 | 71.0 $\pm$ 5.0 | 79.0 $\pm$ 2.9 |
| TQA <sup>*</sup>         | 78.7 $\pm$ 2.9 | 77.4 $\pm$ 1.7 | 74.7 $\pm$ 3.6 | 78.3 $\pm$ 2.2 |

Table 2: **Avg. Approx. Ratio  $\pm$  standard deviation in percentage for MAXCUT for  $P = 1$ .** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Evaluation methods are shown in different colors, with darker colors indicating better approximation ratios.

|              | Approximation Ratio |            |            |            | Num. Instances |              |              |             |
|--------------|---------------------|------------|------------|------------|----------------|--------------|--------------|-------------|
|              | ER                  | HH         | LB         | RR         | ER             | HH           | LB           | RR          |
| MPS (Aer)    |                     |            |            |            |                |              |              |             |
| Fixed Angle* | 76.0 ± 1.2          | -          | 72.4 ± 2.3 | 71.3 ± 1.3 | 10 [50]        | -            | 9 [100]      | 7 [100]     |
| Fixed Angle† | 75.4 ± 1.3          | -          | 72.2 ± 2.3 | 70.5 ± 1.6 | 10 [50]        | -            | 9 [100]      | 7 [100]     |
| Fourier*     | 81.5 ± 1.2          | 75.6 ± 1.2 | 73.1 ± 2.1 | 77.3 ± 2.1 | 20 [40–50]     | 20 [39–144]  | 55 [40–100]  | 17 [40–100] |
| Interp.*     | 81.5 ± 1.2          | 75.6 ± 1.2 | 73.1 ± 2.1 | 77.3 ± 2.1 | 20 [40–50]     | 20 [39–144]  | 55 [40–100]  | 17 [40–100] |
| Linear Ramp  | 75.7 ± 1.3          | 74.6 ± 0.9 | 71.9 ± 2.7 | 71.2 ± 1.3 | 10 [50]        | 10 [144]     | 10 [100]     | 7 [100]     |
| Linear Ramp* | 75.8 ± 1.3          | 74.6 ± 0.9 | 71.9 ± 2.7 | 71.2 ± 1.3 | 10 [50]        | 10 [144]     | 10 [100]     | 7 [100]     |
| TQA          | 75.5 ± 1.3          | 74.5 ± 1.0 | 68.9 ± 4.3 | 71.0 ± 1.4 | 10 [50]        | 10 [144]     | 10 [100]     | 7 [100]     |
| TQA*         | 75.6 ± 1.3          | 74.7 ± 1.0 | 71.8 ± 2.7 | 71.2 ± 1.4 | 10 [50]        | 10 [144]     | 10 [100]     | 7 [100]     |
| MPS (Quimb)  |                     |            |            |            |                |              |              |             |
| Fixed Angle* | 75.6 ± 1.2          | -          | 72.3 ± 2.2 | 69.3 ± 2.6 | 10 [50]        | -            | 9 [100]      | 7 [100]     |
| Fixed Angle† | 74.9 ± 1.1          | -          | 72.3 ± 2.2 | 68.1 ± 2.9 | 10 [50]        | -            | 9 [100]      | 7 [100]     |
| Fourier*     | 81.5 ± 1.2          | 75.4 ± 1.2 | 72.7 ± 1.7 | 77.1 ± 1.8 | 20 [40–50]     | 31 [39–144]  | 130 [40–100] | 58 [30–100] |
| Interp.*     | 81.5 ± 1.2          | 75.4 ± 1.2 | 72.7 ± 1.7 | 77.1 ± 1.8 | 20 [40–50]     | 31 [39–144]  | 130 [40–100] | 58 [30–100] |
| Linear Ramp  | 75.8 ± 1.1          | 68.2 ± 1.1 | 71.9 ± 2.6 | 69.3 ± 2.6 | 10 [50]        | 10 [144]     | 10 [100]     | 7 [100]     |
| Linear Ramp* | 75.8 ± 1.1          | 67.4 ± 1.3 | 71.9 ± 2.6 | 69.3 ± 2.5 | 10 [50]        | 10 [144]     | 10 [100]     | 7 [100]     |
| Recursive TS | 80.1 ± 1.1          | 74.6 ± 0.8 | 71.3 ± 2.0 | 77.2 ± 2.1 | 10 [50]        | 11 [106–144] | 10 [100]     | 9 [40–100]  |
| TQA          | 75.8 ± 1.2          | 67.3 ± 1.2 | 68.6 ± 4.5 | 68.9 ± 3.0 | 10 [50]        | 10 [144]     | 10 [100]     | 7 [100]     |
| TQA*         | 75.8 ± 1.1          | 67.4 ± 1.3 | 71.9 ± 2.6 | 69.4 ± 2.4 | 10 [50]        | 10 [144]     | 10 [100]     | 7 [100]     |
| PP           |                     |            |            |            |                |              |              |             |
| Fixed Angle* | 81.9 ± 1.1          | -          | 73.0 ± 1.4 | 78.8 ± 2.2 | 10 [50]        | -            | 9 [100]      | 7 [100]     |
| Fixed Angle† | 81.6 ± 1.0          | -          | 73.0 ± 1.4 | 78.7 ± 2.2 | 10 [50]        | -            | 9 [100]      | 7 [100]     |
| Fourier*     | 81.5 ± 1.2          | 75.4 ± 1.2 | 72.7 ± 1.7 | 77.1 ± 1.8 | 20 [40–50]     | 31 [39–144]  | 130 [40–100] | 58 [30–100] |
| Interp.*     | 81.5 ± 1.2          | 75.4 ± 1.2 | 72.7 ± 1.7 | 77.1 ± 1.8 | 20 [40–50]     | 31 [39–144]  | 130 [40–100] | 58 [30–100] |
| Linear Ramp  | 81.9 ± 1.1          | 75.3 ± 1.0 | 72.6 ± 1.6 | 78.4 ± 2.3 | 10 [50]        | 20 [105–144] | 20 [40–100]  | 8 [100]     |
| Linear Ramp* | 81.9 ± 1.1          | 75.3 ± 1.0 | 72.6 ± 1.6 | 78.4 ± 2.3 | 10 [50]        | 20 [105–144] | 20 [40–100]  | 8 [100]     |
| Recursive TS | 80.1 ± 1.1          | 74.6 ± 0.8 | 71.3 ± 2.0 | 77.2 ± 2.1 | 10 [50]        | 11 [106–144] | 10 [100]     | 9 [40–100]  |
| TQA          | 77.8 ± 0.9          | 75.1 ± 0.9 | 68.9 ± 4.2 | 76.9 ± 1.0 | 10 [50]        | 10 [144]     | 10 [100]     | 7 [100]     |
| TQA*         | 81.9 ± 1.1          | 75.4 ± 0.9 | 72.9 ± 1.4 | 78.8 ± 2.2 | 10 [50]        | 10 [144]     | 10 [100]     | 7 [100]     |
| SV           |                     |            |            |            |                |              |              |             |
| Fixed Angle* | 79.0 ± 2.8          | -          | 73.4 ± 2.6 | 81.7 ± 4.3 | 64 [10–20]     | -            | 19 [20]      | 273 [10–20] |
| Fixed Angle† | 78.4 ± 2.5          | -          | 73.4 ± 2.6 | 80.7 ± 3.1 | 64 [10–20]     | -            | 19 [20]      | 289 [10–20] |
| Fourier*     | 78.8 ± 2.8          | 78.3 ± 2.6 | 74.7 ± 4.0 | 82.0 ± 4.1 | 92 [10–20]     | 20 [12–21]   | 784 [10–20]  | 570 [10–20] |
| Interp.*     | 78.8 ± 2.8          | 78.3 ± 2.6 | 74.6 ± 3.9 | 81.9 ± 4.2 | 92 [10–20]     | 20 [12–21]   | 828 [10–22]  | 382 [10–20] |
| Linear Ramp  | 78.4 ± 1.9          | 77.4 ± 1.7 | 74.2 ± 1.9 | 79.9 ± 2.8 | 10 [20]        | 10 [21]      | 10 [20]      | 7 [20]      |
| Linear Ramp* | -                   | -          | -          | -          | -              | -            | -            | -           |
| Recursive TS | 78.8 ± 2.8          | 78.3 ± 2.6 | 74.9 ± 4.8 | 81.9 ± 4.2 | 92 [10–20]     | 20 [12–21]   | 505 [10–22]  | 381 [10–20] |
| TQA          | 77.6 ± 2.5          | 75.9 ± 2.7 | 71.0 ± 5.0 | 79.0 ± 2.9 | 74 [10–20]     | 10 [21]      | 611 [10–20]  | 280 [10–20] |
| TQA*         | 78.7 ± 2.9          | 77.4 ± 1.7 | 74.7 ± 3.6 | 78.3 ± 2.2 | 72 [10–20]     | 10 [21]      | 603 [10–20]  | 66 [10–20]  |

Table 3: **Avg. Approx. Ratio and Number of Instances for Various Evaluation Methods for MAX-CUT for  $P = 1$ .** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Evaluation methods are shown in different colors for approximation ratios. Darker colors indicating better approximation ratios or more instances. Approximation ratios are in percentage, with  $\pm$  their standard deviation. The range of graph sizes are shown in brackets, next to the number of instances.

## 2 Tables for depth $P = 2$

|              | ER         | HH           | LB           | RR          |
|--------------|------------|--------------|--------------|-------------|
| MPS (Aer)    |            |              |              |             |
| Fixed Angle* | 10 [50]    | 10 [39]      | 27 [100]     | 17 [40-100] |
| Fixed Angle† | 10 [50]    | 10 [39]      | 27 [100]     | 17 [40-100] |
| Fourier*     | 20 [40-50] | 20 [39-144]  | 55 [40-100]  | 17 [40-100] |
| Interp.*     | 20 [40-50] | 20 [39-144]  | 55 [40-100]  | 17 [40-100] |
| Linear Ramp  | 10 [50]    | 10 [144]     | 19 [100]     | 7 [100]     |
| Linear Ramp* | 10 [50]    | 10 [144]     | 10 [100]     | 7 [100]     |
| TQA          | 20 [40-50] | 20 [39-144]  | 56 [40-100]  | 17 [40-100] |
| TQA*         | 20 [40-50] | 20 [39-144]  | 56 [40-100]  | 17 [40-100] |
| MPS (Quimb)  |            |              |              |             |
| Fixed Angle* | 10 [50]    | 10 [144]     | 90 [100]     | 47 [30-100] |
| Fixed Angle† | 10 [50]    | 10 [144]     | 90 [100]     | 47 [30-100] |
| Fourier*     | 20 [40-50] | 31 [39-144]  | 130 [40-100] | 58 [30-100] |
| Interp.*     | 20 [40-50] | 31 [39-144]  | 130 [40-100] | 58 [30-100] |
| Linear Ramp  | 10 [50]    | 10 [144]     | 10 [100]     | 7 [100]     |
| Linear Ramp* | 10 [50]    | 10 [144]     | 10 [100]     | 7 [100]     |
| Recursive TS | 10 [50]    | 11 [106-144] | 10 [100]     | 9 [40-100]  |
| TQA          | 20 [40-50] | 20 [39-144]  | 130 [40-100] | 57 [30-100] |
| TQA*         | 20 [40-50] | 20 [39-144]  | 130 [40-100] | 57 [30-100] |
| PP           |            |              |              |             |
| Fixed Angle* | 10 [50]    | 30 [39-144]  | 82 [100]     | 47 [30-100] |
| Fixed Angle† | 10 [50]    | 30 [39-144]  | 82 [100]     | 47 [30-100] |
| Fourier*     | 20 [40-50] | 31 [39-144]  | 130 [40-100] | 58 [30-100] |
| Interp.*     | 20 [40-50] | 31 [39-144]  | 130 [40-100] | 58 [30-100] |
| Linear Ramp  | 10 [50]    | 20 [105-144] | 29 [40-100]  | 8 [100]     |
| Linear Ramp* | 10 [50]    | 20 [105-144] | 20 [40-100]  | 8 [100]     |
| Recursive TS | 10 [50]    | 11 [106-144] | 10 [100]     | 9 [40-100]  |
| TQA          | 20 [40-50] | 20 [39-144]  | 130 [40-100] | 57 [30-100] |
| TQA*         | 20 [40-50] | 20 [39-144]  | 130 [40-100] | 57 [30-100] |
| SV           |            |              |              |             |
| Fixed Angle* | 81 [10-20] | -            | 311 [10-22]  | 360 [10-20] |
| Fixed Angle† | 81 [10-20] | -            | 311 [10-22]  | 372 [10-20] |
| Fourier*     | 92 [10-20] | 20 [12-21]   | 784 [10-20]  | 567 [10-20] |
| Interp.*     | 92 [10-20] | 20 [12-21]   | 828 [10-22]  | 382 [10-20] |
| Linear Ramp  | 10 [20]    | 10 [21]      | 10 [20]      | 7 [20]      |
| Linear Ramp* | 10 [20]    | 10 [21]      | 10 [20]      | 7 [20]      |
| Recursive TS | 92 [10-20] | 20 [12-21]   | 505 [10-22]  | 381 [10-20] |
| TQA          | 92 [10-20] | 20 [12-21]   | 822 [10-22]  | 361 [10-20] |
| TQA*         | 92 [10-20] | 20 [12-21]   | 821 [10-22]  | 142 [10-20] |

Table 4: **Number of Instances (num. nodes in brackets) for  $P = 2$ .** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Cells are colored based on the number of instances for the given configuration, with darker colors indicating more instances.

|                          | ER             | HH              | LB             | RR             |
|--------------------------|----------------|-----------------|----------------|----------------|
| MPS (Aer)                |                |                 |                |                |
| Fixed Angle*             | 77.0 $\pm$ 1.9 | 82.2 $\pm$ 1.3  | 75.3 $\pm$ 3.9 | 75.6 $\pm$ 4.9 |
| Fixed Angle <sup>†</sup> | 75.6 $\pm$ 1.3 | 80.9 $\pm$ 1.5  | 70.6 $\pm$ 6.4 | 74.7 $\pm$ 5.1 |
| Fourier*                 | 71.5 $\pm$ 2.3 | 75.6 $\pm$ 3.8  | 69.3 $\pm$ 4.2 | 73.1 $\pm$ 3.5 |
| Interp.*                 | 77.0 $\pm$ 1.6 | 80.4 $\pm$ 2.0  | 73.1 $\pm$ 5.2 | 70.9 $\pm$ 2.2 |
| Linear Ramp              | 76.3 $\pm$ 1.2 | 77.9 $\pm$ 1.6  | 73.0 $\pm$ 5.9 | 70.1 $\pm$ 2.2 |
| Linear Ramp*             | 77.9 $\pm$ 2.1 | 77.9 $\pm$ 1.2  | 78.4 $\pm$ 4.3 | 70.1 $\pm$ 2.2 |
| TQA                      | 69.0 $\pm$ 1.7 | 72.7 $\pm$ 1.4  | 67.1 $\pm$ 3.9 | 72.1 $\pm$ 2.8 |
| TQA*                     | 76.0 $\pm$ 1.2 | 78.8 $\pm$ 2.9  | 72.2 $\pm$ 6.2 | 74.4 $\pm$ 4.6 |
| MPS (Quimb)              |                |                 |                |                |
| Fixed Angle*             | 80.5 $\pm$ 1.9 | 53.9 $\pm$ 0.6  | 79.9 $\pm$ 2.4 | 76.4 $\pm$ 4.9 |
| Fixed Angle <sup>†</sup> | 77.2 $\pm$ 0.9 | 52.5 $\pm$ 0.6  | 79.6 $\pm$ 2.6 | 75.3 $\pm$ 5.5 |
| Fourier*                 | 70.6 $\pm$ 1.8 | 72.1 $\pm$ 8.3  | 76.1 $\pm$ 4.3 | 74.6 $\pm$ 4.7 |
| Interp.*                 | 79.7 $\pm$ 2.0 | 75.9 $\pm$ 6.2  | 79.2 $\pm$ 3.0 | 74.9 $\pm$ 4.4 |
| Linear Ramp              | 78.5 $\pm$ 3.2 | 69.5 $\pm$ 1.3  | 79.0 $\pm$ 3.3 | 70.3 $\pm$ 4.0 |
| Linear Ramp*             | 79.3 $\pm$ 2.5 | 68.7 $\pm$ 1.4  | 79.6 $\pm$ 3.0 | 70.4 $\pm$ 5.3 |
| Recursive TS             | 77.8 $\pm$ 2.1 | 56.7 $\pm$ 5.0  | 78.2 $\pm$ 3.3 | 67.1 $\pm$ 3.9 |
| TQA                      | 69.3 $\pm$ 1.1 | 63.5 $\pm$ 9.8  | 70.0 $\pm$ 3.9 | 72.2 $\pm$ 3.0 |
| TQA*                     | 77.4 $\pm$ 1.3 | 68.6 $\pm$ 14.9 | 79.2 $\pm$ 3.0 | 76.5 $\pm$ 4.8 |
| PP                       |                |                 |                |                |
| Fixed Angle*             | 87.8 $\pm$ 1.0 | 82.7 $\pm$ 1.2  | 79.2 $\pm$ 1.6 | 83.8 $\pm$ 1.5 |
| Fixed Angle <sup>†</sup> | 86.9 $\pm$ 1.0 | 81.2 $\pm$ 1.4  | 79.2 $\pm$ 1.6 | 83.7 $\pm$ 1.5 |
| Fourier*                 | 86.7 $\pm$ 2.1 | 80.8 $\pm$ 2.2  | 79.1 $\pm$ 1.7 | 80.5 $\pm$ 2.7 |
| Interp.*                 | 87.4 $\pm$ 1.2 | 82.7 $\pm$ 1.2  | 79.6 $\pm$ 2.0 | 83.8 $\pm$ 1.4 |
| Linear Ramp              | 86.9 $\pm$ 1.0 | 82.4 $\pm$ 1.0  | 80.5 $\pm$ 1.8 | 83.9 $\pm$ 1.2 |
| Linear Ramp*             | 87.8 $\pm$ 1.0 | 82.5 $\pm$ 1.0  | 80.3 $\pm$ 1.9 | 83.9 $\pm$ 1.1 |
| Recursive TS             | 85.5 $\pm$ 1.1 | 81.3 $\pm$ 1.3  | 79.4 $\pm$ 1.7 | 81.6 $\pm$ 2.5 |
| TQA                      | 78.9 $\pm$ 1.3 | 72.8 $\pm$ 1.4  | 70.6 $\pm$ 3.0 | 77.1 $\pm$ 2.0 |
| TQA*                     | 87.4 $\pm$ 1.2 | 82.8 $\pm$ 1.4  | 79.6 $\pm$ 2.0 | 83.8 $\pm$ 1.4 |
| SV                       |                |                 |                |                |
| Fixed Angle*             | 85.4 $\pm$ 2.2 | -               | 82.5 $\pm$ 2.9 | 88.1 $\pm$ 3.3 |
| Fixed Angle <sup>†</sup> | 84.1 $\pm$ 2.1 | -               | 82.2 $\pm$ 2.9 | 87.2 $\pm$ 3.0 |
| Fourier*                 | 76.4 $\pm$ 8.8 | 75.1 $\pm$ 4.5  | 76.2 $\pm$ 5.7 | 80.6 $\pm$ 8.9 |
| Interp.*                 | 85.3 $\pm$ 2.3 | 84.8 $\pm$ 1.7  | 82.5 $\pm$ 4.0 | 86.4 $\pm$ 4.7 |
| Linear Ramp              | 84.6 $\pm$ 1.7 | 84.5 $\pm$ 1.2  | 81.7 $\pm$ 1.8 | 85.7 $\pm$ 2.4 |
| Linear Ramp*             | 84.9 $\pm$ 1.6 | 84.8 $\pm$ 1.2  | 82.0 $\pm$ 1.8 | 86.0 $\pm$ 2.5 |
| Recursive TS             | 85.2 $\pm$ 2.3 | 84.3 $\pm$ 3.3  | 82.1 $\pm$ 4.8 | 86.1 $\pm$ 4.3 |
| TQA                      | 76.2 $\pm$ 3.1 | 72.9 $\pm$ 4.2  | 72.5 $\pm$ 4.2 | 75.7 $\pm$ 4.4 |
| TQA*                     | 85.5 $\pm$ 2.3 | 85.7 $\pm$ 2.1  | 82.8 $\pm$ 3.1 | 87.3 $\pm$ 3.3 |

Table 5: **Avg. Approx. Ratio  $\pm$  standard deviation in percentage for MAXCUT for  $P = 2$ .** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Evaluation methods are shown in different colors, with darker colors indicating better approximation ratios.

|              | Approximation Ratio |             |            |            | Num. Instances |              |              |             |
|--------------|---------------------|-------------|------------|------------|----------------|--------------|--------------|-------------|
|              | ER                  | HH          | LB         | RR         | ER             | HH           | LB           | RR          |
| MPS (Aer)    |                     |             |            |            |                |              |              |             |
| Fixed Angle* | 77.0 ± 1.9          | 82.2 ± 1.3  | 75.3 ± 3.9 | 75.6 ± 4.9 | 10 [50]        | 10 [39]      | 27 [100]     | 17 [40-100] |
| Fixed Angle† | 75.6 ± 1.3          | 80.9 ± 1.5  | 70.6 ± 6.4 | 74.7 ± 5.1 | 10 [50]        | 10 [39]      | 27 [100]     | 17 [40-100] |
| Fourier*     | 71.5 ± 2.3          | 75.6 ± 3.8  | 69.3 ± 4.2 | 73.1 ± 3.5 | 20 [40-50]     | 20 [39-144]  | 55 [40-100]  | 17 [40-100] |
| Interp.*     | 77.0 ± 1.6          | 80.4 ± 2.0  | 73.1 ± 5.2 | 70.9 ± 2.2 | 20 [40-50]     | 20 [39-144]  | 55 [40-100]  | 17 [40-100] |
| Linear Ramp  | 76.3 ± 1.2          | 77.9 ± 1.6  | 73.0 ± 5.9 | 70.1 ± 2.2 | 10 [50]        | 10 [144]     | 19 [100]     | 7 [100]     |
| Linear Ramp* | 77.9 ± 2.1          | 77.9 ± 1.2  | 78.4 ± 4.3 | 70.1 ± 2.2 | 10 [50]        | 10 [144]     | 10 [100]     | 7 [100]     |
| TQA          | 69.0 ± 1.7          | 72.7 ± 1.4  | 67.1 ± 3.9 | 72.1 ± 2.8 | 20 [40-50]     | 20 [39-144]  | 56 [40-100]  | 17 [40-100] |
| TQA*         | 76.0 ± 1.2          | 78.8 ± 2.9  | 72.2 ± 6.2 | 74.4 ± 4.6 | 20 [40-50]     | 20 [39-144]  | 56 [40-100]  | 17 [40-100] |
| MPS (Quimb)  |                     |             |            |            |                |              |              |             |
| Fixed Angle* | 80.5 ± 1.9          | 53.9 ± 0.6  | 79.9 ± 2.4 | 76.4 ± 4.9 | 10 [50]        | 10 [144]     | 90 [100]     | 47 [30-100] |
| Fixed Angle† | 77.2 ± 0.9          | 52.5 ± 0.6  | 79.6 ± 2.6 | 75.3 ± 5.5 | 10 [50]        | 10 [144]     | 90 [100]     | 47 [30-100] |
| Fourier*     | 70.6 ± 1.8          | 72.1 ± 8.3  | 76.1 ± 4.3 | 74.6 ± 4.7 | 20 [40-50]     | 31 [39-144]  | 130 [40-100] | 58 [30-100] |
| Interp.*     | 79.7 ± 2.0          | 75.9 ± 6.2  | 79.2 ± 3.0 | 74.9 ± 4.4 | 20 [40-50]     | 31 [39-144]  | 130 [40-100] | 58 [30-100] |
| Linear Ramp  | 78.5 ± 3.2          | 69.5 ± 1.3  | 79.0 ± 3.3 | 70.3 ± 4.0 | 10 [50]        | 10 [144]     | 10 [100]     | 7 [100]     |
| Linear Ramp* | 79.3 ± 2.5          | 68.7 ± 1.4  | 79.6 ± 3.0 | 70.4 ± 5.3 | 10 [50]        | 10 [144]     | 10 [100]     | 7 [100]     |
| Recursive TS | 77.8 ± 2.1          | 56.7 ± 5.0  | 78.2 ± 3.3 | 67.1 ± 3.9 | 10 [50]        | 11 [106-144] | 10 [100]     | 9 [40-100]  |
| TQA          | 69.3 ± 1.1          | 63.5 ± 9.8  | 70.0 ± 3.9 | 72.2 ± 3.0 | 20 [40-50]     | 20 [39-144]  | 130 [40-100] | 57 [30-100] |
| TQA*         | 77.4 ± 1.3          | 68.6 ± 14.9 | 79.2 ± 3.0 | 76.5 ± 4.8 | 20 [40-50]     | 20 [39-144]  | 130 [40-100] | 57 [30-100] |
| PP           |                     |             |            |            |                |              |              |             |
| Fixed Angle* | 87.8 ± 1.0          | 82.7 ± 1.2  | 79.2 ± 1.6 | 83.8 ± 1.5 | 10 [50]        | 30 [39-144]  | 82 [100]     | 47 [30-100] |
| Fixed Angle† | 86.9 ± 1.0          | 81.2 ± 1.4  | 79.2 ± 1.6 | 83.7 ± 1.5 | 10 [50]        | 30 [39-144]  | 82 [100]     | 47 [30-100] |
| Fourier*     | 86.7 ± 2.1          | 80.8 ± 2.2  | 79.1 ± 1.7 | 80.5 ± 2.7 | 20 [40-50]     | 31 [39-144]  | 130 [40-100] | 58 [30-100] |
| Interp.*     | 87.4 ± 1.2          | 82.7 ± 1.2  | 79.6 ± 2.0 | 83.8 ± 1.4 | 20 [40-50]     | 31 [39-144]  | 130 [40-100] | 58 [30-100] |
| Linear Ramp  | 86.9 ± 1.0          | 82.4 ± 1.0  | 80.5 ± 1.8 | 83.9 ± 1.2 | 10 [50]        | 20 [105-144] | 29 [40-100]  | 8 [100]     |
| Linear Ramp* | 87.8 ± 1.0          | 82.5 ± 1.0  | 80.3 ± 1.9 | 83.9 ± 1.1 | 10 [50]        | 20 [105-144] | 20 [40-100]  | 8 [100]     |
| Recursive TS | 85.5 ± 1.1          | 81.3 ± 1.3  | 79.4 ± 1.7 | 81.6 ± 2.5 | 10 [50]        | 11 [106-144] | 10 [100]     | 9 [40-100]  |
| TQA          | 78.9 ± 1.3          | 72.8 ± 1.4  | 70.6 ± 3.0 | 77.1 ± 2.0 | 20 [40-50]     | 20 [39-144]  | 130 [40-100] | 57 [30-100] |
| TQA*         | 87.4 ± 1.2          | 82.8 ± 1.4  | 79.6 ± 2.0 | 83.8 ± 1.4 | 20 [40-50]     | 20 [39-144]  | 130 [40-100] | 57 [30-100] |
| SV           |                     |             |            |            |                |              |              |             |
| Fixed Angle* | 85.4 ± 2.2          | -           | 82.5 ± 2.9 | 88.1 ± 3.3 | 81 [10-20]     | -            | 311 [10-22]  | 360 [10-22] |
| Fixed Angle† | 84.1 ± 2.1          | -           | 82.2 ± 2.9 | 87.2 ± 3.0 | 81 [10-20]     | -            | 311 [10-22]  | 372 [10-22] |
| Fourier*     | 76.4 ± 8.8          | 75.1 ± 4.5  | 76.2 ± 5.7 | 80.6 ± 8.9 | 92 [10-20]     | 20 [12-21]   | 784 [10-20]  | 567 [10-22] |
| Interp.*     | 85.3 ± 2.3          | 84.8 ± 1.7  | 82.5 ± 4.0 | 86.4 ± 4.7 | 92 [10-20]     | 20 [12-21]   | 828 [10-22]  | 382 [10-22] |
| Linear Ramp  | 84.6 ± 1.7          | 84.5 ± 1.2  | 81.7 ± 1.8 | 85.7 ± 2.4 | 10 [20]        | 10 [21]      | 10 [20]      | 7 [20]      |
| Linear Ramp* | 84.9 ± 1.6          | 84.8 ± 1.2  | 82.0 ± 1.8 | 86.0 ± 2.5 | 10 [20]        | 10 [21]      | 10 [20]      | 7 [20]      |
| Recursive TS | 85.2 ± 2.3          | 84.3 ± 3.3  | 82.1 ± 4.8 | 86.1 ± 4.3 | 92 [10-20]     | 20 [12-21]   | 505 [10-22]  | 381 [10-22] |
| TQA          | 76.2 ± 3.1          | 72.9 ± 4.2  | 72.5 ± 4.2 | 75.7 ± 4.4 | 92 [10-20]     | 20 [12-21]   | 822 [10-22]  | 361 [10-22] |
| TQA*         | 85.5 ± 2.3          | 85.7 ± 2.1  | 82.8 ± 3.1 | 87.3 ± 3.3 | 92 [10-20]     | 20 [12-21]   | 821 [10-22]  | 142 [10-22] |

Table 6: **Avg. Approx. Ratio and Number of Instances for Various Evaluation Methods for MAX-CUT for  $P = 2$ .** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Evaluation methods are shown in different colors for approximation ratios. Darker colors indicating better approximation ratios or more instances. Approximation ratios are in percentage, with  $\pm$  their standard deviation. The range of graph sizes are shown in brackets, next to the number of instances.

### 3 Tables for depth $P = 3$

|              | ER         | HH           | LB           | RR          |
|--------------|------------|--------------|--------------|-------------|
| MPS (Aer)    |            |              |              |             |
| Fixed Angle* | 7 [50]     | -            | 8 [100]      | 7 [100]     |
| Fixed Angle† | 7 [50]     | -            | 8 [100]      | 7 [100]     |
| Fourier*     | 20 [40–50] | 20 [39–144]  | 55 [40–100]  | 17 [40–100] |
| Interp.*     | 20 [40–50] | 20 [39–144]  | 55 [40–100]  | 17 [40–100] |
| Linear Ramp  | 10 [50]    | 10 [144]     | 10 [100]     | 7 [100]     |
| Linear Ramp* | 10 [50]    | 10 [144]     | 10 [100]     | 7 [100]     |
| TQA          | 10 [50]    | 10 [144]     | 10 [100]     | 7 [100]     |
| TQA*         | 10 [50]    | 10 [144]     | 10 [100]     | 7 [100]     |
| MPS (Quimb)  |            |              |              |             |
| Fixed Angle* | 7 [50]     | 10 [144]     | 8 [100]      | 37 [30–100] |
| Fixed Angle† | 7 [50]     | 10 [144]     | 8 [100]      | 37 [30–100] |
| Fourier*     | 20 [40–50] | 31 [39–144]  | 130 [40–100] | 48 [30–100] |
| Interp.*     | 20 [40–50] | 31 [39–144]  | 130 [40–100] | 48 [30–100] |
| Linear Ramp  | 10 [50]    | 10 [144]     | 10 [100]     | 7 [100]     |
| Linear Ramp* | 10 [50]    | 10 [144]     | 10 [100]     | 7 [100]     |
| Recursive TS | 10 [50]    | 11 [106–144] | 10 [100]     | 9 [40–100]  |
| TQA          | 10 [50]    | 10 [144]     | 20 [40–100]  | 37 [30–100] |
| TQA*         | 10 [50]    | 10 [144]     | 20 [40–100]  | 37 [30–100] |
| PP           |            |              |              |             |
| Fixed Angle* | 7 [50]     | 10 [144]     | 8 [100]      | 37 [30–100] |
| Fixed Angle† | 7 [50]     | 10 [144]     | 8 [100]      | 37 [30–100] |
| Fourier*     | 20 [40–50] | 31 [39–144]  | 130 [40–100] | 48 [30–100] |
| Interp.*     | 20 [40–50] | 31 [39–144]  | 130 [40–100] | 48 [30–100] |
| Linear Ramp  | 10 [50]    | 20 [105–144] | 20 [40–100]  | 8 [100]     |
| Linear Ramp* | 10 [50]    | 20 [105–144] | 20 [40–100]  | 8 [100]     |
| Recursive TS | 10 [50]    | 11 [106–144] | 10 [100]     | 9 [40–100]  |
| TQA          | 10 [50]    | 10 [144]     | 20 [40–100]  | 37 [30–100] |
| TQA*         | 10 [50]    | 10 [144]     | 20 [40–100]  | 37 [30–100] |
| SV           |            |              |              |             |
| Fixed Angle* | 81 [10–20] | -            | 283 [10–20]  | 365 [10–20] |
| Fixed Angle† | 81 [10–20] | -            | 283 [10–20]  | 372 [10–20] |
| Fourier*     | 92 [10–20] | 20 [12–21]   | 784 [10–20]  | 567 [10–20] |
| Interp.*     | 92 [10–20] | 20 [12–21]   | 827 [10–22]  | 382 [10–20] |
| Linear Ramp  | 10 [20]    | 10 [21]      | 10 [20]      | 7 [20]      |
| Linear Ramp* | 10 [20]    | 10 [21]      | 10 [20]      | 7 [20]      |
| Recursive TS | 92 [10–20] | 20 [12–21]   | 505 [10–22]  | 381 [10–20] |
| TQA          | 92 [10–20] | 20 [12–21]   | 821 [10–22]  | 357 [10–20] |
| TQA*         | 91 [10–20] | 20 [12–21]   | 821 [10–22]  | 141 [10–20] |

Table 7: **Number of Instances (num. nodes in brackets) for  $P = 3$ .** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Cells are colored based on the number of instances for the given configuration, with darker colors indicating more instances.

|                          | ER         | HH         | LB         | RR         |
|--------------------------|------------|------------|------------|------------|
| MPS (Aer)                |            |            |            |            |
| Fixed Angle <sup>*</sup> | 78.7 ± 3.5 | -          | 87.9 ± 1.3 | 72.6 ± 4.6 |
| Fixed Angle <sup>†</sup> | 75.2 ± 1.4 | -          | 84.6 ± 3.9 | 68.6 ± 2.3 |
| Fourier <sup>*</sup>     | 79.9 ± 4.0 | 77.9 ± 2.7 | 71.8 ± 7.4 | 74.5 ± 4.0 |
| Interp. <sup>*</sup>     | 81.1 ± 3.7 | 83.4 ± 3.0 | 80.0 ± 7.0 | 72.3 ± 3.3 |
| Linear Ramp              | 77.2 ± 2.1 | 77.5 ± 1.4 | 85.9 ± 1.6 | 70.2 ± 2.8 |
| Linear Ramp <sup>*</sup> | 78.6 ± 3.6 | 79.3 ± 1.2 | 87.9 ± 1.4 | 70.9 ± 3.2 |
| TQA                      | 69.4 ± 3.9 | 76.7 ± 2.0 | 78.5 ± 5.8 | 69.8 ± 2.5 |
| TQA <sup>*</sup>         | 75.9 ± 1.2 | 78.6 ± 2.5 | 82.9 ± 4.7 | 70.9 ± 3.3 |
| MPS (Quimb)              |            |            |            |            |
| Fixed Angle <sup>*</sup> | 84.5 ± 2.2 | 56.5 ± 1.5 | 86.4 ± 1.7 | 81.9 ± 6.0 |
| Fixed Angle <sup>†</sup> | 79.0 ± 2.1 | 54.1 ± 0.8 | 85.7 ± 2.3 | 79.1 ± 7.8 |
| Fourier <sup>*</sup>     | 84.2 ± 6.6 | 75.6 ± 6.8 | 80.1 ± 4.4 | 77.0 ± 5.6 |
| Interp. <sup>*</sup>     | 87.9 ± 1.5 | 79.0 ± 7.0 | 85.7 ± 1.6 | 80.5 ± 6.9 |
| Linear Ramp              | 86.0 ± 2.0 | 71.4 ± 1.2 | 84.1 ± 1.4 | 75.4 ± 5.4 |
| Linear Ramp <sup>*</sup> | 88.7 ± 2.5 | 70.7 ± 1.2 | 86.9 ± 1.4 | 76.3 ± 5.1 |
| Recursive TS             | 86.8 ± 4.0 | 58.1 ± 4.2 | 83.8 ± 1.4 | 70.6 ± 6.5 |
| TQA                      | 75.3 ± 3.6 | 57.0 ± 0.9 | 80.5 ± 3.3 | 74.7 ± 4.7 |
| TQA <sup>*</sup>         | 81.3 ± 5.0 | 57.5 ± 1.0 | 86.4 ± 3.7 | 79.5 ± 7.3 |
| PP                       |            |            |            |            |
| Fixed Angle <sup>*</sup> | 90.8 ± 1.2 | 86.4 ± 1.0 | 83.8 ± 1.6 | 87.4 ± 1.4 |
| Fixed Angle <sup>†</sup> | 89.7 ± 1.2 | 83.7 ± 1.5 | 83.0 ± 1.2 | 87.4 ± 1.3 |
| Fourier <sup>*</sup>     | 86.7 ± 1.9 | 83.9 ± 2.7 | 81.0 ± 2.6 | 84.7 ± 2.1 |
| Interp. <sup>*</sup>     | 89.0 ± 1.7 | 86.7 ± 1.1 | 83.3 ± 1.9 | 87.2 ± 1.4 |
| Linear Ramp              | 89.6 ± 1.0 | 85.4 ± 0.8 | 82.7 ± 1.8 | 85.8 ± 1.2 |
| Linear Ramp <sup>*</sup> | 91.2 ± 1.1 | 86.5 ± 0.9 | 84.2 ± 1.9 | 86.8 ± 0.9 |
| Recursive TS             | 87.0 ± 1.1 | 83.6 ± 0.9 | 82.7 ± 1.2 | 83.5 ± 1.6 |
| TQA                      | 80.7 ± 0.8 | 79.1 ± 1.0 | 71.4 ± 4.6 | 81.0 ± 1.5 |
| TQA <sup>*</sup>         | 89.4 ± 1.4 | 83.5 ± 0.8 | 80.0 ± 2.1 | 86.9 ± 1.6 |
| SV                       |            |            |            |            |
| Fixed Angle <sup>*</sup> | 89.1 ± 1.8 | -          | 87.7 ± 2.6 | 91.6 ± 2.8 |
| Fixed Angle <sup>†</sup> | 87.1 ± 1.7 | -          | 87.2 ± 2.5 | 89.9 ± 2.2 |
| Fourier <sup>*</sup>     | 83.8 ± 5.6 | 79.8 ± 5.3 | 78.7 ± 7.5 | 88.5 ± 4.0 |
| Interp. <sup>*</sup>     | 88.8 ± 1.9 | 89.0 ± 1.6 | 87.4 ± 3.8 | 89.6 ± 4.1 |
| Linear Ramp              | 87.7 ± 1.5 | 86.8 ± 1.8 | 84.9 ± 1.8 | 88.5 ± 2.3 |
| Linear Ramp <sup>*</sup> | 88.5 ± 1.4 | 88.8 ± 1.3 | 86.9 ± 1.9 | 89.5 ± 2.0 |
| Recursive TS             | 87.8 ± 2.4 | 87.3 ± 3.3 | 85.7 ± 4.9 | 88.5 ± 3.8 |
| TQA                      | 81.3 ± 3.8 | 79.5 ± 4.1 | 77.3 ± 5.1 | 80.1 ± 5.5 |
| TQA <sup>*</sup>         | 87.7 ± 2.2 | 87.0 ± 2.1 | 85.1 ± 3.8 | 89.8 ± 2.6 |

Table 8: **Avg. Approx. Ratio ± standard deviation in percentage for MAXCUT for  $P = 3$ .** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Evaluation methods are shown in different colors, with darker colors indicating better approximation ratios.

|              | Approximation Ratio |            |            |            | Num. Instances |              |              |             |
|--------------|---------------------|------------|------------|------------|----------------|--------------|--------------|-------------|
|              | ER                  | HH         | LB         | RR         | ER             | HH           | LB           | RR          |
| MPS (Aer)    |                     |            |            |            |                |              |              |             |
| Fixed Angle* | 78.7 ± 3.5          | -          | 87.9 ± 1.3 | 72.6 ± 4.6 | 7 [50]         | -            | 8 [100]      | 7 [100]     |
| Fixed Angle† | 75.2 ± 1.4          | -          | 84.6 ± 3.9 | 68.6 ± 2.3 | 7 [50]         | -            | 8 [100]      | 7 [100]     |
| Fourier*     | 79.9 ± 4.0          | 77.9 ± 2.7 | 71.8 ± 7.4 | 74.5 ± 4.0 | 20 [40–50]     | 20 [39–144]  | 55 [40–100]  | 17 [40–100] |
| Interp.*     | 81.1 ± 3.7          | 83.4 ± 3.0 | 80.0 ± 7.0 | 72.3 ± 3.3 | 20 [40–50]     | 20 [39–144]  | 55 [40–100]  | 17 [40–100] |
| Linear Ramp  | 77.2 ± 2.1          | 77.5 ± 1.4 | 85.9 ± 1.6 | 70.2 ± 2.8 | 10 [50]        | 10 [144]     | 10 [100]     | 7 [100]     |
| Linear Ramp* | 78.6 ± 3.6          | 79.3 ± 1.2 | 87.9 ± 1.4 | 70.9 ± 3.2 | 10 [50]        | 10 [144]     | 10 [100]     | 7 [100]     |
| TQA          | 69.4 ± 3.9          | 76.7 ± 2.0 | 78.5 ± 5.8 | 69.8 ± 2.5 | 10 [50]        | 10 [144]     | 10 [100]     | 7 [100]     |
| TQA*         | 75.9 ± 1.2          | 78.6 ± 2.5 | 82.9 ± 4.7 | 70.9 ± 3.3 | 10 [50]        | 10 [144]     | 10 [100]     | 7 [100]     |
| MPS (Quimb)  |                     |            |            |            |                |              |              |             |
| Fixed Angle* | 84.5 ± 2.2          | 56.5 ± 1.5 | 86.4 ± 1.7 | 81.9 ± 6.0 | 7 [50]         | 10 [144]     | 8 [100]      | 37 [30–100] |
| Fixed Angle† | 79.0 ± 2.1          | 54.1 ± 0.8 | 85.7 ± 2.3 | 79.1 ± 7.8 | 7 [50]         | 10 [144]     | 8 [100]      | 37 [30–100] |
| Fourier*     | 84.2 ± 6.6          | 75.6 ± 6.8 | 80.1 ± 4.4 | 77.0 ± 5.6 | 20 [40–50]     | 31 [39–144]  | 130 [40–100] | 48 [30–100] |
| Interp.*     | 87.9 ± 1.5          | 79.0 ± 7.0 | 85.7 ± 1.6 | 80.5 ± 6.9 | 20 [40–50]     | 31 [39–144]  | 130 [40–100] | 48 [30–100] |
| Linear Ramp  | 86.0 ± 2.0          | 71.4 ± 1.2 | 84.1 ± 1.4 | 75.4 ± 5.4 | 10 [50]        | 10 [144]     | 10 [100]     | 7 [100]     |
| Linear Ramp* | 88.7 ± 2.5          | 70.7 ± 1.2 | 86.9 ± 1.4 | 76.3 ± 5.1 | 10 [50]        | 10 [144]     | 10 [100]     | 7 [100]     |
| Recursive TS | 86.8 ± 4.0          | 58.1 ± 4.2 | 83.8 ± 1.4 | 70.6 ± 6.5 | 10 [50]        | 11 [106–144] | 10 [100]     | 9 [40–100]  |
| TQA          | 75.3 ± 3.6          | 57.0 ± 0.9 | 80.5 ± 3.3 | 74.7 ± 4.7 | 10 [50]        | 10 [144]     | 20 [40–100]  | 37 [30–100] |
| TQA*         | 81.3 ± 5.0          | 57.5 ± 1.0 | 86.4 ± 3.7 | 79.5 ± 7.3 | 10 [50]        | 10 [144]     | 20 [40–100]  | 37 [30–100] |
| PP           |                     |            |            |            |                |              |              |             |
| Fixed Angle* | 90.8 ± 1.2          | 86.4 ± 1.0 | 83.8 ± 1.6 | 87.4 ± 1.4 | 7 [50]         | 10 [144]     | 8 [100]      | 37 [30–100] |
| Fixed Angle† | 89.7 ± 1.2          | 83.7 ± 1.5 | 83.0 ± 1.2 | 87.4 ± 1.3 | 7 [50]         | 10 [144]     | 8 [100]      | 37 [30–100] |
| Fourier*     | 86.7 ± 1.9          | 83.9 ± 2.7 | 81.0 ± 2.6 | 84.7 ± 2.1 | 20 [40–50]     | 31 [39–144]  | 130 [40–100] | 48 [30–100] |
| Interp.*     | 89.0 ± 1.7          | 86.7 ± 1.1 | 83.3 ± 1.9 | 87.2 ± 1.4 | 20 [40–50]     | 31 [39–144]  | 130 [40–100] | 48 [30–100] |
| Linear Ramp  | 89.6 ± 1.0          | 85.4 ± 0.8 | 82.7 ± 1.8 | 85.8 ± 1.2 | 10 [50]        | 20 [105–144] | 20 [40–100]  | 8 [100]     |
| Linear Ramp* | 91.2 ± 1.1          | 86.5 ± 0.9 | 84.2 ± 1.9 | 86.8 ± 0.9 | 10 [50]        | 20 [105–144] | 20 [40–100]  | 8 [100]     |
| Recursive TS | 87.0 ± 1.1          | 83.6 ± 0.9 | 82.7 ± 1.2 | 83.5 ± 1.6 | 10 [50]        | 11 [106–144] | 10 [100]     | 9 [40–100]  |
| TQA          | 80.7 ± 0.8          | 79.1 ± 1.0 | 71.4 ± 4.6 | 81.0 ± 1.5 | 10 [50]        | 10 [144]     | 20 [40–100]  | 37 [30–100] |
| TQA*         | 89.4 ± 1.4          | 83.5 ± 0.8 | 80.0 ± 2.1 | 86.9 ± 1.6 | 10 [50]        | 10 [144]     | 20 [40–100]  | 37 [30–100] |
| SV           |                     |            |            |            |                |              |              |             |
| Fixed Angle* | 89.1 ± 1.8          | -          | 87.7 ± 2.6 | 91.6 ± 2.8 | 81 [10–20]     | -            | 283 [10–20]  | 365 [10–20] |
| Fixed Angle† | 87.1 ± 1.7          | -          | 87.2 ± 2.5 | 89.9 ± 2.2 | 81 [10–20]     | -            | 283 [10–20]  | 372 [10–20] |
| Fourier*     | 83.8 ± 5.6          | 79.8 ± 5.3 | 78.7 ± 7.5 | 88.5 ± 4.0 | 92 [10–20]     | 20 [12–21]   | 784 [10–20]  | 567 [10–20] |
| Interp.*     | 88.8 ± 1.9          | 89.0 ± 1.6 | 87.4 ± 3.8 | 89.6 ± 4.1 | 92 [10–20]     | 20 [12–21]   | 827 [10–22]  | 382 [10–20] |
| Linear Ramp  | 87.7 ± 1.5          | 86.8 ± 1.8 | 84.9 ± 1.8 | 88.5 ± 2.3 | 10 [20]        | 10 [21]      | 10 [20]      | 7 [20]      |
| Linear Ramp* | 88.5 ± 1.4          | 88.8 ± 1.3 | 86.9 ± 1.9 | 89.5 ± 2.0 | 10 [20]        | 10 [21]      | 10 [20]      | 7 [20]      |
| Recursive TS | 87.8 ± 2.4          | 87.3 ± 3.3 | 85.7 ± 4.9 | 88.5 ± 3.8 | 92 [10–20]     | 20 [12–21]   | 505 [10–22]  | 381 [10–20] |
| TQA          | 81.3 ± 3.8          | 79.5 ± 4.1 | 77.3 ± 5.1 | 80.1 ± 5.5 | 92 [10–20]     | 20 [12–21]   | 821 [10–22]  | 357 [10–20] |
| TQA*         | 87.7 ± 2.2          | 87.0 ± 2.1 | 85.1 ± 3.8 | 89.8 ± 2.6 | 91 [10–20]     | 20 [12–21]   | 821 [10–22]  | 141 [10–20] |

Table 9: **Avg. Approx. Ratio and Number of Instances for Various Evaluation Methods for MAX-CUT for  $P = 3$ .** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Evaluation methods are shown in different colors for approximation ratios. Darker colors indicating better approximation ratios or more instances. Approximation ratios are in percentage, with  $\pm$  their standard deviation. The range of graph sizes are shown in brackets, next to the number of instances.

## 4 Tables for depth $P = 4$

|              | ER         | HH           | LB           | RR          |
|--------------|------------|--------------|--------------|-------------|
| MPS (Aer)    |            |              |              |             |
| Fixed Angle* | -          | 10 [39]      | 21 [100]     | 13 [40-100] |
| Fixed Angle† | -          | 10 [39]      | 21 [100]     | 13 [40-100] |
| Fourier*     | 20 [40-50] | 20 [39-144]  | 55 [40-100]  | 17 [40-100] |
| Interp.*     | 20 [40-50] | 20 [39-144]  | 55 [40-100]  | 17 [40-100] |
| Linear Ramp  | 10 [50]    | 10 [144]     | 10 [100]     | 7 [100]     |
| Linear Ramp* | 10 [50]    | 10 [144]     | 10 [100]     | 7 [100]     |
| TQA          | 20 [40-50] | 20 [39-144]  | 56 [40-100]  | 17 [40-100] |
| TQA*         | 20 [40-50] | 20 [39-144]  | 56 [40-100]  | 17 [40-100] |
| MPS (Quimb)  |            |              |              |             |
| Fixed Angle* | -          | 10 [144]     | 30 [100]     | 13 [40-100] |
| Fixed Angle† | -          | 10 [144]     | 30 [100]     | 13 [40-100] |
| Fourier*     | 20 [40-50] | 31 [39-144]  | 130 [40-100] | 28 [40-100] |
| Interp.*     | 20 [40-50] | 31 [39-144]  | 130 [40-100] | 28 [40-100] |
| Linear Ramp  | 10 [50]    | 10 [144]     | 10 [100]     | 7 [100]     |
| Linear Ramp* | 10 [50]    | 10 [144]     | 10 [100]     | 7 [100]     |
| Recursive TS | 10 [50]    | 11 [106-144] | 10 [100]     | 9 [40-100]  |
| TQA          | 20 [40-50] | 20 [39-144]  | 130 [40-100] | 27 [40-100] |
| TQA*         | 20 [40-50] | 20 [39-144]  | 130 [40-100] | 27 [40-100] |
| PP           |            |              |              |             |
| Fixed Angle* | -          | 30 [39-144]  | 21 [100]     | 13 [40-100] |
| Fixed Angle† | -          | 30 [39-144]  | 21 [100]     | 13 [40-100] |
| Fourier*     | 20 [40-50] | 31 [39-144]  | 130 [40-100] | 28 [40-100] |
| Interp.*     | 20 [40-50] | 31 [39-144]  | 130 [40-100] | 28 [40-100] |
| Linear Ramp  | 10 [50]    | 20 [105-144] | 20 [40-100]  | 8 [100]     |
| Linear Ramp* | 10 [50]    | 20 [105-144] | 20 [40-100]  | 8 [100]     |
| Recursive TS | 10 [50]    | 11 [106-144] | 10 [100]     | 9 [40-100]  |
| TQA          | 20 [40-50] | 20 [39-144]  | 130 [40-100] | 27 [40-100] |
| TQA*         | 20 [40-50] | 20 [39-144]  | 130 [40-100] | 27 [40-100] |
| SV           |            |              |              |             |
| Fixed Angle* | 61 [10-20] | -            | 106 [10-20]  | 155 [10-20] |
| Fixed Angle† | 61 [10-20] | -            | 106 [10-20]  | 158 [10-20] |
| Fourier*     | 92 [10-20] | 20 [12-21]   | 784 [10-20]  | 567 [10-20] |
| Interp.*     | 92 [10-20] | 20 [12-21]   | 827 [10-22]  | 382 [10-20] |
| Linear Ramp  | 10 [20]    | 10 [21]      | 10 [20]      | 7 [20]      |
| Linear Ramp* | 10 [20]    | 10 [21]      | 10 [20]      | 7 [20]      |
| Recursive TS | 92 [10-20] | 20 [12-21]   | 504 [10-22]  | 381 [10-20] |
| TQA          | 92 [10-20] | 20 [12-21]   | 821 [10-22]  | 355 [10-20] |
| TQA*         | 91 [10-20] | 20 [12-21]   | 821 [10-22]  | 142 [10-20] |

Table 10: **Number of Instances (num. nodes in brackets) for  $P = 4$ .** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Cells are colored based on the number of instances for the given configuration, with darker colors indicating more instances.

|                          | ER              | HH              | LB             | RR              |
|--------------------------|-----------------|-----------------|----------------|-----------------|
| MPS (Aer)                |                 |                 |                |                 |
| Fixed Angle*             | -               | $88.8 \pm 1.3$  | $82.8 \pm 6.9$ | $88.6 \pm 8.8$  |
| Fixed Angle <sup>†</sup> | -               | $87.9 \pm 1.5$  | $77.1 \pm 8.3$ | $85.4 \pm 11.0$ |
| Fourier*                 | $75.6 \pm 4.5$  | $80.9 \pm 2.7$  | $74.4 \pm 8.1$ | $76.6 \pm 5.5$  |
| Interp.*                 | $87.0 \pm 2.6$  | $85.7 \pm 3.4$  | $86.1 \pm 6.2$ | $77.7 \pm 6.7$  |
| Linear Ramp              | $81.7 \pm 4.7$  | $79.9 \pm 1.7$  | $88.8 \pm 1.8$ | $74.5 \pm 5.8$  |
| Linear Ramp*             | $85.8 \pm 4.1$  | $81.4 \pm 1.0$  | $91.1 \pm 1.2$ | $75.0 \pm 6.0$  |
| TQA                      | $69.6 \pm 1.8$  | $79.8 \pm 2.1$  | $74.4 \pm 9.2$ | $75.4 \pm 5.1$  |
| TQA*                     | $77.8 \pm 4.9$  | $82.5 \pm 2.7$  | $78.7 \pm 9.6$ | $79.2 \pm 6.3$  |
| MPS (Quimb)              |                 |                 |                |                 |
| Fixed Angle*             | -               | $52.8 \pm 1.1$  | $90.5 \pm 1.4$ | $89.5 \pm 5.9$  |
| Fixed Angle <sup>†</sup> | -               | $51.0 \pm 0.9$  | $90.4 \pm 1.5$ | $84.8 \pm 10.7$ |
| Fourier*                 | $77.4 \pm 6.4$  | $76.0 \pm 8.2$  | $83.0 \pm 4.4$ | $83.6 \pm 9.6$  |
| Interp.*                 | $92.1 \pm 1.1$  | $79.5 \pm 8.6$  | $89.9 \pm 1.1$ | $87.3 \pm 8.4$  |
| Linear Ramp              | $90.7 \pm 1.5$  | $69.4 \pm 2.0$  | $87.7 \pm 1.6$ | $76.7 \pm 5.8$  |
| Linear Ramp*             | $92.4 \pm 1.9$  | $68.8 \pm 2.0$  | $90.2 \pm 1.0$ | $77.8 \pm 7.4$  |
| Recursive TS             | $92.3 \pm 2.0$  | $57.6 \pm 5.6$  | $88.7 \pm 1.9$ | $73.9 \pm 7.7$  |
| TQA                      | $70.5 \pm 3.4$  | $68.1 \pm 14.3$ | $84.2 \pm 2.4$ | $77.4 \pm 5.5$  |
| TQA*                     | $79.0 \pm 7.0$  | $70.7 \pm 16.6$ | $88.5 \pm 1.9$ | $84.2 \pm 7.9$  |
| PP                       |                 |                 |                |                 |
| Fixed Angle*             | -               | $89.3 \pm 1.1$  | $86.2 \pm 1.7$ | $89.6 \pm 1.5$  |
| Fixed Angle <sup>†</sup> | -               | $88.2 \pm 1.5$  | $84.2 \pm 1.4$ | $89.4 \pm 1.6$  |
| Fourier*                 | $88.0 \pm 2.1$  | $86.4 \pm 2.7$  | $82.4 \pm 2.9$ | $87.2 \pm 1.8$  |
| Interp.*                 | $90.7 \pm 1.9$  | $89.3 \pm 1.1$  | $85.3 \pm 1.8$ | $89.1 \pm 1.5$  |
| Linear Ramp              | $91.4 \pm 1.0$  | $87.5 \pm 0.7$  | $85.2 \pm 2.3$ | $87.9 \pm 0.7$  |
| Linear Ramp*             | $92.9 \pm 1.1$  | $88.9 \pm 0.8$  | $86.6 \pm 2.0$ | $88.6 \pm 0.7$  |
| Recursive TS             | $88.6 \pm 1.1$  | $85.6 \pm 0.8$  | $84.6 \pm 1.4$ | $85.8 \pm 1.7$  |
| TQA                      | $83.7 \pm 1.4$  | $81.9 \pm 1.3$  | $75.7 \pm 4.8$ | $85.1 \pm 1.8$  |
| TQA*                     | $89.2 \pm 1.9$  | $86.8 \pm 1.2$  | $83.5 \pm 1.9$ | $87.8 \pm 1.4$  |
| SV                       |                 |                 |                |                 |
| Fixed Angle*             | $91.4 \pm 1.6$  | -               | $91.7 \pm 2.4$ | $93.4 \pm 2.2$  |
| Fixed Angle <sup>†</sup> | $89.0 \pm 2.1$  | -               | $90.9 \pm 2.4$ | $92.8 \pm 2.2$  |
| Fourier*                 | $78.4 \pm 10.6$ | $81.3 \pm 6.1$  | $80.7 \pm 7.8$ | $90.3 \pm 4.4$  |
| Interp.*                 | $91.2 \pm 1.6$  | $91.3 \pm 1.4$  | $90.3 \pm 3.8$ | $90.9 \pm 4.9$  |
| Linear Ramp              | $89.4 \pm 1.4$  | $88.7 \pm 2.5$  | $87.4 \pm 2.1$ | $90.4 \pm 2.2$  |
| Linear Ramp*             | $90.9 \pm 1.2$  | $90.5 \pm 1.4$  | $89.9 \pm 2.0$ | $91.8 \pm 1.5$  |
| Recursive TS             | $90.1 \pm 2.2$  | $89.0 \pm 2.8$  | $88.3 \pm 4.8$ | $90.1 \pm 3.5$  |
| TQA                      | $84.8 \pm 3.8$  | $81.8 \pm 4.3$  | $79.9 \pm 6.0$ | $84.0 \pm 5.6$  |
| TQA*                     | $89.4 \pm 1.9$  | $88.8 \pm 1.8$  | $87.8 \pm 2.7$ | $91.0 \pm 2.5$  |

Table 11: **Avg. Approx. Ratio  $\pm$  standard deviation in percentage for MAXCUT for  $P = 4$ .** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Evaluation methods are shown in different colors, with darker colors indicating better approximation ratios.

|              | ER              | Approximation Ratio |                |                 |            | Num. Instances |              |          |  |
|--------------|-----------------|---------------------|----------------|-----------------|------------|----------------|--------------|----------|--|
|              | ER              | HH                  | LB             | RR              | ER         | HH             | LB           | RR       |  |
| MPS (Aer)    |                 |                     |                |                 |            |                |              |          |  |
| Fixed Angle* | -               | $88.8 \pm 1.3$      | $82.8 \pm 6.9$ | $88.6 \pm 8.8$  | -          | 10 [39]        | 21 [100]     | 13 [40]  |  |
| Fixed Angle† | -               | $87.9 \pm 1.5$      | $77.1 \pm 8.3$ | $85.4 \pm 11.0$ | -          | 10 [39]        | 21 [100]     | 13 [40]  |  |
| Fourier*     | $75.6 \pm 4.5$  | $80.9 \pm 2.7$      | $74.4 \pm 8.1$ | $76.6 \pm 5.5$  | 20 [40–50] | 20 [39–144]    | 55 [40–100]  | 17 [40]  |  |
| Interp.*     | $87.0 \pm 2.6$  | $85.7 \pm 3.4$      | $86.1 \pm 6.2$ | $77.7 \pm 6.7$  | 20 [40–50] | 20 [39–144]    | 55 [40–100]  | 17 [40]  |  |
| Linear Ramp  | $81.7 \pm 4.7$  | $79.9 \pm 1.7$      | $88.8 \pm 1.8$ | $74.5 \pm 5.8$  | 10 [50]    | 10 [144]       | 10 [100]     | 7 [10]   |  |
| Linear Ramp* | $85.8 \pm 4.1$  | $81.4 \pm 1.0$      | $91.1 \pm 1.2$ | $75.0 \pm 6.0$  | 10 [50]    | 10 [144]       | 10 [100]     | 7 [10]   |  |
| TQA          | $69.6 \pm 1.8$  | $79.8 \pm 2.1$      | $74.4 \pm 9.2$ | $75.4 \pm 5.1$  | 20 [40–50] | 20 [39–144]    | 56 [40–100]  | 17 [40]  |  |
| TQA*         | $77.8 \pm 4.9$  | $82.5 \pm 2.7$      | $78.7 \pm 9.6$ | $79.2 \pm 6.3$  | 20 [40–50] | 20 [39–144]    | 56 [40–100]  | 17 [40]  |  |
| MPS (Quimb)  |                 |                     |                |                 |            |                |              |          |  |
| Fixed Angle* | -               | $52.8 \pm 1.1$      | $90.5 \pm 1.4$ | $89.5 \pm 5.9$  | -          | 10 [144]       | 30 [100]     | 13 [40]  |  |
| Fixed Angle† | -               | $51.0 \pm 0.9$      | $90.4 \pm 1.5$ | $84.8 \pm 10.7$ | -          | 10 [144]       | 30 [100]     | 13 [40]  |  |
| Fourier*     | $77.4 \pm 6.4$  | $76.0 \pm 8.2$      | $83.0 \pm 4.4$ | $83.6 \pm 9.6$  | 20 [40–50] | 31 [39–144]    | 130 [40–100] | 28 [40]  |  |
| Interp.*     | $92.1 \pm 1.1$  | $79.5 \pm 8.6$      | $89.9 \pm 1.1$ | $87.3 \pm 8.4$  | 20 [40–50] | 31 [39–144]    | 130 [40–100] | 28 [40]  |  |
| Linear Ramp  | $90.7 \pm 1.5$  | $69.4 \pm 2.0$      | $87.7 \pm 1.6$ | $76.7 \pm 5.8$  | 10 [50]    | 10 [144]       | 10 [100]     | 7 [10]   |  |
| Linear Ramp* | $92.4 \pm 1.9$  | $68.8 \pm 2.0$      | $90.2 \pm 1.0$ | $77.8 \pm 7.4$  | 10 [50]    | 10 [144]       | 10 [100]     | 7 [10]   |  |
| Recursive TS | $92.3 \pm 2.0$  | $57.6 \pm 5.6$      | $88.7 \pm 1.9$ | $73.9 \pm 7.7$  | 10 [50]    | 11 [106–144]   | 10 [100]     | 9 [40]   |  |
| TQA          | $70.5 \pm 3.4$  | $68.1 \pm 14.3$     | $84.2 \pm 2.4$ | $77.4 \pm 5.5$  | 20 [40–50] | 20 [39–144]    | 130 [40–100] | 27 [40]  |  |
| TQA*         | $79.0 \pm 7.0$  | $70.7 \pm 16.6$     | $88.5 \pm 1.9$ | $84.2 \pm 7.9$  | 20 [40–50] | 20 [39–144]    | 130 [40–100] | 27 [40]  |  |
| PP           |                 |                     |                |                 |            |                |              |          |  |
| Fixed Angle* | -               | $89.3 \pm 1.1$      | $86.2 \pm 1.7$ | $89.6 \pm 1.5$  | -          | 30 [39–144]    | 21 [100]     | 13 [40]  |  |
| Fixed Angle† | -               | $88.2 \pm 1.5$      | $84.2 \pm 1.4$ | $89.4 \pm 1.6$  | -          | 30 [39–144]    | 21 [100]     | 13 [40]  |  |
| Fourier*     | $88.0 \pm 2.1$  | $86.4 \pm 2.7$      | $82.4 \pm 2.9$ | $87.2 \pm 1.8$  | 20 [40–50] | 31 [39–144]    | 130 [40–100] | 28 [40]  |  |
| Interp.*     | $90.7 \pm 1.9$  | $89.3 \pm 1.1$      | $85.3 \pm 1.8$ | $89.1 \pm 1.5$  | 20 [40–50] | 31 [39–144]    | 130 [40–100] | 28 [40]  |  |
| Linear Ramp  | $91.4 \pm 1.0$  | $87.5 \pm 0.7$      | $85.2 \pm 2.3$ | $87.9 \pm 0.7$  | 10 [50]    | 20 [105–144]   | 20 [40–100]  | 8 [10]   |  |
| Linear Ramp* | $92.9 \pm 1.1$  | $88.9 \pm 0.8$      | $86.6 \pm 2.0$ | $88.6 \pm 0.7$  | 10 [50]    | 20 [105–144]   | 20 [40–100]  | 8 [10]   |  |
| Recursive TS | $88.6 \pm 1.1$  | $85.6 \pm 0.8$      | $84.6 \pm 1.4$ | $85.8 \pm 1.7$  | 10 [50]    | 11 [106–144]   | 10 [100]     | 9 [40]   |  |
| TQA          | $83.7 \pm 1.4$  | $81.9 \pm 1.3$      | $75.7 \pm 4.8$ | $85.1 \pm 1.8$  | 20 [40–50] | 20 [39–144]    | 130 [40–100] | 27 [40]  |  |
| TQA*         | $89.2 \pm 1.9$  | $86.8 \pm 1.2$      | $83.5 \pm 1.9$ | $87.8 \pm 1.4$  | 20 [40–50] | 20 [39–144]    | 130 [40–100] | 27 [40]  |  |
| SV           |                 |                     |                |                 |            |                |              |          |  |
| Fixed Angle* | $91.4 \pm 1.6$  | -                   | $91.7 \pm 2.4$ | $93.4 \pm 2.2$  | 61 [10–20] | -              | 106 [10–20]  | 155 [10] |  |
| Fixed Angle† | $89.0 \pm 2.1$  | -                   | $90.9 \pm 2.4$ | $92.8 \pm 2.2$  | 61 [10–20] | -              | 106 [10–20]  | 158 [10] |  |
| Fourier*     | $78.4 \pm 10.6$ | $81.3 \pm 6.1$      | $80.7 \pm 7.8$ | $90.3 \pm 4.4$  | 92 [10–20] | 20 [12–21]     | 784 [10–20]  | 567 [10] |  |
| Interp.*     | $91.2 \pm 1.6$  | $91.3 \pm 1.4$      | $90.3 \pm 3.8$ | $90.9 \pm 4.9$  | 92 [10–20] | 20 [12–21]     | 827 [10–22]  | 382 [10] |  |
| Linear Ramp  | $89.4 \pm 1.4$  | $88.7 \pm 2.5$      | $87.4 \pm 2.1$ | $90.4 \pm 2.2$  | 10 [20]    | 10 [21]        | 10 [20]      | 7 [2]    |  |
| Linear Ramp* | $90.9 \pm 1.2$  | $90.5 \pm 1.4$      | $89.9 \pm 2.0$ | $91.8 \pm 1.5$  | 10 [20]    | 10 [21]        | 10 [20]      | 7 [2]    |  |
| Recursive TS | $90.1 \pm 2.2$  | $89.0 \pm 2.8$      | $88.3 \pm 4.8$ | $90.1 \pm 3.5$  | 92 [10–20] | 20 [12–21]     | 504 [10–22]  | 381 [10] |  |
| TQA          | $84.8 \pm 3.8$  | $81.8 \pm 4.3$      | $79.9 \pm 6.0$ | $84.0 \pm 5.6$  | 92 [10–20] | 20 [12–21]     | 821 [10–22]  | 355 [10] |  |
| TQA*         | $89.4 \pm 1.9$  | $88.8 \pm 1.8$      | $87.8 \pm 2.7$ | $91.0 \pm 2.5$  | 91 [10–20] | 20 [12–21]     | 821 [10–22]  | 142 [10] |  |

Table 12: **Avg. Approx. Ratio and Number of Instances for Various Evaluation Methods for MAX-CUT for  $P = 4$ .** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Evaluation methods are shown in different colors for approximation ratios. Darker colors indicating better approximation ratios or more instances. Approximation ratios are in percentage, with  $\pm$  their standard deviation. The range of graph sizes are shown in brackets, next to the number of instances.

## 5 Tables for depth $P = 5$

|              | ER         | HH           | LB           | RR          |
|--------------|------------|--------------|--------------|-------------|
| MPS (Aer)    |            |              |              |             |
| Fixed Angle* | -          | 10 [144]     | 2 [100]      | 2 [100]     |
| Fixed Angle† | -          | 10 [144]     | 2 [100]      | 2 [100]     |
| Fourier*     | 20 [40–50] | 20 [39–144]  | 55 [40–100]  | 17 [40–100] |
| Interp.*     | 20 [40–50] | 20 [39–144]  | 55 [40–100]  | 17 [40–100] |
| Linear Ramp  | 10 [50]    | 10 [144]     | 19 [100]     | 7 [100]     |
| Linear Ramp* | 10 [50]    | 10 [144]     | 10 [100]     | 7 [100]     |
| TQA          | 10 [50]    | 10 [144]     | 10 [100]     | 7 [100]     |
| TQA*         | 10 [50]    | 10 [144]     | 10 [100]     | 7 [100]     |
| MPS (Quimb)  |            |              |              |             |
| Fixed Angle* | -          | -            | 2 [100]      | 12 [40–100] |
| Fixed Angle† | -          | -            | 2 [100]      | 12 [40–100] |
| Fourier*     | 20 [40–50] | 31 [39–144]  | 130 [40–100] | 28 [40–100] |
| Interp.*     | 20 [40–50] | 31 [39–144]  | 130 [40–100] | 28 [40–100] |
| Linear Ramp  | 10 [50]    | 10 [144]     | 10 [100]     | 7 [100]     |
| Linear Ramp* | 10 [50]    | 10 [144]     | 10 [100]     | 7 [100]     |
| Recursive TS | 10 [50]    | 11 [106–144] | 10 [100]     | 9 [40–100]  |
| TQA          | 10 [50]    | 20 [105–144] | 20 [40–100]  | 17 [40–100] |
| TQA*         | 10 [50]    | 20 [105–144] | 20 [40–100]  | 17 [40–100] |
| PP           |            |              |              |             |
| Fixed Angle* | -          | 10 [144]     | 2 [100]      | 12 [40–100] |
| Fixed Angle† | -          | 10 [144]     | 2 [100]      | 12 [40–100] |
| Fourier*     | 20 [40–50] | 31 [39–144]  | 130 [40–100] | 28 [40–100] |
| Interp.*     | 20 [40–50] | 31 [39–144]  | 130 [40–100] | 28 [40–100] |
| Linear Ramp  | 10 [50]    | 20 [105–144] | 29 [40–100]  | 8 [100]     |
| Linear Ramp* | 10 [50]    | 20 [105–144] | 20 [40–100]  | 8 [100]     |
| Recursive TS | 10 [50]    | 11 [106–144] | 10 [100]     | 9 [40–100]  |
| TQA          | 10 [50]    | 20 [105–144] | 20 [40–100]  | 17 [40–100] |
| TQA*         | 10 [50]    | 20 [105–144] | 20 [40–100]  | 17 [40–100] |
| SV           |            |              |              |             |
| Fixed Angle* | 46 [10–20] | -            | 60 [10–20]   | 115 [10–20] |
| Fixed Angle† | 46 [10–20] | -            | 61 [10–20]   | 121 [10–20] |
| Fourier*     | 92 [10–20] | 20 [12–21]   | 784 [10–20]  | 382 [10–20] |
| Interp.*     | 92 [10–20] | 20 [12–21]   | 781 [10–20]  | 382 [10–20] |
| Linear Ramp  | 10 [20]    | 10 [21]      | 10 [20]      | 7 [20]      |
| Linear Ramp* | 10 [20]    | 10 [21]      | 10 [20]      | 7 [20]      |
| Recursive TS | 92 [10–20] | 20 [12–21]   | 416 [10–20]  | 381 [10–20] |
| TQA          | 92 [10–20] | 20 [12–21]   | 677 [10–20]  | 359 [10–20] |
| TQA*         | 90 [10–20] | 20 [12–21]   | 672 [10–20]  | 142 [10–20] |

Table 13: **Number of Instances (num. nodes in brackets) for  $P = 5$ .** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Cells are colored based on the number of instances for the given configuration, with darker colors indicating more instances.

|                          | ER              | HH             | LB              | RR              |
|--------------------------|-----------------|----------------|-----------------|-----------------|
| MPS (Aer)                |                 |                |                 |                 |
| Fixed Angle*             | -               | $84.3 \pm 3.3$ | $94.5 \pm 0.7$  | $79.1 \pm 2.5$  |
| Fixed Angle <sup>†</sup> | -               | $83.3 \pm 3.4$ | $94.4 \pm 0.6$  | $68.1 \pm 0.6$  |
| Fourier*                 | $79.9 \pm 6.1$  | $82.4 \pm 3.3$ | $78.1 \pm 10.5$ | $76.8 \pm 6.3$  |
| Interp.*                 | $90.5 \pm 2.0$  | $87.2 \pm 3.8$ | $89.3 \pm 5.3$  | $80.8 \pm 9.1$  |
| Linear Ramp              | $85.1 \pm 4.6$  | $85.7 \pm 1.0$ | $78.4 \pm 14.1$ | $79.7 \pm 5.5$  |
| Linear Ramp*             | $88.8 \pm 2.7$  | $82.0 \pm 2.0$ | $93.4 \pm 0.8$  | $82.3 \pm 5.4$  |
| TQA                      | $67.9 \pm 7.7$  | $79.2 \pm 1.2$ | $89.2 \pm 1.8$  | $74.9 \pm 7.2$  |
| TQA*                     | $75.4 \pm 2.6$  | $81.5 \pm 1.3$ | $91.0 \pm 1.2$  | $80.1 \pm 8.9$  |
| MPS (Quimb)              |                 |                |                 |                 |
| Fixed Angle*             | -               | -              | $94.4 \pm 0.7$  | $93.2 \pm 4.2$  |
| Fixed Angle <sup>†</sup> | -               | -              | $94.3 \pm 0.7$  | $90.0 \pm 8.9$  |
| Fourier*                 | $82.0 \pm 7.5$  | $77.1 \pm 8.9$ | $85.8 \pm 4.6$  | $86.4 \pm 10.4$ |
| Interp.*                 | $94.5 \pm 1.2$  | $81.5 \pm 8.7$ | $92.4 \pm 1.1$  | $90.4 \pm 7.1$  |
| Linear Ramp              | $93.1 \pm 1.3$  | $69.3 \pm 1.6$ | $90.0 \pm 1.5$  | $84.8 \pm 4.0$  |
| Linear Ramp*             | $94.2 \pm 1.5$  | $70.2 \pm 1.3$ | $92.4 \pm 1.0$  | $86.3 \pm 4.9$  |
| Recursive TS             | $94.4 \pm 1.7$  | $58.6 \pm 4.3$ | $91.0 \pm 1.4$  | $77.5 \pm 7.5$  |
| TQA                      | $66.5 \pm 5.3$  | $73.4 \pm 4.6$ | $87.8 \pm 1.5$  | $79.0 \pm 7.8$  |
| TQA*                     | $77.8 \pm 6.5$  | $75.2 \pm 4.8$ | $90.9 \pm 1.2$  | $85.4 \pm 8.1$  |
| PP                       |                 |                |                 |                 |
| Fixed Angle*             | -               | $90.5 \pm 0.8$ | $89.1 \pm 0.1$  | $91.4 \pm 1.6$  |
| Fixed Angle <sup>†</sup> | -               | $88.7 \pm 1.8$ | $86.3 \pm 0.9$  | $91.1 \pm 1.7$  |
| Fourier*                 | $89.6 \pm 1.9$  | $88.4 \pm 2.7$ | $83.5 \pm 2.5$  | $88.6 \pm 1.6$  |
| Interp.*                 | $91.9 \pm 2.0$  | $91.1 \pm 1.0$ | $86.6 \pm 1.8$  | $90.3 \pm 1.8$  |
| Linear Ramp              | $92.5 \pm 1.0$  | $89.0 \pm 0.8$ | $86.8 \pm 2.1$  | $88.9 \pm 0.8$  |
| Linear Ramp*             | $94.0 \pm 1.0$  | $90.2 \pm 0.6$ | $88.1 \pm 2.1$  | $89.6 \pm 0.8$  |
| Recursive TS             | $89.2 \pm 1.2$  | $87.1 \pm 1.0$ | $85.5 \pm 1.2$  | $86.7 \pm 2.1$  |
| TQA                      | $83.6 \pm 0.9$  | $83.8 \pm 1.1$ | $72.8 \pm 5.9$  | $85.6 \pm 1.5$  |
| TQA*                     | $93.3 \pm 1.4$  | $88.2 \pm 0.8$ | $84.1 \pm 2.0$  | $89.2 \pm 1.4$  |
| SV                       |                 |                |                 |                 |
| Fixed Angle*             | $93.1 \pm 1.4$  | -              | $94.8 \pm 2.1$  | $95.3 \pm 2.0$  |
| Fixed Angle <sup>†</sup> | $90.1 \pm 2.2$  | -              | $93.7 \pm 2.1$  | $94.9 \pm 2.0$  |
| Fourier*                 | $80.5 \pm 10.2$ | $79.6 \pm 6.2$ | $81.2 \pm 8.2$  | $84.9 \pm 10.6$ |
| Interp.*                 | $92.9 \pm 1.4$  | $92.2 \pm 1.0$ | $92.4 \pm 3.9$  | $92.3 \pm 4.5$  |
| Linear Ramp              | $90.7 \pm 1.3$  | $90.8 \pm 0.7$ | $89.0 \pm 1.9$  | $91.5 \pm 2.1$  |
| Linear Ramp*             | $92.5 \pm 1.1$  | $92.2 \pm 1.0$ | $92.0 \pm 1.9$  | $93.4 \pm 1.3$  |
| Recursive TS             | $91.7 \pm 2.0$  | $90.5 \pm 2.5$ | $90.0 \pm 5.1$  | $91.2 \pm 3.5$  |
| TQA                      | $86.9 \pm 3.6$  | $83.6 \pm 4.3$ | $81.9 \pm 6.3$  | $86.6 \pm 5.3$  |
| TQA*                     | $91.6 \pm 1.5$  | $90.2 \pm 1.8$ | $90.6 \pm 3.3$  | $93.1 \pm 2.2$  |

Table 14: **Avg. Approx. Ratio  $\pm$  standard deviation in percentage for MAXCUT for  $P = 5$ .** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Evaluation methods are shown in different colors, with darker colors indicating better approximation ratios.

|                          | ER              | Approximation Ratio |                 | RR              | ER         | Num. Instances |              | RR          |
|--------------------------|-----------------|---------------------|-----------------|-----------------|------------|----------------|--------------|-------------|
|                          |                 | HH                  | LB              |                 |            | HH             | LB           |             |
| MPS (Aer)                |                 |                     |                 |                 |            |                |              |             |
| Fixed Angle*             | -               | 84.3 $\pm$ 3.3      | 94.5 $\pm$ 0.7  | 79.1 $\pm$ 2.5  | -          | 10 [144]       | 2 [100]      | 2 [10]      |
| Fixed Angle <sup>†</sup> | -               | 83.3 $\pm$ 3.4      | 94.4 $\pm$ 0.6  | 68.1 $\pm$ 0.6  | -          | 10 [144]       | 2 [100]      | 2 [10]      |
| Fourier*                 | 79.9 $\pm$ 6.1  | 82.4 $\pm$ 3.3      | 78.1 $\pm$ 10.5 | 76.8 $\pm$ 6.3  | 20 [40–50] | 20 [39–144]    | 55 [40–100]  | 17 [40–100] |
| Interp.*                 | 90.5 $\pm$ 2.0  | 87.2 $\pm$ 3.8      | 89.3 $\pm$ 5.3  | 80.8 $\pm$ 9.1  | 20 [40–50] | 20 [39–144]    | 55 [40–100]  | 17 [40–100] |
| Linear Ramp              | 85.1 $\pm$ 4.6  | 85.7 $\pm$ 1.0      | 78.4 $\pm$ 14.1 | 79.7 $\pm$ 5.5  | 10 [50]    | 10 [144]       | 19 [100]     | 7 [10]      |
| Linear Ramp*             | 88.8 $\pm$ 2.7  | 82.0 $\pm$ 2.0      | 93.4 $\pm$ 0.8  | 82.3 $\pm$ 5.4  | 10 [50]    | 10 [144]       | 10 [100]     | 7 [10]      |
| TQA                      | 67.9 $\pm$ 7.7  | 79.2 $\pm$ 1.2      | 89.2 $\pm$ 1.8  | 74.9 $\pm$ 7.2  | 10 [50]    | 10 [144]       | 10 [100]     | 7 [10]      |
| TQA*                     | 75.4 $\pm$ 2.6  | 81.5 $\pm$ 1.3      | 91.0 $\pm$ 1.2  | 80.1 $\pm$ 8.9  | 10 [50]    | 10 [144]       | 10 [100]     | 7 [10]      |
| MPS (Quimb)              |                 |                     |                 |                 |            |                |              |             |
| Fixed Angle*             | -               | -                   | 94.4 $\pm$ 0.7  | 93.2 $\pm$ 4.2  | -          | -              | 2 [100]      | 12 [40–100] |
| Fixed Angle <sup>†</sup> | -               | -                   | 94.3 $\pm$ 0.7  | 90.0 $\pm$ 8.9  | -          | -              | 2 [100]      | 12 [40–100] |
| Fourier*                 | 82.0 $\pm$ 7.5  | 77.1 $\pm$ 8.9      | 85.8 $\pm$ 4.6  | 86.4 $\pm$ 10.4 | 20 [40–50] | 31 [39–144]    | 130 [40–100] | 28 [40–100] |
| Interp.*                 | 94.5 $\pm$ 1.2  | 81.5 $\pm$ 8.7      | 92.4 $\pm$ 1.1  | 90.4 $\pm$ 7.1  | 20 [40–50] | 31 [39–144]    | 130 [40–100] | 28 [40–100] |
| Linear Ramp              | 93.1 $\pm$ 1.3  | 69.3 $\pm$ 1.6      | 90.0 $\pm$ 1.5  | 84.8 $\pm$ 4.0  | 10 [50]    | 10 [144]       | 10 [100]     | 7 [10]      |
| Linear Ramp*             | 94.2 $\pm$ 1.5  | 70.2 $\pm$ 1.3      | 92.4 $\pm$ 1.0  | 86.3 $\pm$ 4.9  | 10 [50]    | 10 [144]       | 10 [100]     | 7 [10]      |
| Recursive TS             | 94.4 $\pm$ 1.7  | 58.6 $\pm$ 4.3      | 91.0 $\pm$ 1.4  | 77.5 $\pm$ 7.5  | 10 [50]    | 11 [106–144]   | 10 [100]     | 9 [40–100]  |
| TQA                      | 66.5 $\pm$ 5.3  | 73.4 $\pm$ 4.6      | 87.8 $\pm$ 1.5  | 79.0 $\pm$ 7.8  | 10 [50]    | 20 [105–144]   | 20 [40–100]  | 17 [40–100] |
| TQA*                     | 77.8 $\pm$ 6.5  | 75.2 $\pm$ 4.8      | 90.9 $\pm$ 1.2  | 85.4 $\pm$ 8.1  | 10 [50]    | 20 [105–144]   | 20 [40–100]  | 17 [40–100] |
| PP                       |                 |                     |                 |                 |            |                |              |             |
| Fixed Angle*             | -               | 90.5 $\pm$ 0.8      | 89.1 $\pm$ 0.1  | 91.4 $\pm$ 1.6  | -          | 10 [144]       | 2 [100]      | 12 [40–100] |
| Fixed Angle <sup>†</sup> | -               | 88.7 $\pm$ 1.8      | 86.3 $\pm$ 0.9  | 91.1 $\pm$ 1.7  | -          | 10 [144]       | 2 [100]      | 12 [40–100] |
| Fourier*                 | 89.6 $\pm$ 1.9  | 88.4 $\pm$ 2.7      | 83.5 $\pm$ 2.5  | 88.6 $\pm$ 1.6  | 20 [40–50] | 31 [39–144]    | 130 [40–100] | 28 [40–100] |
| Interp.*                 | 91.9 $\pm$ 2.0  | 91.1 $\pm$ 1.0      | 86.6 $\pm$ 1.8  | 90.3 $\pm$ 1.8  | 20 [40–50] | 31 [39–144]    | 130 [40–100] | 28 [40–100] |
| Linear Ramp              | 92.5 $\pm$ 1.0  | 89.0 $\pm$ 0.8      | 86.8 $\pm$ 2.1  | 88.9 $\pm$ 0.8  | 10 [50]    | 20 [105–144]   | 29 [40–100]  | 8 [10]      |
| Linear Ramp*             | 94.0 $\pm$ 1.0  | 90.2 $\pm$ 0.6      | 88.1 $\pm$ 2.1  | 89.6 $\pm$ 0.8  | 10 [50]    | 20 [105–144]   | 20 [40–100]  | 8 [10]      |
| Recursive TS             | 89.2 $\pm$ 1.2  | 87.1 $\pm$ 1.0      | 85.5 $\pm$ 1.2  | 86.7 $\pm$ 2.1  | 10 [50]    | 11 [106–144]   | 10 [100]     | 9 [40–100]  |
| TQA                      | 83.6 $\pm$ 0.9  | 83.8 $\pm$ 1.1      | 72.8 $\pm$ 5.9  | 85.6 $\pm$ 1.5  | 10 [50]    | 20 [105–144]   | 20 [40–100]  | 17 [40–100] |
| TQA*                     | 93.3 $\pm$ 1.4  | 88.2 $\pm$ 0.8      | 84.1 $\pm$ 2.0  | 89.2 $\pm$ 1.4  | 10 [50]    | 20 [105–144]   | 20 [40–100]  | 17 [40–100] |
| SV                       |                 |                     |                 |                 |            |                |              |             |
| Fixed Angle*             | 93.1 $\pm$ 1.4  | -                   | 94.8 $\pm$ 2.1  | 95.3 $\pm$ 2.0  | 46 [10–20] | -              | 60 [10–20]   | 115 [10–20] |
| Fixed Angle <sup>†</sup> | 90.1 $\pm$ 2.2  | -                   | 93.7 $\pm$ 2.1  | 94.9 $\pm$ 2.0  | 46 [10–20] | -              | 61 [10–20]   | 121 [10–20] |
| Fourier*                 | 80.5 $\pm$ 10.2 | 79.6 $\pm$ 6.2      | 81.2 $\pm$ 8.2  | 84.9 $\pm$ 10.6 | 92 [10–20] | 20 [12–21]     | 784 [10–20]  | 382 [10–20] |
| Interp.*                 | 92.9 $\pm$ 1.4  | 92.2 $\pm$ 1.0      | 92.4 $\pm$ 3.9  | 92.3 $\pm$ 4.5  | 92 [10–20] | 20 [12–21]     | 781 [10–20]  | 382 [10–20] |
| Linear Ramp              | 90.7 $\pm$ 1.3  | 90.8 $\pm$ 0.7      | 89.0 $\pm$ 1.9  | 91.5 $\pm$ 2.1  | 10 [20]    | 10 [21]        | 10 [20]      | 7 [20]      |
| Linear Ramp*             | 92.5 $\pm$ 1.1  | 92.2 $\pm$ 1.0      | 92.0 $\pm$ 1.9  | 93.4 $\pm$ 1.3  | 10 [20]    | 10 [21]        | 10 [20]      | 7 [20]      |
| Recursive TS             | 91.7 $\pm$ 2.0  | 90.5 $\pm$ 2.5      | 90.0 $\pm$ 5.1  | 91.2 $\pm$ 3.5  | 92 [10–20] | 20 [12–21]     | 416 [10–20]  | 381 [10–20] |
| TQA                      | 86.9 $\pm$ 3.6  | 83.6 $\pm$ 4.3      | 81.9 $\pm$ 6.3  | 86.6 $\pm$ 5.3  | 92 [10–20] | 20 [12–21]     | 677 [10–20]  | 359 [10–20] |
| TQA*                     | 91.6 $\pm$ 1.5  | 90.2 $\pm$ 1.8      | 90.6 $\pm$ 3.3  | 93.1 $\pm$ 2.2  | 90 [10–20] | 20 [12–21]     | 672 [10–20]  | 142 [10–20] |

Table 15: **Avg. Approx. Ratio and Number of Instances for Various Evaluation Methods for MAX-CUT for  $P = 5$ .** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Evaluation methods are shown in different colors for approximation ratios. Darker colors indicating better approximation ratios or more instances. Approximation ratios are in percentage, with  $\pm$  their standard deviation. The range of graph sizes are shown in brackets, next to the number of instances.

## 6 Tables for depth $P = 6$

|              | ER         | HH           | LB           | RR          |
|--------------|------------|--------------|--------------|-------------|
| MPS (Aer)    |            |              |              |             |
| Fixed Angle* | -          | 10 [39]      | 20 [100]     | 11 [40-100] |
| Fixed Angle† | -          | 10 [39]      | 20 [100]     | 11 [40-100] |
| Fourier*     | 20 [40-50] | 20 [39-144]  | 55 [40-100]  | 17 [40-100] |
| Interp.*     | 20 [40-50] | 20 [39-144]  | 55 [40-100]  | 17 [40-100] |
| Linear Ramp  | 10 [50]    | 10 [144]     | 19 [100]     | 7 [100]     |
| Linear Ramp* | 10 [50]    | 10 [144]     | 10 [100]     | 7 [100]     |
| TQA          | 20 [40-50] | 20 [39-144]  | 56 [40-100]  | 17 [40-100] |
| TQA*         | 20 [40-50] | 20 [39-144]  | 56 [40-100]  | 17 [40-100] |
| MPS (Quimb)  |            |              |              |             |
| Fixed Angle* | -          | 10 [144]     | 10 [100]     | 11 [40-100] |
| Fixed Angle† | -          | 10 [144]     | 10 [100]     | 11 [40-100] |
| Fourier*     | 20 [40-50] | 31 [39-144]  | 130 [40-100] | 28 [40-100] |
| Interp.*     | 20 [40-50] | 31 [39-144]  | 130 [40-100] | 28 [40-100] |
| Linear Ramp  | 10 [50]    | 10 [144]     | 10 [100]     | 7 [100]     |
| Linear Ramp* | 10 [50]    | 10 [144]     | 10 [100]     | 7 [100]     |
| Recursive TS | 10 [50]    | 10 [144]     | 10 [100]     | 7 [100]     |
| TQA          | 20 [40-50] | 20 [39-144]  | 130 [40-100] | 27 [40-100] |
| TQA*         | 20 [40-50] | 20 [39-144]  | 130 [40-100] | 27 [40-100] |
| PP           |            |              |              |             |
| Fixed Angle* | -          | 30 [39-144]  | 10 [100]     | 11 [40-100] |
| Fixed Angle† | -          | 30 [39-144]  | 10 [100]     | 11 [40-100] |
| Fourier*     | 20 [40-50] | 31 [39-144]  | 130 [40-100] | 28 [40-100] |
| Interp.*     | 20 [40-50] | 31 [39-144]  | 130 [40-100] | 28 [40-100] |
| Linear Ramp  | 10 [50]    | 20 [105-144] | 29 [40-100]  | 8 [100]     |
| Linear Ramp* | 10 [50]    | 20 [105-144] | 20 [40-100]  | 8 [100]     |
| Recursive TS | 10 [50]    | 10 [144]     | 10 [100]     | 7 [100]     |
| TQA          | 20 [40-50] | 20 [39-144]  | 130 [40-100] | 27 [40-100] |
| TQA*         | 20 [40-50] | 20 [39-144]  | 130 [40-100] | 27 [40-100] |
| SV           |            |              |              |             |
| Fixed Angle* | 17 [11-20] | -            | 20 [10-20]   | 59 [10-20]  |
| Fixed Angle† | 17 [11-20] | -            | 20 [10-20]   | 59 [10-20]  |
| Fourier*     | 92 [10-20] | 20 [12-21]   | 781 [10-20]  | 382 [10-20] |
| Interp.*     | 92 [10-20] | 20 [12-21]   | 781 [10-20]  | 382 [10-20] |
| Linear Ramp  | 10 [20]    | 10 [21]      | 10 [20]      | 7 [20]      |
| Linear Ramp* | 10 [20]    | 10 [21]      | 10 [20]      | 7 [20]      |
| Recursive TS | 92 [10-20] | 20 [12-21]   | 413 [10-20]  | 381 [10-20] |
| TQA          | 92 [10-20] | 20 [12-21]   | 677 [10-20]  | 360 [10-20] |
| TQA*         | 90 [10-20] | 20 [12-21]   | 672 [10-20]  | 142 [10-20] |

Table 16: **Number of Instances (num. nodes in brackets) for  $P = 6$ .** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Cells are colored based on the number of instances for the given configuration, with darker colors indicating more instances.

|                          | ER              | HH              | LB              | RR              |
|--------------------------|-----------------|-----------------|-----------------|-----------------|
| MPS (Aer)                |                 |                 |                 |                 |
| Fixed Angle*             | -               | 91.6 $\pm$ 0.9  | 90.1 $\pm$ 2.9  | 95.6 $\pm$ 5.3  |
| Fixed Angle <sup>†</sup> | -               | 89.9 $\pm$ 1.7  | 86.0 $\pm$ 5.5  | 94.3 $\pm$ 7.6  |
| Fourier*                 | 78.1 $\pm$ 5.7  | 83.1 $\pm$ 3.0  | 80.1 $\pm$ 9.2  | 81.4 $\pm$ 7.6  |
| Interp.*                 | 92.2 $\pm$ 1.8  | 88.3 $\pm$ 4.4  | 91.8 $\pm$ 3.8  | 84.7 $\pm$ 7.0  |
| Linear Ramp              | 89.2 $\pm$ 1.8  | 81.4 $\pm$ 2.7  | 79.7 $\pm$ 14.0 | 84.5 $\pm$ 3.1  |
| Linear Ramp*             | 92.0 $\pm$ 1.8  | 81.4 $\pm$ 1.7  | 94.6 $\pm$ 0.9  | 86.4 $\pm$ 2.2  |
| TQA                      | 74.3 $\pm$ 4.5  | 82.9 $\pm$ 2.5  | 80.2 $\pm$ 8.6  | 82.2 $\pm$ 7.4  |
| TQA*                     | 80.9 $\pm$ 3.8  | 84.6 $\pm$ 3.1  | 84.5 $\pm$ 8.8  | 87.8 $\pm$ 8.6  |
| MPS (Quimb)              |                 |                 |                 |                 |
| Fixed Angle*             | -               | 53.9 $\pm$ 2.0  | 95.1 $\pm$ 1.1  | 96.1 $\pm$ 2.4  |
| Fixed Angle <sup>†</sup> | -               | 50.9 $\pm$ 1.1  | 95.1 $\pm$ 1.0  | 94.7 $\pm$ 4.5  |
| Fourier*                 | 78.9 $\pm$ 7.6  | 76.6 $\pm$ 10.0 | 87.9 $\pm$ 3.5  | 88.9 $\pm$ 9.6  |
| Interp.*                 | 95.8 $\pm$ 1.1  | 81.9 $\pm$ 9.7  | 93.9 $\pm$ 1.0  | 92.0 $\pm$ 6.2  |
| Linear Ramp              | 94.1 $\pm$ 1.3  | 67.9 $\pm$ 1.8  | 91.4 $\pm$ 1.3  | 84.3 $\pm$ 7.6  |
| Linear Ramp*             | 95.2 $\pm$ 1.6  | 69.2 $\pm$ 1.6  | 93.9 $\pm$ 1.3  | 86.4 $\pm$ 7.3  |
| Recursive TS             | 95.6 $\pm$ 1.0  | 56.0 $\pm$ 1.7  | 92.2 $\pm$ 1.4  | 77.9 $\pm$ 7.2  |
| TQA                      | 71.7 $\pm$ 6.0  | 70.0 $\pm$ 16.1 | 88.0 $\pm$ 2.1  | 85.9 $\pm$ 8.9  |
| TQA*                     | 85.5 $\pm$ 6.7  | 71.7 $\pm$ 17.6 | 91.1 $\pm$ 1.4  | 90.6 $\pm$ 7.1  |
| PP                       |                 |                 |                 |                 |
| Fixed Angle*             | -               | 92.0 $\pm$ 0.7  | 89.1 $\pm$ 1.8  | 92.3 $\pm$ 1.6  |
| Fixed Angle <sup>†</sup> | -               | 90.2 $\pm$ 1.5  | 85.5 $\pm$ 1.6  | 91.9 $\pm$ 1.7  |
| Fourier*                 | 90.7 $\pm$ 1.9  | 89.3 $\pm$ 2.8  | 84.8 $\pm$ 2.2  | 89.5 $\pm$ 1.8  |
| Interp.*                 | 92.7 $\pm$ 2.1  | 92.2 $\pm$ 0.9  | 87.6 $\pm$ 1.8  | 91.1 $\pm$ 2.0  |
| Linear Ramp              | 93.2 $\pm$ 1.0  | 90.0 $\pm$ 0.7  | 87.8 $\pm$ 2.0  | 89.5 $\pm$ 0.8  |
| Linear Ramp*             | 94.7 $\pm$ 1.0  | 91.3 $\pm$ 0.7  | 89.2 $\pm$ 2.2  | 90.4 $\pm$ 0.7  |
| Recursive TS             | 89.6 $\pm$ 1.2  | 87.9 $\pm$ 1.1  | 86.3 $\pm$ 1.6  | 86.8 $\pm$ 1.6  |
| TQA                      | 84.6 $\pm$ 1.9  | 85.6 $\pm$ 1.2  | 76.5 $\pm$ 5.9  | 87.5 $\pm$ 2.1  |
| TQA*                     | 91.3 $\pm$ 2.4  | 89.6 $\pm$ 0.9  | 85.2 $\pm$ 2.2  | 90.3 $\pm$ 1.6  |
| SV                       |                 |                 |                 |                 |
| Fixed Angle*             | 94.4 $\pm$ 1.5  | -               | 96.5 $\pm$ 1.4  | 96.2 $\pm$ 1.4  |
| Fixed Angle <sup>†</sup> | 90.3 $\pm$ 2.5  | -               | 95.6 $\pm$ 1.3  | 95.9 $\pm$ 1.3  |
| Fourier*                 | 80.3 $\pm$ 10.6 | 78.5 $\pm$ 9.3  | 80.9 $\pm$ 9.2  | 84.6 $\pm$ 11.5 |
| Interp.*                 | 94.1 $\pm$ 1.3  | 93.2 $\pm$ 1.2  | 93.7 $\pm$ 3.8  | 92.9 $\pm$ 4.9  |
| Linear Ramp              | 91.8 $\pm$ 1.3  | 91.4 $\pm$ 1.3  | 90.1 $\pm$ 1.7  | 92.4 $\pm$ 1.9  |
| Linear Ramp*             | 93.5 $\pm$ 1.0  | 93.0 $\pm$ 0.9  | 93.2 $\pm$ 1.9  | 94.5 $\pm$ 1.0  |
| Recursive TS             | 92.9 $\pm$ 1.9  | 91.2 $\pm$ 2.4  | 91.3 $\pm$ 5.0  | 92.2 $\pm$ 3.4  |
| TQA                      | 88.3 $\pm$ 3.3  | 85.4 $\pm$ 3.9  | 82.5 $\pm$ 7.3  | 88.4 $\pm$ 5.0  |
| TQA*                     | 92.9 $\pm$ 1.4  | 90.8 $\pm$ 1.9  | 91.9 $\pm$ 3.0  | 94.2 $\pm$ 2.1  |

Table 17: **Avg. Approx. Ratio  $\pm$  standard deviation in percentage for MAXCUT for  $P = 6$ .** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Evaluation methods are shown in different colors, with darker colors indicating better approximation ratios.

|              | Approximation Ratio |             |             |             | Num. Instances |              |              |             |
|--------------|---------------------|-------------|-------------|-------------|----------------|--------------|--------------|-------------|
|              | ER                  | HH          | LB          | RR          | ER             | HH           | LB           | RR          |
| MPS (Aer)    |                     |             |             |             |                |              |              |             |
| Fixed Angle* | -                   | 91.6 ± 0.9  | 90.1 ± 2.9  | 95.6 ± 5.3  | -              | 10 [39]      | 20 [100]     | 11 [40]     |
| Fixed Angle† | -                   | 89.9 ± 1.7  | 86.0 ± 5.5  | 94.3 ± 7.6  | -              | 10 [39]      | 20 [100]     | 11 [40]     |
| Fourier*     | 78.1 ± 5.7          | 83.1 ± 3.0  | 80.1 ± 9.2  | 81.4 ± 7.6  | 20 [40–50]     | 20 [39–144]  | 55 [40–100]  | 17 [40]     |
| Interp.*     | 92.2 ± 1.8          | 88.3 ± 4.4  | 91.8 ± 3.8  | 84.7 ± 7.0  | 20 [40–50]     | 20 [39–144]  | 55 [40–100]  | 17 [40]     |
| Linear Ramp  | 89.2 ± 1.8          | 81.4 ± 2.7  | 79.7 ± 14.0 | 84.5 ± 3.1  | 10 [50]        | 10 [144]     | 19 [100]     | 7 [40]      |
| Linear Ramp* | 92.0 ± 1.8          | 81.4 ± 1.7  | 94.6 ± 0.9  | 86.4 ± 2.2  | 10 [50]        | 10 [144]     | 10 [100]     | 7 [40]      |
| TQA          | 74.3 ± 4.5          | 82.9 ± 2.5  | 80.2 ± 8.6  | 82.2 ± 7.4  | 20 [40–50]     | 20 [39–144]  | 56 [40–100]  | 17 [40]     |
| TQA*         | 80.9 ± 3.8          | 84.6 ± 3.1  | 84.5 ± 8.8  | 87.8 ± 8.6  | 20 [40–50]     | 20 [39–144]  | 56 [40–100]  | 17 [40]     |
| MPS (Quimb)  |                     |             |             |             |                |              |              |             |
| Fixed Angle* | -                   | 53.9 ± 2.0  | 95.1 ± 1.1  | 96.1 ± 2.4  | -              | 10 [144]     | 10 [100]     | 11 [40]     |
| Fixed Angle† | -                   | 50.9 ± 1.1  | 95.1 ± 1.0  | 94.7 ± 4.5  | -              | 10 [144]     | 10 [100]     | 11 [40]     |
| Fourier*     | 78.9 ± 7.6          | 76.6 ± 10.0 | 87.9 ± 3.5  | 88.9 ± 9.6  | 20 [40–50]     | 31 [39–144]  | 130 [40–100] | 28 [40]     |
| Interp.*     | 95.8 ± 1.1          | 81.9 ± 9.7  | 93.9 ± 1.0  | 92.0 ± 6.2  | 20 [40–50]     | 31 [39–144]  | 130 [40–100] | 28 [40]     |
| Linear Ramp  | 94.1 ± 1.3          | 67.9 ± 1.8  | 91.4 ± 1.3  | 84.3 ± 7.6  | 10 [50]        | 10 [144]     | 10 [100]     | 7 [40]      |
| Linear Ramp* | 95.2 ± 1.6          | 69.2 ± 1.6  | 93.9 ± 1.3  | 86.4 ± 7.3  | 10 [50]        | 10 [144]     | 10 [100]     | 7 [40]      |
| Recursive TS | 95.6 ± 1.0          | 56.0 ± 1.7  | 92.2 ± 1.4  | 77.9 ± 7.2  | 10 [50]        | 10 [144]     | 10 [100]     | 7 [40]      |
| TQA          | 71.7 ± 6.0          | 70.0 ± 16.1 | 88.0 ± 2.1  | 85.9 ± 8.9  | 20 [40–50]     | 20 [39–144]  | 130 [40–100] | 27 [40]     |
| TQA*         | 85.5 ± 6.7          | 71.7 ± 17.6 | 91.1 ± 1.4  | 90.6 ± 7.1  | 20 [40–50]     | 20 [39–144]  | 130 [40–100] | 27 [40]     |
| PP           |                     |             |             |             |                |              |              |             |
| Fixed Angle* | -                   | 92.0 ± 0.7  | 89.1 ± 1.8  | 92.3 ± 1.6  | -              | 30 [39–144]  | 10 [100]     | 11 [40]     |
| Fixed Angle† | -                   | 90.2 ± 1.5  | 85.5 ± 1.6  | 91.9 ± 1.7  | -              | 30 [39–144]  | 10 [100]     | 11 [40]     |
| Fourier*     | 90.7 ± 1.9          | 89.3 ± 2.8  | 84.8 ± 2.2  | 89.5 ± 1.8  | 20 [40–50]     | 31 [39–144]  | 130 [40–100] | 28 [40]     |
| Interp.*     | 92.7 ± 2.1          | 92.2 ± 0.9  | 87.6 ± 1.8  | 91.1 ± 2.0  | 20 [40–50]     | 31 [39–144]  | 130 [40–100] | 28 [40]     |
| Linear Ramp  | 93.2 ± 1.0          | 90.0 ± 0.7  | 87.8 ± 2.0  | 89.5 ± 0.8  | 10 [50]        | 20 [105–144] | 29 [40–100]  | 8 [40]      |
| Linear Ramp* | 94.7 ± 1.0          | 91.3 ± 0.7  | 89.2 ± 2.2  | 90.4 ± 0.7  | 10 [50]        | 20 [105–144] | 20 [40–100]  | 8 [40]      |
| Recursive TS | 89.6 ± 1.2          | 87.9 ± 1.1  | 86.3 ± 1.6  | 86.8 ± 1.6  | 10 [50]        | 10 [144]     | 10 [100]     | 7 [40]      |
| TQA          | 84.6 ± 1.9          | 85.6 ± 1.2  | 76.5 ± 5.9  | 87.5 ± 2.1  | 20 [40–50]     | 20 [39–144]  | 130 [40–100] | 27 [40]     |
| TQA*         | 91.3 ± 2.4          | 89.6 ± 0.9  | 85.2 ± 2.2  | 90.3 ± 1.6  | 20 [40–50]     | 20 [39–144]  | 130 [40–100] | 27 [40]     |
| SV           |                     |             |             |             |                |              |              |             |
| Fixed Angle* | 94.4 ± 1.5          | -           | 96.5 ± 1.4  | 96.2 ± 1.4  | 17 [11–20]     | -            | 20 [10–20]   | 59 [10–20]  |
| Fixed Angle† | 90.3 ± 2.5          | -           | 95.6 ± 1.3  | 95.9 ± 1.3  | 17 [11–20]     | -            | 20 [10–20]   | 59 [10–20]  |
| Fourier*     | 80.3 ± 10.6         | 78.5 ± 9.3  | 80.9 ± 9.2  | 84.6 ± 11.5 | 92 [10–20]     | 20 [12–21]   | 781 [10–20]  | 382 [10–20] |
| Interp.*     | 94.1 ± 1.3          | 93.2 ± 1.2  | 93.7 ± 3.8  | 92.9 ± 4.9  | 92 [10–20]     | 20 [12–21]   | 781 [10–20]  | 382 [10–20] |
| Linear Ramp  | 91.8 ± 1.3          | 91.4 ± 1.3  | 90.1 ± 1.7  | 92.4 ± 1.9  | 10 [20]        | 10 [21]      | 10 [20]      | 7 [20]      |
| Linear Ramp* | 93.5 ± 1.0          | 93.0 ± 0.9  | 93.2 ± 1.9  | 94.5 ± 1.0  | 10 [20]        | 10 [21]      | 10 [20]      | 7 [20]      |
| Recursive TS | 92.9 ± 1.9          | 91.2 ± 2.4  | 91.3 ± 5.0  | 92.2 ± 3.4  | 92 [10–20]     | 20 [12–21]   | 413 [10–20]  | 381 [10–20] |
| TQA          | 88.3 ± 3.3          | 85.4 ± 3.9  | 82.5 ± 7.3  | 88.4 ± 5.0  | 92 [10–20]     | 20 [12–21]   | 677 [10–20]  | 360 [10–20] |
| TQA*         | 92.9 ± 1.4          | 90.8 ± 1.9  | 91.9 ± 3.0  | 94.2 ± 2.1  | 90 [10–20]     | 20 [12–21]   | 672 [10–20]  | 142 [10–20] |

Table 18: **Avg. Approx. Ratio and Number of Instances for Various Evaluation Methods for MAX-CUT for  $P = 6$ .** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Evaluation methods are shown in different colors for approximation ratios. Darker colors indicating better approximation ratios or more instances. Approximation ratios are in percentage, with  $\pm$  their standard deviation. The range of graph sizes are shown in brackets, next to the number of instances.

## 7 Tables for depth $P = 7$

|              | ER         | HH           | LB           | RR          |
|--------------|------------|--------------|--------------|-------------|
| MPS (Aer)    |            |              |              |             |
| Fixed Angle* | -          | -            | 1 [100]      | 1 [100]     |
| Fixed Angle† | -          | -            | 1 [100]      | 1 [100]     |
| Fourier*     | 20 [40–50] | 20 [39–144]  | 49 [40–100]  | 17 [40–100] |
| Interp.*     | 20 [40–50] | 20 [39–144]  | 49 [40–100]  | 17 [40–100] |
| Linear Ramp  | 10 [50]    | 10 [144]     | 10 [100]     | 7 [100]     |
| Linear Ramp* | 10 [50]    | 10 [144]     | 10 [100]     | 7 [100]     |
| TQA          | 10 [50]    | 10 [144]     | 10 [100]     | 7 [100]     |
| TQA*         | 10 [50]    | 10 [144]     | 10 [100]     | 7 [100]     |
| MPS (Quimb)  |            |              |              |             |
| Fixed Angle* | -          | 10 [144]     | 1 [100]      | 1 [100]     |
| Fixed Angle† | -          | 10 [144]     | 1 [100]      | 1 [100]     |
| Fourier*     | 20 [40–50] | 31 [39–144]  | 120 [40–100] | 28 [40–100] |
| Interp.*     | 20 [40–50] | 31 [39–144]  | 120 [40–100] | 28 [40–100] |
| Linear Ramp  | 10 [50]    | 10 [144]     | 10 [100]     | 7 [100]     |
| Linear Ramp* | 10 [50]    | 10 [144]     | 10 [100]     | 7 [100]     |
| Recursive TS | 10 [50]    | 10 [144]     | 10 [100]     | 7 [100]     |
| TQA          | 10 [50]    | 10 [144]     | 10 [100]     | 7 [100]     |
| TQA*         | 10 [50]    | 10 [144]     | 10 [100]     | 7 [100]     |
| PP           |            |              |              |             |
| Fixed Angle* | -          | 10 [144]     | 1 [100]      | 1 [100]     |
| Fixed Angle† | -          | 10 [144]     | 1 [100]      | 1 [100]     |
| Fourier*     | 20 [40–50] | 31 [39–144]  | 120 [40–100] | 28 [40–100] |
| Interp.*     | 20 [40–50] | 31 [39–144]  | 120 [40–100] | 28 [40–100] |
| Linear Ramp  | 10 [50]    | 20 [105–144] | 20 [40–100]  | 8 [100]     |
| Linear Ramp* | 10 [50]    | 20 [105–144] | 20 [40–100]  | 8 [100]     |
| Recursive TS | 10 [50]    | 10 [144]     | 10 [100]     | 7 [100]     |
| TQA          | 10 [50]    | 10 [144]     | 10 [100]     | 7 [100]     |
| TQA*         | 10 [50]    | 10 [144]     | 10 [100]     | 7 [100]     |
| SV           |            |              |              |             |
| Fixed Angle* | 17 [11–20] | -            | 19 [10–20]   | 59 [10–20]  |
| Fixed Angle† | 17 [11–20] | -            | 19 [10–20]   | 60 [10–20]  |
| Fourier*     | 92 [10–20] | 20 [12–21]   | 780 [10–20]  | 382 [10–20] |
| Interp.*     | 92 [10–20] | 20 [12–21]   | 780 [10–20]  | 382 [10–20] |
| Linear Ramp  | 10 [20]    | 10 [21]      | 10 [20]      | 7 [20]      |
| Linear Ramp* | 10 [20]    | 10 [21]      | 10 [20]      | 7 [20]      |
| Recursive TS | 92 [10–20] | 20 [12–21]   | 412 [10–20]  | 379 [10–20] |
| TQA          | 92 [10–20] | 20 [12–21]   | 677 [10–20]  | 361 [10–20] |
| TQA*         | 90 [10–20] | 20 [12–21]   | 672 [10–20]  | 141 [10–20] |

Table 19: **Number of Instances (num. nodes in brackets) for  $P = 7$ .** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Cells are colored based on the number of instances for the given configuration, with darker colors indicating more instances.

|                          | ER          | HH          | LB         | RR          |
|--------------------------|-------------|-------------|------------|-------------|
| MPS (Aer)                |             |             |            |             |
| Fixed Angle*             | -           | -           | 97.0       | 82.6        |
| Fixed Angle <sup>†</sup> | -           | -           | 96.8       | 71.9        |
| Fourier*                 | 79.4 ± 5.7  | 84.4 ± 3.2  | 80.3 ± 9.4 | 83.7 ± 8.2  |
| Interp.*                 | 93.7 ± 2.1  | 89.2 ± 4.7  | 93.4 ± 3.0 | 85.0 ± 8.3  |
| Linear Ramp              | 89.8 ± 2.9  | 82.5 ± 2.2  | 93.4 ± 1.0 | 86.0 ± 2.4  |
| Linear Ramp*             | 92.9 ± 2.0  | 83.5 ± 2.5  | 95.2 ± 0.8 | 87.1 ± 3.3  |
| TQA                      | 78.1 ± 5.3  | 82.0 ± 1.5  | 90.7 ± 1.1 | 79.2 ± 7.8  |
| TQA*                     | 81.7 ± 2.5  | 83.2 ± 1.9  | 92.5 ± 1.2 | 86.8 ± 4.5  |
| MPS (Quimb)              |             |             |            |             |
| Fixed Angle*             | -           | 56.3 ± 2.6  | 97.0       | 91.9        |
| Fixed Angle <sup>†</sup> | -           | 52.3 ± 2.6  | 96.7       | 84.8        |
| Fourier*                 | 81.1 ± 7.1  | 77.6 ± 10.1 | 88.1 ± 4.1 | 89.0 ± 8.8  |
| Interp.*                 | 96.4 ± 1.3  | 83.7 ± 9.4  | 94.9 ± 1.0 | 93.0 ± 6.0  |
| Linear Ramp              | 94.8 ± 1.3  | 69.2 ± 1.5  | 92.5 ± 1.3 | 88.6 ± 3.4  |
| Linear Ramp*             | 95.9 ± 1.4  | 71.7 ± 1.7  | 94.6 ± 0.8 | 90.2 ± 2.5  |
| Recursive TS             | 96.4 ± 0.9  | 57.8 ± 1.2  | 93.0 ± 1.1 | 81.7 ± 6.2  |
| TQA                      | 71.9 ± 9.7  | 57.3 ± 1.0  | 89.6 ± 1.0 | 76.8 ± 11.9 |
| TQA*                     | 86.8 ± 4.8  | 57.5 ± 1.0  | 92.4 ± 0.8 | 83.2 ± 8.5  |
| PP                       |             |             |            |             |
| Fixed Angle*             | -           | 92.5 ± 0.6  | 90.6       | 90.7        |
| Fixed Angle <sup>†</sup> | -           | 90.2 ± 1.6  | 86.4       | 89.8        |
| Fourier*                 | 90.0 ± 2.8  | 89.9 ± 2.7  | 85.2 ± 2.1 | 89.9 ± 1.7  |
| Interp.*                 | 93.3 ± 2.1  | 93.1 ± 0.9  | 88.3 ± 1.8 | 91.2 ± 2.1  |
| Linear Ramp              | 93.8 ± 1.0  | 90.9 ± 0.6  | 88.5 ± 2.2 | 90.0 ± 0.8  |
| Linear Ramp*             | 95.1 ± 1.0  | 92.1 ± 0.6  | 89.8 ± 2.3 | 90.8 ± 0.8  |
| Recursive TS             | 90.1 ± 1.1  | 88.7 ± 1.2  | 86.9 ± 1.8 | 87.1 ± 1.6  |
| TQA                      | 83.8 ± 0.9  | 87.0 ± 0.8  | 77.2 ± 6.9 | 86.3 ± 2.3  |
| TQA*                     | 94.5 ± 1.5  | 90.2 ± 0.6  | 85.6 ± 2.3 | 89.6 ± 1.1  |
| SV                       |             |             |            |             |
| Fixed Angle*             | 95.4 ± 1.5  | -           | 97.4 ± 1.2 | 97.2 ± 1.2  |
| Fixed Angle <sup>†</sup> | 90.7 ± 3.0  | -           | 96.3 ± 1.0 | 96.8 ± 1.2  |
| Fourier*                 | 80.4 ± 10.8 | 77.9 ± 10.7 | 80.8 ± 9.7 | 84.1 ± 12.4 |
| Interp.*                 | 95.1 ± 1.2  | 93.9 ± 1.2  | 94.5 ± 4.0 | 93.1 ± 5.4  |
| Linear Ramp              | 92.5 ± 1.2  | 92.0 ± 1.3  | 91.1 ± 1.6 | 93.1 ± 1.8  |
| Linear Ramp*             | 94.3 ± 1.0  | 93.6 ± 0.8  | 94.0 ± 1.6 | 95.2 ± 0.9  |
| Recursive TS             | 93.8 ± 1.8  | 91.8 ± 2.3  | 92.3 ± 4.9 | 92.9 ± 3.3  |
| TQA                      | 89.4 ± 3.3  | 86.8 ± 3.7  | 83.1 ± 7.8 | 89.3 ± 4.8  |
| TQA*                     | 93.8 ± 1.3  | 91.4 ± 1.9  | 92.5 ± 3.2 | 95.0 ± 1.8  |

Table 20: **Avg. Approx. Ratio ± standard deviation in percentage for MAXCUT for  $P = 7$ .** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Evaluation methods are shown in different colors, with darker colors indicating better approximation ratios.

|              | Approximation Ratio |             |            |             | Num. Instances |              |              |             |
|--------------|---------------------|-------------|------------|-------------|----------------|--------------|--------------|-------------|
|              | ER                  | HH          | LB         | RR          | ER             | HH           | LB           | RR          |
| MPS (Aer)    |                     |             |            |             |                |              |              |             |
| Fixed Angle* | -                   | -           | 97.0       | 82.6        | -              | -            | 1 [100]      | 1 [10]      |
| Fixed Angle† | -                   | -           | 96.8       | 71.9        | -              | -            | 1 [100]      | 1 [10]      |
| Fourier*     | 79.4 ± 5.7          | 84.4 ± 3.2  | 80.3 ± 9.4 | 83.7 ± 8.2  | 20 [40–50]     | 20 [39–144]  | 49 [40–100]  | 17 [40–100] |
| Interp.*     | 93.7 ± 2.1          | 89.2 ± 4.7  | 93.4 ± 3.0 | 85.0 ± 8.3  | 20 [40–50]     | 20 [39–144]  | 49 [40–100]  | 17 [40–100] |
| Linear Ramp  | 89.8 ± 2.9          | 82.5 ± 2.2  | 93.4 ± 1.0 | 86.0 ± 2.4  | 10 [50]        | 10 [144]     | 10 [100]     | 7 [10]      |
| Linear Ramp* | 92.9 ± 2.0          | 83.5 ± 2.5  | 95.2 ± 0.8 | 87.1 ± 3.3  | 10 [50]        | 10 [144]     | 10 [100]     | 7 [10]      |
| TQA          | 78.1 ± 5.3          | 82.0 ± 1.5  | 90.7 ± 1.1 | 79.2 ± 7.8  | 10 [50]        | 10 [144]     | 10 [100]     | 7 [10]      |
| TQA*         | 81.7 ± 2.5          | 83.2 ± 1.9  | 92.5 ± 1.2 | 86.8 ± 4.5  | 10 [50]        | 10 [144]     | 10 [100]     | 7 [10]      |
| MPS (Quimb)  |                     |             |            |             |                |              |              |             |
| Fixed Angle* | -                   | 56.3 ± 2.6  | 97.0       | 91.9        | -              | 10 [144]     | 1 [100]      | 1 [10]      |
| Fixed Angle† | -                   | 52.3 ± 2.6  | 96.7       | 84.8        | -              | 10 [144]     | 1 [100]      | 1 [10]      |
| Fourier*     | 81.1 ± 7.1          | 77.6 ± 10.1 | 88.1 ± 4.1 | 89.0 ± 8.8  | 20 [40–50]     | 31 [39–144]  | 120 [40–100] | 28 [40–100] |
| Interp.*     | 96.4 ± 1.3          | 83.7 ± 9.4  | 94.9 ± 1.0 | 93.0 ± 6.0  | 20 [40–50]     | 31 [39–144]  | 120 [40–100] | 28 [40–100] |
| Linear Ramp  | 94.8 ± 1.3          | 69.2 ± 1.5  | 92.5 ± 1.3 | 88.6 ± 3.4  | 10 [50]        | 10 [144]     | 10 [100]     | 7 [10]      |
| Linear Ramp* | 95.9 ± 1.4          | 71.7 ± 1.7  | 94.6 ± 0.8 | 90.2 ± 2.5  | 10 [50]        | 10 [144]     | 10 [100]     | 7 [10]      |
| Recursive TS | 96.4 ± 0.9          | 57.8 ± 1.2  | 93.0 ± 1.1 | 81.7 ± 6.2  | 10 [50]        | 10 [144]     | 10 [100]     | 7 [10]      |
| TQA          | 71.9 ± 9.7          | 57.3 ± 1.0  | 89.6 ± 1.0 | 76.8 ± 11.9 | 10 [50]        | 10 [144]     | 10 [100]     | 7 [10]      |
| TQA*         | 86.8 ± 4.8          | 57.5 ± 1.0  | 92.4 ± 0.8 | 83.2 ± 8.5  | 10 [50]        | 10 [144]     | 10 [100]     | 7 [10]      |
| PP           |                     |             |            |             |                |              |              |             |
| Fixed Angle* | -                   | 92.5 ± 0.6  | 90.6       | 90.7        | -              | 10 [144]     | 1 [100]      | 1 [10]      |
| Fixed Angle† | -                   | 90.2 ± 1.6  | 86.4       | 89.8        | -              | 10 [144]     | 1 [100]      | 1 [10]      |
| Fourier*     | 90.0 ± 2.8          | 89.9 ± 2.7  | 85.2 ± 2.1 | 89.9 ± 1.7  | 20 [40–50]     | 31 [39–144]  | 120 [40–100] | 28 [40–100] |
| Interp.*     | 93.3 ± 2.1          | 93.1 ± 0.9  | 88.3 ± 1.8 | 91.2 ± 2.1  | 20 [40–50]     | 31 [39–144]  | 120 [40–100] | 28 [40–100] |
| Linear Ramp  | 93.8 ± 1.0          | 90.9 ± 0.6  | 88.5 ± 2.2 | 90.0 ± 0.8  | 10 [50]        | 20 [105–144] | 20 [40–100]  | 8 [10]      |
| Linear Ramp* | 95.1 ± 1.0          | 92.1 ± 0.6  | 89.8 ± 2.3 | 90.8 ± 0.8  | 10 [50]        | 20 [105–144] | 20 [40–100]  | 8 [10]      |
| Recursive TS | 90.1 ± 1.1          | 88.7 ± 1.2  | 86.9 ± 1.8 | 87.1 ± 1.6  | 10 [50]        | 10 [144]     | 10 [100]     | 7 [10]      |
| TQA          | 83.8 ± 0.9          | 87.0 ± 0.8  | 77.2 ± 6.9 | 86.3 ± 2.3  | 10 [50]        | 10 [144]     | 10 [100]     | 7 [10]      |
| TQA*         | 94.5 ± 1.5          | 90.2 ± 0.6  | 85.6 ± 2.3 | 89.6 ± 1.1  | 10 [50]        | 10 [144]     | 10 [100]     | 7 [10]      |
| SV           |                     |             |            |             |                |              |              |             |
| Fixed Angle* | 95.4 ± 1.5          | -           | 97.4 ± 1.2 | 97.2 ± 1.2  | 17 [11–20]     | -            | 19 [10–20]   | 59 [10–20]  |
| Fixed Angle† | 90.7 ± 3.0          | -           | 96.3 ± 1.0 | 96.8 ± 1.2  | 17 [11–20]     | -            | 19 [10–20]   | 60 [10–20]  |
| Fourier*     | 80.4 ± 10.8         | 77.9 ± 10.7 | 80.8 ± 9.7 | 84.1 ± 12.4 | 92 [10–20]     | 20 [12–21]   | 780 [10–20]  | 382 [10–20] |
| Interp.*     | 95.1 ± 1.2          | 93.9 ± 1.2  | 94.5 ± 4.0 | 93.1 ± 5.4  | 92 [10–20]     | 20 [12–21]   | 780 [10–20]  | 382 [10–20] |
| Linear Ramp  | 92.5 ± 1.2          | 92.0 ± 1.3  | 91.1 ± 1.6 | 93.1 ± 1.8  | 10 [20]        | 10 [21]      | 10 [20]      | 7 [20]      |
| Linear Ramp* | 94.3 ± 1.0          | 93.6 ± 0.8  | 94.0 ± 1.6 | 95.2 ± 0.9  | 10 [20]        | 10 [21]      | 10 [20]      | 7 [20]      |
| Recursive TS | 93.8 ± 1.8          | 91.8 ± 2.3  | 92.3 ± 4.9 | 92.9 ± 3.3  | 92 [10–20]     | 20 [12–21]   | 412 [10–20]  | 379 [10–20] |
| TQA          | 89.4 ± 3.3          | 86.8 ± 3.7  | 83.1 ± 7.8 | 89.3 ± 4.8  | 92 [10–20]     | 20 [12–21]   | 677 [10–20]  | 361 [10–20] |
| TQA*         | 93.8 ± 1.3          | 91.4 ± 1.9  | 92.5 ± 3.2 | 95.0 ± 1.8  | 90 [10–20]     | 20 [12–21]   | 672 [10–20]  | 141 [10–20] |

Table 21: **Avg. Approx. Ratio and Number of Instances for Various Evaluation Methods for MAX-CUT for  $P = 7$ .** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Evaluation methods are shown in different colors for approximation ratios. Darker colors indicating better approximation ratios or more instances. Approximation ratios are in percentage, with  $\pm$  their standard deviation. The range of graph sizes are shown in brackets, next to the number of instances.

## 8 Tables for depth $P = 8$

|              | ER         | HH           | LB           | RR          |
|--------------|------------|--------------|--------------|-------------|
| MPS (Aer)    |            |              |              |             |
| Fixed Angle* | -          | 10 [39]      | 11 [100]     | 11 [40-100] |
| Fixed Angle† | -          | 10 [39]      | 11 [100]     | 11 [40-100] |
| Fourier*     | 20 [40-50] | 20 [39-144]  | 49 [40-100]  | 17 [40-100] |
| Interp.*     | 20 [40-50] | 20 [39-144]  | 49 [40-100]  | 17 [40-100] |
| Linear Ramp  | 10 [50]    | 10 [144]     | 10 [100]     | 7 [100]     |
| Linear Ramp* | 10 [50]    | 10 [144]     | 10 [100]     | 7 [100]     |
| TQA          | 20 [40-50] | 20 [39-144]  | 50 [40-100]  | 17 [40-100] |
| TQA*         | 20 [40-50] | 20 [39-144]  | 50 [40-100]  | 17 [40-100] |
| MPS (Quimb)  |            |              |              |             |
| Fixed Angle* | -          | 10 [144]     | 10 [100]     | 11 [40-100] |
| Fixed Angle† | -          | 10 [144]     | 10 [100]     | 11 [40-100] |
| Fourier*     | 20 [40-50] | 31 [39-144]  | 120 [40-100] | 28 [40-100] |
| Interp.*     | 20 [40-50] | 31 [39-144]  | 120 [40-100] | 28 [40-100] |
| Linear Ramp  | 10 [50]    | 10 [144]     | 10 [100]     | 7 [100]     |
| Linear Ramp* | 10 [50]    | 10 [144]     | 10 [100]     | 7 [100]     |
| Recursive TS | 10 [50]    | 10 [144]     | 10 [100]     | 7 [100]     |
| TQA          | 20 [40-50] | 20 [39-144]  | 120 [40-100] | 27 [40-100] |
| TQA*         | 20 [40-50] | 20 [39-144]  | 120 [40-100] | 27 [40-100] |
| PP           |            |              |              |             |
| Fixed Angle* | -          | 30 [39-144]  | 10 [100]     | 11 [40-100] |
| Fixed Angle† | -          | 30 [39-144]  | 10 [100]     | 11 [40-100] |
| Fourier*     | 20 [40-50] | 31 [39-144]  | 120 [40-100] | 28 [40-100] |
| Interp.*     | 20 [40-50] | 31 [39-144]  | 120 [40-100] | 28 [40-100] |
| Linear Ramp  | 10 [50]    | 20 [105-144] | 20 [40-100]  | 8 [100]     |
| Linear Ramp* | 10 [50]    | 20 [105-144] | 20 [40-100]  | 8 [100]     |
| Recursive TS | 10 [50]    | 10 [144]     | 10 [100]     | 7 [100]     |
| TQA          | 20 [40-50] | 20 [39-144]  | 120 [40-100] | 27 [40-100] |
| TQA*         | 20 [40-50] | 20 [39-144]  | 120 [40-100] | 27 [40-100] |
| SV           |            |              |              |             |
| Fixed Angle* | 17 [11-20] | -            | 19 [10-20]   | 50 [10-20]  |
| Fixed Angle† | 17 [11-20] | -            | 19 [10-20]   | 50 [10-20]  |
| Fourier*     | 92 [10-20] | 20 [12-21]   | 778 [10-20]  | 382 [10-20] |
| Interp.*     | 92 [10-20] | 20 [12-21]   | 779 [10-20]  | 382 [10-20] |
| Linear Ramp  | 10 [20]    | 10 [21]      | 10 [20]      | 7 [20]      |
| Linear Ramp* | 10 [20]    | 10 [21]      | 10 [20]      | 7 [20]      |
| Recursive TS | 92 [10-20] | 20 [12-21]   | 407 [10-20]  | 379 [10-20] |
| TQA          | 92 [10-20] | 20 [12-21]   | 677 [10-20]  | 356 [10-20] |
| TQA*         | 90 [10-20] | 20 [12-21]   | 672 [10-20]  | 142 [10-20] |

Table 22: **Number of Instances (num. nodes in brackets) for  $P = 8$ .** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Cells are colored based on the number of instances for the given configuration, with darker colors indicating more instances.

|              | ER              | HH              | LB              | RR              |
|--------------|-----------------|-----------------|-----------------|-----------------|
| MPS (Aer)    |                 |                 |                 |                 |
| Fixed Angle* | -               | $92.5 \pm 0.9$  | $94.0 \pm 1.4$  | $97.2 \pm 3.9$  |
| Fixed Angle† | -               | $90.8 \pm 1.8$  | $90.2 \pm 2.9$  | $96.8 \pm 4.3$  |
| Fourier*     | $77.9 \pm 5.5$  | $85.2 \pm 3.2$  | $82.3 \pm 8.7$  | $81.8 \pm 9.6$  |
| Interp.*     | $94.2 \pm 2.1$  | $90.0 \pm 4.7$  | $94.6 \pm 2.6$  | $86.6 \pm 6.8$  |
| Linear Ramp  | $92.4 \pm 1.2$  | $83.9 \pm 2.2$  | $94.0 \pm 1.0$  | $87.7 \pm 2.1$  |
| Linear Ramp* | $94.5 \pm 1.8$  | $84.2 \pm 1.4$  | $95.6 \pm 0.7$  | $89.2 \pm 2.4$  |
| TQA          | $81.5 \pm 3.5$  | $85.3 \pm 2.7$  | $83.8 \pm 7.1$  | $87.3 \pm 6.6$  |
| TQA*         | $85.8 \pm 2.2$  | $86.5 \pm 3.2$  | $88.7 \pm 5.9$  | $92.2 \pm 6.2$  |
| MPS (Quimb)  |                 |                 |                 |                 |
| Fixed Angle* | -               | $56.5 \pm 3.5$  | $96.7 \pm 0.8$  | $97.9 \pm 1.3$  |
| Fixed Angle† | -               | $51.9 \pm 2.6$  | $96.6 \pm 0.8$  | $96.9 \pm 3.3$  |
| Fourier*     | $81.3 \pm 6.6$  | $76.8 \pm 10.2$ | $88.2 \pm 5.2$  | $91.1 \pm 8.2$  |
| Interp.*     | $96.9 \pm 1.3$  | $84.3 \pm 10.1$ | $95.6 \pm 1.0$  | $93.9 \pm 6.5$  |
| Linear Ramp  | $95.3 \pm 1.2$  | $69.4 \pm 1.9$  | $93.4 \pm 1.2$  | $90.6 \pm 3.3$  |
| Linear Ramp* | $96.7 \pm 0.8$  | $69.0 \pm 2.7$  | $94.9 \pm 1.2$  | $93.8 \pm 0.8$  |
| Recursive TS | $96.9 \pm 1.0$  | $56.3 \pm 1.4$  | $93.3 \pm 1.1$  | $84.4 \pm 4.5$  |
| TQA          | $82.2 \pm 6.9$  | $71.3 \pm 17.4$ | $89.3 \pm 2.0$  | $90.1 \pm 8.0$  |
| TQA*         | $91.1 \pm 3.7$  | $72.6 \pm 18.2$ | $92.2 \pm 1.3$  | $93.4 \pm 6.7$  |
| PP           |                 |                 |                 |                 |
| Fixed Angle* | -               | $93.4 \pm 0.6$  | $90.3 \pm 1.9$  | $93.2 \pm 1.6$  |
| Fixed Angle† | -               | $91.0 \pm 1.4$  | $86.2 \pm 1.7$  | $92.6 \pm 1.7$  |
| Fourier*     | $91.0 \pm 1.9$  | $90.1 \pm 3.0$  | $85.7 \pm 2.0$  | $90.1 \pm 2.1$  |
| Interp.*     | $93.7 \pm 2.0$  | $93.9 \pm 0.9$  | $88.8 \pm 1.8$  | $91.5 \pm 2.2$  |
| Linear Ramp  | $94.1 \pm 1.0$  | $91.6 \pm 0.6$  | $89.0 \pm 2.3$  | $90.4 \pm 0.8$  |
| Linear Ramp* | $95.5 \pm 1.0$  | $92.8 \pm 0.6$  | $90.3 \pm 2.2$  | $91.0 \pm 0.8$  |
| Recursive TS | $90.5 \pm 1.1$  | $89.3 \pm 1.1$  | $87.2 \pm 1.9$  | $87.3 \pm 1.6$  |
| TQA          | $84.6 \pm 2.2$  | $88.2 \pm 1.0$  | $77.4 \pm 6.7$  | $89.1 \pm 2.4$  |
| TQA*         | $91.7 \pm 2.3$  | $90.8 \pm 0.7$  | $85.6 \pm 2.5$  | $91.5 \pm 1.5$  |
| SV           |                 |                 |                 |                 |
| Fixed Angle* | $96.3 \pm 1.4$  | -               | $98.0 \pm 1.0$  | $97.8 \pm 1.1$  |
| Fixed Angle† | $91.5 \pm 3.3$  | -               | $96.7 \pm 1.1$  | $97.2 \pm 1.1$  |
| Fourier*     | $78.9 \pm 12.3$ | $76.0 \pm 10.2$ | $79.7 \pm 10.9$ | $83.5 \pm 13.1$ |
| Interp.*     | $95.7 \pm 1.2$  | $94.3 \pm 1.1$  | $95.2 \pm 3.8$  | $93.5 \pm 5.5$  |
| Linear Ramp  | $93.2 \pm 1.2$  | $93.1 \pm 0.7$  | $92.0 \pm 1.7$  | $93.8 \pm 1.7$  |
| Linear Ramp* | $95.0 \pm 0.9$  | $94.3 \pm 0.6$  | $94.4 \pm 1.5$  | $95.5 \pm 0.7$  |
| Recursive TS | $94.4 \pm 1.8$  | $92.2 \pm 2.4$  | $93.0 \pm 4.9$  | $93.5 \pm 3.2$  |
| TQA          | $90.3 \pm 3.4$  | $88.0 \pm 3.6$  | $83.6 \pm 8.5$  | $90.0 \pm 4.9$  |
| TQA*         | $94.4 \pm 1.3$  | $91.9 \pm 1.7$  | $93.1 \pm 3.3$  | $95.6 \pm 1.7$  |

Table 23: **Avg. Approx. Ratio  $\pm$  standard deviation in percentage for MAXCUT for  $P = 8$ .** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Evaluation methods are shown in different colors, with darker colors indicating better approximation ratios.

|              | ER          | Approximation Ratio |             |             |            | Num. Instances |              |          |  |
|--------------|-------------|---------------------|-------------|-------------|------------|----------------|--------------|----------|--|
|              |             | HH                  | LB          | RR          | ER         | HH             | LB           | RR       |  |
| MPS (Aer)    |             |                     |             |             |            |                |              |          |  |
| Fixed Angle* | -           | 92.5 ± 0.9          | 94.0 ± 1.4  | 97.2 ± 3.9  | -          | 10 [39]        | 11 [100]     | 11 [40]  |  |
| Fixed Angle† | -           | 90.8 ± 1.8          | 90.2 ± 2.9  | 96.8 ± 4.3  | -          | 10 [39]        | 11 [100]     | 11 [40]  |  |
| Fourier*     | 77.9 ± 5.5  | 85.2 ± 3.2          | 82.3 ± 8.7  | 81.8 ± 9.6  | 20 [40–50] | 20 [39–144]    | 49 [40–100]  | 17 [40]  |  |
| Interp.*     | 94.2 ± 2.1  | 90.0 ± 4.7          | 94.6 ± 2.6  | 86.6 ± 6.8  | 20 [40–50] | 20 [39–144]    | 49 [40–100]  | 17 [40]  |  |
| Linear Ramp  | 92.4 ± 1.2  | 83.9 ± 2.2          | 94.0 ± 1.0  | 87.7 ± 2.1  | 10 [50]    | 10 [144]       | 10 [100]     | 7 [40]   |  |
| Linear Ramp* | 94.5 ± 1.8  | 84.2 ± 1.4          | 95.6 ± 0.7  | 89.2 ± 2.4  | 10 [50]    | 10 [144]       | 10 [100]     | 7 [40]   |  |
| TQA          | 81.5 ± 3.5  | 85.3 ± 2.7          | 83.8 ± 7.1  | 87.3 ± 6.6  | 20 [40–50] | 20 [39–144]    | 50 [40–100]  | 17 [40]  |  |
| TQA*         | 85.8 ± 2.2  | 86.5 ± 3.2          | 88.7 ± 5.9  | 92.2 ± 6.2  | 20 [40–50] | 20 [39–144]    | 50 [40–100]  | 17 [40]  |  |
| MPS (Quimb)  |             |                     |             |             |            |                |              |          |  |
| Fixed Angle* | -           | 56.5 ± 3.5          | 96.7 ± 0.8  | 97.9 ± 1.3  | -          | 10 [144]       | 10 [100]     | 11 [40]  |  |
| Fixed Angle† | -           | 51.9 ± 2.6          | 96.6 ± 0.8  | 96.9 ± 3.3  | -          | 10 [144]       | 10 [100]     | 11 [40]  |  |
| Fourier*     | 81.3 ± 6.6  | 76.8 ± 10.2         | 88.2 ± 5.2  | 91.1 ± 8.2  | 20 [40–50] | 31 [39–144]    | 120 [40–100] | 28 [40]  |  |
| Interp.*     | 96.9 ± 1.3  | 84.3 ± 10.1         | 95.6 ± 1.0  | 93.9 ± 6.5  | 20 [40–50] | 31 [39–144]    | 120 [40–100] | 28 [40]  |  |
| Linear Ramp  | 95.3 ± 1.2  | 69.4 ± 1.9          | 93.4 ± 1.2  | 90.6 ± 3.3  | 10 [50]    | 10 [144]       | 10 [100]     | 7 [40]   |  |
| Linear Ramp* | 96.7 ± 0.8  | 69.0 ± 2.7          | 94.9 ± 1.2  | 93.8 ± 0.8  | 10 [50]    | 10 [144]       | 10 [100]     | 7 [40]   |  |
| Recursive TS | 96.9 ± 1.0  | 56.3 ± 1.4          | 93.3 ± 1.1  | 84.4 ± 4.5  | 10 [50]    | 10 [144]       | 10 [100]     | 7 [40]   |  |
| TQA          | 82.2 ± 6.9  | 71.3 ± 17.4         | 89.3 ± 2.0  | 90.1 ± 8.0  | 20 [40–50] | 20 [39–144]    | 120 [40–100] | 27 [40]  |  |
| TQA*         | 91.1 ± 3.7  | 72.6 ± 18.2         | 92.2 ± 1.3  | 93.4 ± 6.7  | 20 [40–50] | 20 [39–144]    | 120 [40–100] | 27 [40]  |  |
| PP           |             |                     |             |             |            |                |              |          |  |
| Fixed Angle* | -           | 93.4 ± 0.6          | 90.3 ± 1.9  | 93.2 ± 1.6  | -          | 30 [39–144]    | 10 [100]     | 11 [40]  |  |
| Fixed Angle† | -           | 91.0 ± 1.4          | 86.2 ± 1.7  | 92.6 ± 1.7  | -          | 30 [39–144]    | 10 [100]     | 11 [40]  |  |
| Fourier*     | 91.0 ± 1.9  | 90.1 ± 3.0          | 85.7 ± 2.0  | 90.1 ± 2.1  | 20 [40–50] | 31 [39–144]    | 120 [40–100] | 28 [40]  |  |
| Interp.*     | 93.7 ± 2.0  | 93.9 ± 0.9          | 88.8 ± 1.8  | 91.5 ± 2.2  | 20 [40–50] | 31 [39–144]    | 120 [40–100] | 28 [40]  |  |
| Linear Ramp  | 94.1 ± 1.0  | 91.6 ± 0.6          | 89.0 ± 2.3  | 90.4 ± 0.8  | 10 [50]    | 20 [105–144]   | 20 [40–100]  | 8 [40]   |  |
| Linear Ramp* | 95.5 ± 1.0  | 92.8 ± 0.6          | 90.3 ± 2.2  | 91.0 ± 0.8  | 10 [50]    | 20 [105–144]   | 20 [40–100]  | 8 [40]   |  |
| Recursive TS | 90.5 ± 1.1  | 89.3 ± 1.1          | 87.2 ± 1.9  | 87.3 ± 1.6  | 10 [50]    | 10 [144]       | 10 [100]     | 7 [40]   |  |
| TQA          | 84.6 ± 2.2  | 88.2 ± 1.0          | 77.4 ± 6.7  | 89.1 ± 2.4  | 20 [40–50] | 20 [39–144]    | 120 [40–100] | 27 [40]  |  |
| TQA*         | 91.7 ± 2.3  | 90.8 ± 0.7          | 85.6 ± 2.5  | 91.5 ± 1.5  | 20 [40–50] | 20 [39–144]    | 120 [40–100] | 27 [40]  |  |
| SV           |             |                     |             |             |            |                |              |          |  |
| Fixed Angle* | 96.3 ± 1.4  | -                   | 98.0 ± 1.0  | 97.8 ± 1.1  | 17 [11–20] | -              | 19 [10–20]   | 50 [10]  |  |
| Fixed Angle† | 91.5 ± 3.3  | -                   | 96.7 ± 1.1  | 97.2 ± 1.1  | 17 [11–20] | -              | 19 [10–20]   | 50 [10]  |  |
| Fourier*     | 78.9 ± 12.3 | 76.0 ± 10.2         | 79.7 ± 10.9 | 83.5 ± 13.1 | 92 [10–20] | 20 [12–21]     | 778 [10–20]  | 382 [10] |  |
| Interp.*     | 95.7 ± 1.2  | 94.3 ± 1.1          | 95.2 ± 3.8  | 93.5 ± 5.5  | 92 [10–20] | 20 [12–21]     | 779 [10–20]  | 382 [10] |  |
| Linear Ramp  | 93.2 ± 1.2  | 93.1 ± 0.7          | 92.0 ± 1.7  | 93.8 ± 1.7  | 10 [20]    | 10 [21]        | 10 [20]      | 7 [10]   |  |
| Linear Ramp* | 95.0 ± 0.9  | 94.3 ± 0.6          | 94.4 ± 1.5  | 95.5 ± 0.7  | 10 [20]    | 10 [21]        | 10 [20]      | 7 [10]   |  |
| Recursive TS | 94.4 ± 1.8  | 92.2 ± 2.4          | 93.0 ± 4.9  | 93.5 ± 3.2  | 92 [10–20] | 20 [12–21]     | 407 [10–20]  | 379 [10] |  |
| TQA          | 90.3 ± 3.4  | 88.0 ± 3.6          | 83.6 ± 8.5  | 90.0 ± 4.9  | 92 [10–20] | 20 [12–21]     | 677 [10–20]  | 356 [10] |  |
| TQA*         | 94.4 ± 1.3  | 91.9 ± 1.7          | 93.1 ± 3.3  | 95.6 ± 1.7  | 90 [10–20] | 20 [12–21]     | 672 [10–20]  | 142 [10] |  |

Table 24: **Avg. Approx. Ratio and Number of Instances for Various Evaluation Methods for MAX-CUT for  $P = 8$ .** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Evaluation methods are shown in different colors for approximation ratios. Darker colors indicating better approximation ratios or more instances. Approximation ratios are in percentage, with  $\pm$  their standard deviation. The range of graph sizes are shown in brackets, next to the number of instances.

## 9 Tables for depth $P = 9$

|              | ER         | HH           | LB           | RR          |
|--------------|------------|--------------|--------------|-------------|
| MPS (Aer)    |            |              |              |             |
| Fixed Angle* | -          | -            | 1 [100]      | 1 [100]     |
| Fixed Angle† | -          | -            | 1 [100]      | 1 [100]     |
| Fourier*     | 20 [40–50] | 20 [39–144]  | 49 [40–100]  | 17 [40–100] |
| Interp.*     | 20 [40–50] | 20 [39–144]  | 49 [40–100]  | 17 [40–100] |
| Linear Ramp  | 10 [50]    | 10 [144]     | 10 [100]     | 7 [100]     |
| Linear Ramp* | 10 [50]    | 10 [144]     | 10 [100]     | 7 [100]     |
| TQA          | 10 [50]    | 10 [144]     | 10 [100]     | 7 [100]     |
| TQA*         | 10 [50]    | 10 [144]     | 10 [100]     | 7 [100]     |
| MPS (Quimb)  |            |              |              |             |
| Fixed Angle* | -          | 10 [144]     | 1 [100]      | 1 [100]     |
| Fixed Angle† | -          | 10 [144]     | 1 [100]      | 1 [100]     |
| Fourier*     | 20 [40–50] | 31 [39–144]  | 111 [40–100] | 18 [40–100] |
| Interp.*     | 20 [40–50] | 31 [39–144]  | 111 [40–100] | 18 [40–100] |
| Linear Ramp  | 10 [50]    | 10 [144]     | 10 [100]     | 7 [100]     |
| Linear Ramp* | 10 [50]    | 10 [144]     | 10 [100]     | 7 [100]     |
| Recursive TS | 10 [50]    | 10 [144]     | 10 [100]     | 7 [100]     |
| TQA          | 10 [50]    | 10 [144]     | 10 [100]     | 7 [100]     |
| TQA*         | 10 [50]    | 10 [144]     | 10 [100]     | 7 [100]     |
| PP           |            |              |              |             |
| Fixed Angle* | -          | 10 [144]     | 1 [100]      | 1 [100]     |
| Fixed Angle† | -          | 10 [144]     | 1 [100]      | 1 [100]     |
| Fourier*     | 20 [40–50] | 31 [39–144]  | 111 [40–100] | 18 [40–100] |
| Interp.*     | 20 [40–50] | 31 [39–144]  | 111 [40–100] | 18 [40–100] |
| Linear Ramp  | 10 [50]    | 20 [105–144] | 20 [40–100]  | 8 [100]     |
| Linear Ramp* | 10 [50]    | 20 [105–144] | 20 [40–100]  | 8 [100]     |
| Recursive TS | 10 [50]    | 10 [144]     | 10 [100]     | 7 [100]     |
| TQA          | 10 [50]    | 10 [144]     | 10 [100]     | 7 [100]     |
| TQA*         | 10 [50]    | 10 [144]     | 10 [100]     | 7 [100]     |
| SV           |            |              |              |             |
| Fixed Angle* | 17 [11–20] | -            | 19 [10–20]   | 50 [10–20]  |
| Fixed Angle† | 17 [11–20] | -            | 19 [10–20]   | 50 [10–20]  |
| Fourier*     | 92 [10–20] | 20 [12–21]   | 776 [10–20]  | 380 [10–20] |
| Interp.*     | 92 [10–20] | 20 [12–21]   | 777 [10–20]  | 381 [10–20] |
| Linear Ramp  | 10 [20]    | 10 [21]      | 10 [20]      | 7 [20]      |
| Linear Ramp* | 10 [20]    | 10 [21]      | 10 [20]      | 7 [20]      |
| Recursive TS | 92 [10–20] | 20 [12–21]   | 407 [10–20]  | 376 [10–20] |
| TQA          | 92 [10–20] | 20 [12–21]   | 676 [10–20]  | 356 [10–20] |
| TQA*         | 90 [10–20] | 20 [12–21]   | 671 [10–20]  | 141 [10–20] |

Table 25: **Number of Instances (num. nodes in brackets) for  $P = 9$ .** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Cells are colored based on the number of instances for the given configuration, with darker colors indicating more instances.

|                          | ER          | HH          | LB          | RR          |
|--------------------------|-------------|-------------|-------------|-------------|
| MPS (Aer)                |             |             |             |             |
| Fixed Angle <sup>*</sup> | -           | -           | 97.8        | 90.2        |
| Fixed Angle <sup>†</sup> | -           | -           | 97.5        | 85.1        |
| Fourier <sup>*</sup>     | 78.1 ± 5.3  | 85.8 ± 4.0  | 79.4 ± 8.3  | 84.0 ± 9.3  |
| Interp. <sup>*</sup>     | 94.3 ± 3.6  | 90.6 ± 4.8  | 94.7 ± 2.5  | 88.7 ± 7.1  |
| Linear Ramp              | 93.1 ± 1.4  | 84.4 ± 3.0  | 94.6 ± 1.0  | 88.6 ± 3.0  |
| Linear Ramp <sup>*</sup> | 95.4 ± 1.8  | 84.2 ± 2.0  | 96.0 ± 0.7  | 91.3 ± 2.1  |
| TQA                      | 85.4 ± 1.5  | 83.6 ± 1.7  | 91.3 ± 2.1  | 83.8 ± 7.3  |
| TQA <sup>*</sup>         | 86.5 ± 1.5  | 84.2 ± 1.9  | 93.8 ± 1.4  | 88.5 ± 3.9  |
| MPS (Quimb)              |             |             |             |             |
| Fixed Angle <sup>*</sup> | -           | 59.1 ± 3.3  | 97.8        | 96.1        |
| Fixed Angle <sup>†</sup> | -           | 52.4 ± 3.3  | 97.4        | 90.9        |
| Fourier <sup>*</sup>     | 82.0 ± 7.8  | 78.3 ± 9.7  | 88.6 ± 4.3  | 88.8 ± 8.8  |
| Interp. <sup>*</sup>     | 97.2 ± 1.4  | 86.2 ± 9.3  | 96.2 ± 0.9  | 90.9 ± 7.3  |
| Linear Ramp              | 95.5 ± 1.3  | 70.4 ± 1.5  | 94.1 ± 1.1  | 91.5 ± 2.0  |
| Linear Ramp <sup>*</sup> | 97.0 ± 1.1  | 72.1 ± 1.9  | 95.4 ± 1.0  | 92.9 ± 3.0  |
| Recursive TS             | 97.1 ± 1.1  | 57.7 ± 1.1  | 93.6 ± 1.3  | 85.8 ± 4.0  |
| TQA                      | 83.7 ± 7.3  | 57.4 ± 1.1  | 90.5 ± 1.8  | 81.6 ± 12.0 |
| TQA <sup>*</sup>         | 90.0 ± 2.3  | 57.8 ± 1.2  | 93.3 ± 1.0  | 90.1 ± 1.5  |
| PP                       |             |             |             |             |
| Fixed Angle <sup>*</sup> | -           | 93.7 ± 0.4  | 91.5        | 91.2        |
| Fixed Angle <sup>†</sup> | -           | 90.9 ± 1.5  | 87.2        | 90.7        |
| Fourier <sup>*</sup>     | 91.7 ± 1.6  | 90.6 ± 3.2  | 86.2 ± 1.9  | 90.5 ± 2.3  |
| Interp. <sup>*</sup>     | 94.1 ± 2.0  | 94.4 ± 0.8  | 89.2 ± 1.7  | 90.8 ± 2.3  |
| Linear Ramp              | 94.4 ± 1.0  | 92.2 ± 0.6  | 89.5 ± 2.2  | 90.8 ± 0.8  |
| Linear Ramp <sup>*</sup> | 95.6 ± 1.0  | 93.3 ± 0.5  | 90.7 ± 2.2  | 91.4 ± 0.8  |
| Recursive TS             | 90.6 ± 1.2  | 89.7 ± 1.0  | 87.3 ± 1.9  | 87.3 ± 1.7  |
| TQA                      | 83.6 ± 0.9  | 89.1 ± 0.7  | 77.1 ± 7.9  | 86.9 ± 3.0  |
| TQA <sup>*</sup>         | 92.9 ± 1.4  | 91.0 ± 0.6  | 85.6 ± 2.9  | 90.4 ± 0.9  |
| SV                       |             |             |             |             |
| Fixed Angle <sup>*</sup> | 96.9 ± 1.4  | -           | 98.4 ± 0.8  | 98.3 ± 1.1  |
| Fixed Angle <sup>†</sup> | 91.7 ± 3.5  | -           | 97.0 ± 1.3  | 97.7 ± 1.0  |
| Fourier <sup>*</sup>     | 78.5 ± 12.6 | 73.8 ± 10.4 | 78.9 ± 11.5 | 83.1 ± 13.8 |
| Interp. <sup>*</sup>     | 96.2 ± 1.1  | 94.5 ± 1.1  | 95.5 ± 4.1  | 93.4 ± 6.2  |
| Linear Ramp              | 93.7 ± 1.2  | 93.8 ± 0.5  | 92.8 ± 1.5  | 94.3 ± 1.5  |
| Linear Ramp <sup>*</sup> | 95.4 ± 0.8  | 94.7 ± 0.6  | 94.8 ± 1.5  | 95.9 ± 0.9  |
| Recursive TS             | 94.8 ± 1.8  | 92.5 ± 2.4  | 93.1 ± 4.4  | 93.8 ± 3.1  |
| TQA                      | 91.0 ± 3.6  | 88.9 ± 3.4  | 83.8 ± 9.2  | 90.8 ± 4.9  |
| TQA <sup>*</sup>         | 94.9 ± 1.3  | 92.5 ± 1.6  | 93.6 ± 3.3  | 96.1 ± 1.7  |

Table 26: **Avg. Approx. Ratio ± standard deviation in percentage for MAXCUT for  $P = 9$ .** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Evaluation methods are shown in different colors, with darker colors indicating better approximation ratios.

|              | Approximation Ratio |             |             |             | Num. Instances |              |              |             |
|--------------|---------------------|-------------|-------------|-------------|----------------|--------------|--------------|-------------|
|              | ER                  | HH          | LB          | RR          | ER             | HH           | LB           | RR          |
| MPS (Aer)    |                     |             |             |             |                |              |              |             |
| Fixed Angle* | -                   | -           | 97.8        | 90.2        | -              | -            | 1 [100]      | 1 [100]     |
| Fixed Angle† | -                   | -           | 97.5        | 85.1        | -              | -            | 1 [100]      | 1 [100]     |
| Fourier*     | 78.1 ± 5.3          | 85.8 ± 4.0  | 79.4 ± 8.3  | 84.0 ± 9.3  | 20 [40–50]     | 20 [39–144]  | 49 [40–100]  | 17 [40–100] |
| Interp.*     | 94.3 ± 3.6          | 90.6 ± 4.8  | 94.7 ± 2.5  | 88.7 ± 7.1  | 20 [40–50]     | 20 [39–144]  | 49 [40–100]  | 17 [40–100] |
| Linear Ramp  | 93.1 ± 1.4          | 84.4 ± 3.0  | 94.6 ± 1.0  | 88.6 ± 3.0  | 10 [50]        | 10 [144]     | 10 [100]     | 7 [100]     |
| Linear Ramp* | 95.4 ± 1.8          | 84.2 ± 2.0  | 96.0 ± 0.7  | 91.3 ± 2.1  | 10 [50]        | 10 [144]     | 10 [100]     | 7 [100]     |
| TQA          | 85.4 ± 1.5          | 83.6 ± 1.7  | 91.3 ± 2.1  | 83.8 ± 7.3  | 10 [50]        | 10 [144]     | 10 [100]     | 7 [100]     |
| TQA*         | 86.5 ± 1.5          | 84.2 ± 1.9  | 93.8 ± 1.4  | 88.5 ± 3.9  | 10 [50]        | 10 [144]     | 10 [100]     | 7 [100]     |
| MPS (Quimb)  |                     |             |             |             |                |              |              |             |
| Fixed Angle* | -                   | 59.1 ± 3.3  | 97.8        | 96.1        | -              | 10 [144]     | 1 [100]      | 1 [100]     |
| Fixed Angle† | -                   | 52.4 ± 3.3  | 97.4        | 90.9        | -              | 10 [144]     | 1 [100]      | 1 [100]     |
| Fourier*     | 82.0 ± 7.8          | 78.3 ± 9.7  | 88.6 ± 4.3  | 88.8 ± 8.8  | 20 [40–50]     | 31 [39–144]  | 111 [40–100] | 18 [40–100] |
| Interp.*     | 97.2 ± 1.4          | 86.2 ± 9.3  | 96.2 ± 0.9  | 90.9 ± 7.3  | 20 [40–50]     | 31 [39–144]  | 111 [40–100] | 18 [40–100] |
| Linear Ramp  | 95.5 ± 1.3          | 70.4 ± 1.5  | 94.1 ± 1.1  | 91.5 ± 2.0  | 10 [50]        | 10 [144]     | 10 [100]     | 7 [100]     |
| Linear Ramp* | 97.0 ± 1.1          | 72.1 ± 1.9  | 95.4 ± 1.0  | 92.9 ± 3.0  | 10 [50]        | 10 [144]     | 10 [100]     | 7 [100]     |
| Recursive TS | 97.1 ± 1.1          | 57.7 ± 1.1  | 93.6 ± 1.3  | 85.8 ± 4.0  | 10 [50]        | 10 [144]     | 10 [100]     | 7 [100]     |
| TQA          | 83.7 ± 7.3          | 57.4 ± 1.1  | 90.5 ± 1.8  | 81.6 ± 12.0 | 10 [50]        | 10 [144]     | 10 [100]     | 7 [100]     |
| TQA*         | 90.0 ± 2.3          | 57.8 ± 1.2  | 93.3 ± 1.0  | 90.1 ± 1.5  | 10 [50]        | 10 [144]     | 10 [100]     | 7 [100]     |
| PP           |                     |             |             |             |                |              |              |             |
| Fixed Angle* | -                   | 93.7 ± 0.4  | 91.5        | 91.2        | -              | 10 [144]     | 1 [100]      | 1 [100]     |
| Fixed Angle† | -                   | 90.9 ± 1.5  | 87.2        | 90.7        | -              | 10 [144]     | 1 [100]      | 1 [100]     |
| Fourier*     | 91.7 ± 1.6          | 90.6 ± 3.2  | 86.2 ± 1.9  | 90.5 ± 2.3  | 20 [40–50]     | 31 [39–144]  | 111 [40–100] | 18 [40–100] |
| Interp.*     | 94.1 ± 2.0          | 94.4 ± 0.8  | 89.2 ± 1.7  | 90.8 ± 2.3  | 20 [40–50]     | 31 [39–144]  | 111 [40–100] | 18 [40–100] |
| Linear Ramp  | 94.4 ± 1.0          | 92.2 ± 0.6  | 89.5 ± 2.2  | 90.8 ± 0.8  | 10 [50]        | 20 [105–144] | 20 [40–100]  | 8 [100]     |
| Linear Ramp* | 95.6 ± 1.0          | 93.3 ± 0.5  | 90.7 ± 2.2  | 91.4 ± 0.8  | 10 [50]        | 20 [105–144] | 20 [40–100]  | 8 [100]     |
| Recursive TS | 90.6 ± 1.2          | 89.7 ± 1.0  | 87.3 ± 1.9  | 87.3 ± 1.7  | 10 [50]        | 10 [144]     | 10 [100]     | 7 [100]     |
| TQA          | 83.6 ± 0.9          | 89.1 ± 0.7  | 77.1 ± 7.9  | 86.9 ± 3.0  | 10 [50]        | 10 [144]     | 10 [100]     | 7 [100]     |
| TQA*         | 92.9 ± 1.4          | 91.0 ± 0.6  | 85.6 ± 2.9  | 90.4 ± 0.9  | 10 [50]        | 10 [144]     | 10 [100]     | 7 [100]     |
| SV           |                     |             |             |             |                |              |              |             |
| Fixed Angle* | 96.9 ± 1.4          | -           | 98.4 ± 0.8  | 98.3 ± 1.1  | 17 [11–20]     | -            | 19 [10–20]   | 50 [10–20]  |
| Fixed Angle† | 91.7 ± 3.5          | -           | 97.0 ± 1.3  | 97.7 ± 1.0  | 17 [11–20]     | -            | 19 [10–20]   | 50 [10–20]  |
| Fourier*     | 78.5 ± 12.6         | 73.8 ± 10.4 | 78.9 ± 11.5 | 83.1 ± 13.8 | 92 [10–20]     | 20 [12–21]   | 776 [10–20]  | 380 [10–20] |
| Interp.*     | 96.2 ± 1.1          | 94.5 ± 1.1  | 95.5 ± 4.1  | 93.4 ± 6.2  | 92 [10–20]     | 20 [12–21]   | 777 [10–20]  | 381 [10–20] |
| Linear Ramp  | 93.7 ± 1.2          | 93.8 ± 0.5  | 92.8 ± 1.5  | 94.3 ± 1.5  | 10 [20]        | 10 [21]      | 10 [20]      | 7 [20]      |
| Linear Ramp* | 95.4 ± 0.8          | 94.7 ± 0.6  | 94.8 ± 1.5  | 95.9 ± 0.9  | 10 [20]        | 10 [21]      | 10 [20]      | 7 [20]      |
| Recursive TS | 94.8 ± 1.8          | 92.5 ± 2.4  | 93.1 ± 4.4  | 93.8 ± 3.1  | 92 [10–20]     | 20 [12–21]   | 407 [10–20]  | 376 [10–20] |
| TQA          | 91.0 ± 3.6          | 88.9 ± 3.4  | 83.8 ± 9.2  | 90.8 ± 4.9  | 92 [10–20]     | 20 [12–21]   | 676 [10–20]  | 356 [10–20] |
| TQA*         | 94.9 ± 1.3          | 92.5 ± 1.6  | 93.6 ± 3.3  | 96.1 ± 1.7  | 90 [10–20]     | 20 [12–21]   | 671 [10–20]  | 141 [10–20] |

Table 27: **Avg. Approx. Ratio and Number of Instances for Various Evaluation Methods for MAX-CUT for  $P = 9$ .** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Evaluation methods are shown in different colors for approximation ratios. Darker colors indicating better approximation ratios or more instances. Approximation ratios are in percentage, with  $\pm$  their standard deviation. The range of graph sizes are shown in brackets, next to the number of instances.

## 10 Tables for depth $P = 10$

|              | ER         | HH           | LB           | RR          |
|--------------|------------|--------------|--------------|-------------|
| MPS (Aer)    |            |              |              |             |
| Fixed Angle* | -          | 20 [39–144]  | 11 [100]     | 11 [40–100] |
| Fixed Angle† | -          | 20 [39–144]  | 11 [100]     | 11 [40–100] |
| Fourier*     | 20 [40–50] | 20 [39–144]  | 49 [40–100]  | 11 [40–100] |
| Interp.*     | 20 [40–50] | 20 [39–144]  | 49 [40–100]  | 11 [40–100] |
| Linear Ramp  | 20 [40–50] | 20 [39–144]  | 50 [40–100]  | 11 [40–100] |
| Linear Ramp* | 10 [50]    | 10 [144]     | 10 [100]     | 1 [100]     |
| TQA          | 20 [40–50] | 20 [39–144]  | 50 [40–100]  | 11 [40–100] |
| TQA*         | 20 [40–50] | 20 [39–144]  | 50 [40–100]  | 11 [40–100] |
| MPS (Quimb)  |            |              |              |             |
| Fixed Angle* | -          | -            | 1 [100]      | 12 [40–100] |
| Fixed Angle† | -          | -            | 1 [100]      | 12 [40–100] |
| Fourier*     | 20 [40–50] | 31 [39–144]  | 10 [100]     | 12 [40–100] |
| Interp.*     | 20 [40–50] | 31 [39–144]  | 10 [100]     | 12 [40–100] |
| Linear Ramp  | 20 [40–50] | 20 [39–144]  | 10 [100]     | 1 [100]     |
| Linear Ramp* | 10 [50]    | 10 [144]     | 10 [100]     | 1 [100]     |
| Recursive TS | 10 [50]    | 10 [144]     | 10 [100]     | 1 [100]     |
| TQA          | 20 [40–50] | 31 [39–144]  | 10 [100]     | 12 [40–100] |
| TQA*         | 20 [40–50] | 31 [39–144]  | 10 [100]     | 12 [40–100] |
| PP           |            |              |              |             |
| Fixed Angle* | -          | 30 [39–144]  | 10 [100]     | 12 [40–100] |
| Fixed Angle† | -          | 30 [39–144]  | 10 [100]     | 12 [40–100] |
| Fourier*     | 20 [40–50] | 30 [39–144]  | 111 [40–100] | 18 [40–100] |
| Interp.*     | 20 [40–50] | 30 [39–144]  | 111 [40–100] | 18 [40–100] |
| Linear Ramp  | 20 [40–50] | 30 [39–144]  | 111 [40–100] | 18 [40–100] |
| Linear Ramp* | 10 [50]    | 20 [105–144] | 20 [40–100]  | 8 [100]     |
| Recursive TS | 10 [50]    | 10 [144]     | 10 [100]     | 7 [100]     |
| TQA          | 20 [40–50] | 30 [39–144]  | 111 [40–100] | 18 [40–100] |
| TQA*         | 20 [40–50] | 30 [39–144]  | 111 [40–100] | 18 [40–100] |
| SV           |            |              |              |             |
| Fixed Angle* | 17 [11–20] | -            | 16 [10–20]   | 60 [10–20]  |
| Fixed Angle† | 17 [11–20] | -            | 16 [10–20]   | 60 [10–20]  |
| Fourier*     | 92 [10–20] | 20 [12–21]   | 180 [10–20]  | 184 [10–20] |
| Interp.*     | 92 [10–20] | 20 [12–21]   | 180 [10–20]  | 184 [10–20] |
| Linear Ramp  | 92 [10–20] | 20 [12–21]   | 180 [10–20]  | 184 [10–20] |
| Linear Ramp* | 92 [10–20] | 20 [12–21]   | 180 [10–20]  | 184 [10–20] |
| Recursive TS | 92 [10–20] | 20 [12–21]   | 180 [10–20]  | 184 [10–20] |
| TQA          | 92 [10–20] | 20 [12–21]   | 180 [10–20]  | 184 [10–20] |
| TQA*         | 92 [10–20] | 20 [12–21]   | 180 [10–20]  | 184 [10–20] |

Table 28: **Number of Instances (num. nodes in brackets) for  $P = 10$ .** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Cells are colored based on the number of instances for the given configuration, with darker colors indicating more instances.

|              | ER              | HH              | LB              | RR              |
|--------------|-----------------|-----------------|-----------------|-----------------|
| MPS (Aer)    |                 |                 |                 |                 |
| Fixed Angle* | -               | $89.7 \pm 3.9$  | $96.2 \pm 1.0$  | $98.5 \pm 1.4$  |
| Fixed Angle† | -               | $88.3 \pm 3.6$  | $93.8 \pm 2.1$  | $97.8 \pm 3.3$  |
| Fourier*     | $81.2 \pm 6.1$  | $85.1 \pm 4.8$  | $80.6 \pm 8.9$  | $90.1 \pm 5.9$  |
| Interp.*     | $94.2 \pm 4.0$  | $91.3 \pm 4.6$  | $95.4 \pm 2.4$  | $91.1 \pm 3.3$  |
| Linear Ramp  | $94.4 \pm 1.8$  | $92.1 \pm 1.8$  | $91.4 \pm 4.8$  | $95.6 \pm 1.9$  |
| Linear Ramp* | $96.2 \pm 0.8$  | $84.6 \pm 1.3$  | $96.2 \pm 0.7$  | 91.0            |
| TQA          | $86.5 \pm 2.1$  | $87.1 \pm 2.5$  | $85.4 \pm 4.4$  | $92.7 \pm 4.6$  |
| TQA*         | $89.4 \pm 2.0$  | $87.9 \pm 2.9$  | $89.8 \pm 3.0$  | $95.5 \pm 3.5$  |
| MPS (Quimb)  |                 |                 |                 |                 |
| Fixed Angle* | -               | -               | 98.1            | $98.4 \pm 1.2$  |
| Fixed Angle† | -               | -               | 97.7            | $97.7 \pm 2.2$  |
| Fourier*     | $79.6 \pm 6.1$  | $77.4 \pm 10.9$ | $88.3 \pm 5.6$  | $91.8 \pm 9.3$  |
| Interp.*     | $97.4 \pm 1.5$  | $86.6 \pm 10.1$ | $97.1 \pm 0.6$  | $92.9 \pm 5.3$  |
| Linear Ramp  | $96.4 \pm 0.9$  | $82.2 \pm 11.8$ | $94.6 \pm 1.1$  | 88.7            |
| Linear Ramp* | $97.4 \pm 0.7$  | $69.5 \pm 2.0$  | $95.9 \pm 1.0$  | 92.7            |
| Recursive TS | $97.3 \pm 1.1$  | $56.5 \pm 1.6$  | $93.8 \pm 1.4$  | 88.4            |
| TQA          | $87.7 \pm 1.8$  | $79.6 \pm 9.2$  | $90.5 \pm 2.6$  | $92.8 \pm 5.2$  |
| TQA*         | $91.8 \pm 2.9$  | $80.8 \pm 9.3$  | $92.9 \pm 1.8$  | $96.1 \pm 2.2$  |
| PP           |                 |                 |                 |                 |
| Fixed Angle* | -               | $94.2 \pm 0.6$  | $90.7 \pm 1.9$  | $93.5 \pm 1.6$  |
| Fixed Angle† | -               | $91.5 \pm 1.3$  | $86.7 \pm 1.7$  | $92.9 \pm 1.6$  |
| Fourier*     | $91.5 \pm 1.9$  | $91.3 \pm 3.1$  | $86.0 \pm 2.2$  | $90.7 \pm 2.4$  |
| Interp.*     | $94.4 \pm 1.9$  | $95.0 \pm 0.7$  | $89.6 \pm 1.7$  | $90.4 \pm 2.5$  |
| Linear Ramp  | $94.7 \pm 1.2$  | $93.0 \pm 0.7$  | $88.1 \pm 1.9$  | $91.9 \pm 1.4$  |
| Linear Ramp* | $95.7 \pm 1.1$  | $93.6 \pm 0.5$  | $91.0 \pm 2.3$  | $91.7 \pm 0.8$  |
| Recursive TS | $90.6 \pm 1.1$  | $89.8 \pm 1.1$  | $87.4 \pm 2.0$  | $87.4 \pm 1.8$  |
| TQA          | $84.4 \pm 2.4$  | $90.0 \pm 0.9$  | $77.5 \pm 7.8$  | $89.4 \pm 3.0$  |
| TQA*         | $91.3 \pm 2.5$  | $91.8 \pm 0.7$  | $85.6 \pm 3.0$  | $91.6 \pm 1.6$  |
| SV           |                 |                 |                 |                 |
| Fixed Angle* | $97.4 \pm 1.4$  | -               | $97.5 \pm 1.3$  | $98.7 \pm 1.0$  |
| Fixed Angle† | $92.0 \pm 3.6$  | -               | $96.3 \pm 1.6$  | $98.0 \pm 1.0$  |
| Fourier*     | $78.1 \pm 13.1$ | $74.2 \pm 12.3$ | $76.8 \pm 11.3$ | $80.7 \pm 14.0$ |
| Interp.*     | $96.5 \pm 1.1$  | $94.7 \pm 1.0$  | $95.4 \pm 1.9$  | $92.1 \pm 6.8$  |
| Linear Ramp  | $95.0 \pm 1.4$  | $94.6 \pm 0.9$  | $93.5 \pm 1.9$  | $96.2 \pm 1.7$  |
| Linear Ramp* | $96.7 \pm 1.1$  | $95.3 \pm 0.8$  | $95.5 \pm 1.6$  | $97.8 \pm 1.2$  |
| Recursive TS | $95.0 \pm 2.0$  | $92.8 \pm 2.4$  | $92.5 \pm 2.2$  | $93.8 \pm 3.1$  |
| TQA          | $91.4 \pm 3.7$  | $89.8 \pm 3.3$  | $84.1 \pm 8.2$  | $91.9 \pm 4.6$  |
| TQA*         | $95.3 \pm 1.3$  | $92.8 \pm 1.9$  | $93.2 \pm 2.5$  | $96.7 \pm 1.5$  |

Table 29: **Avg. Approx. Ratio  $\pm$  standard deviation in percentage for MAXCUT for  $P = 10$ .** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Evaluation methods are shown in different colors, with darker colors indicating better approximation ratios.

|              | Approximation Ratio |                 |                 |                 | Num. Instances |              |              |             |
|--------------|---------------------|-----------------|-----------------|-----------------|----------------|--------------|--------------|-------------|
|              | ER                  | HH              | LB              | RR              | ER             | HH           | LB           | RR          |
| MPS (Aer)    |                     |                 |                 |                 |                |              |              |             |
| Fixed Angle* | -                   | $89.7 \pm 3.9$  | $96.2 \pm 1.0$  | $98.5 \pm 1.4$  | -              | 20 [39–144]  | 11 [100]     | 11 [40–100] |
| Fixed Angle† | -                   | $88.3 \pm 3.6$  | $93.8 \pm 2.1$  | $97.8 \pm 3.3$  | -              | 20 [39–144]  | 11 [100]     | 11 [40–100] |
| Fourier*     | $81.2 \pm 6.1$      | $85.1 \pm 4.8$  | $80.6 \pm 8.9$  | $90.1 \pm 5.9$  | 20 [40–50]     | 20 [39–144]  | 49 [40–100]  | 11 [40–100] |
| Interp.*     | $94.2 \pm 4.0$      | $91.3 \pm 4.6$  | $95.4 \pm 2.4$  | $91.1 \pm 3.3$  | 20 [40–50]     | 20 [39–144]  | 49 [40–100]  | 11 [40–100] |
| Linear Ramp  | $94.4 \pm 1.8$      | $92.1 \pm 1.8$  | $91.4 \pm 4.8$  | $95.6 \pm 1.9$  | 20 [40–50]     | 20 [39–144]  | 50 [40–100]  | 11 [40–100] |
| Linear Ramp* | $96.2 \pm 0.8$      | $84.6 \pm 1.3$  | $96.2 \pm 0.7$  | 91.0            | 10 [50]        | 10 [144]     | 10 [100]     | 1 [40–100]  |
| TQA          | $86.5 \pm 2.1$      | $87.1 \pm 2.5$  | $85.4 \pm 4.4$  | $92.7 \pm 4.6$  | 20 [40–50]     | 20 [39–144]  | 50 [40–100]  | 11 [40–100] |
| TQA*         | $89.4 \pm 2.0$      | $87.9 \pm 2.9$  | $89.8 \pm 3.0$  | $95.5 \pm 3.5$  | 20 [40–50]     | 20 [39–144]  | 50 [40–100]  | 11 [40–100] |
| MPS (Quimb)  |                     |                 |                 |                 |                |              |              |             |
| Fixed Angle* | -                   | -               | 98.1            | $98.4 \pm 1.2$  | -              | -            | 1 [100]      | 12 [40–100] |
| Fixed Angle† | -                   | -               | 97.7            | $97.7 \pm 2.2$  | -              | -            | 1 [100]      | 12 [40–100] |
| Fourier*     | $79.6 \pm 6.1$      | $77.4 \pm 10.9$ | $88.3 \pm 5.6$  | $91.8 \pm 9.3$  | 20 [40–50]     | 31 [39–144]  | 10 [100]     | 12 [40–100] |
| Interp.*     | $97.4 \pm 1.5$      | $86.6 \pm 10.1$ | $97.1 \pm 0.6$  | $92.9 \pm 5.3$  | 20 [40–50]     | 31 [39–144]  | 10 [100]     | 12 [40–100] |
| Linear Ramp  | $96.4 \pm 0.9$      | $82.2 \pm 11.8$ | $94.6 \pm 1.1$  | 88.7            | 20 [40–50]     | 20 [39–144]  | 10 [100]     | 1 [40–100]  |
| Linear Ramp* | $97.4 \pm 0.7$      | $69.5 \pm 2.0$  | $95.9 \pm 1.0$  | 92.7            | 10 [50]        | 10 [144]     | 10 [100]     | 1 [40–100]  |
| Recursive TS | $97.3 \pm 1.1$      | $56.5 \pm 1.6$  | $93.8 \pm 1.4$  | 88.4            | 10 [50]        | 10 [144]     | 10 [100]     | 1 [40–100]  |
| TQA          | $87.7 \pm 1.8$      | $79.6 \pm 9.2$  | $90.5 \pm 2.6$  | $92.8 \pm 5.2$  | 20 [40–50]     | 31 [39–144]  | 10 [100]     | 12 [40–100] |
| TQA*         | $91.8 \pm 2.9$      | $80.8 \pm 9.3$  | $92.9 \pm 1.8$  | $96.1 \pm 2.2$  | 20 [40–50]     | 31 [39–144]  | 10 [100]     | 12 [40–100] |
| PP           |                     |                 |                 |                 |                |              |              |             |
| Fixed Angle* | -                   | $94.2 \pm 0.6$  | $90.7 \pm 1.9$  | $93.5 \pm 1.6$  | -              | 30 [39–144]  | 10 [100]     | 12 [40–100] |
| Fixed Angle† | -                   | $91.5 \pm 1.3$  | $86.7 \pm 1.7$  | $92.9 \pm 1.6$  | -              | 30 [39–144]  | 10 [100]     | 12 [40–100] |
| Fourier*     | $91.5 \pm 1.9$      | $91.3 \pm 3.1$  | $86.0 \pm 2.2$  | $90.7 \pm 2.4$  | 20 [40–50]     | 30 [39–144]  | 111 [40–100] | 18 [40–100] |
| Interp.*     | $94.4 \pm 1.9$      | $95.0 \pm 0.7$  | $89.6 \pm 1.7$  | $90.4 \pm 2.5$  | 20 [40–50]     | 30 [39–144]  | 111 [40–100] | 18 [40–100] |
| Linear Ramp  | $94.7 \pm 1.2$      | $93.0 \pm 0.7$  | $88.1 \pm 1.9$  | $91.9 \pm 1.4$  | 20 [40–50]     | 30 [39–144]  | 111 [40–100] | 18 [40–100] |
| Linear Ramp* | $95.7 \pm 1.1$      | $93.6 \pm 0.5$  | $91.0 \pm 2.3$  | $91.7 \pm 0.8$  | 10 [50]        | 20 [105–144] | 20 [40–100]  | 8 [40–100]  |
| Recursive TS | $90.6 \pm 1.1$      | $89.8 \pm 1.1$  | $87.4 \pm 2.0$  | $87.4 \pm 1.8$  | 10 [50]        | 10 [144]     | 10 [100]     | 7 [40–100]  |
| TQA          | $84.4 \pm 2.4$      | $90.0 \pm 0.9$  | $77.5 \pm 7.8$  | $89.4 \pm 3.0$  | 20 [40–50]     | 30 [39–144]  | 111 [40–100] | 18 [40–100] |
| TQA*         | $91.3 \pm 2.5$      | $91.8 \pm 0.7$  | $85.6 \pm 3.0$  | $91.6 \pm 1.6$  | 20 [40–50]     | 30 [39–144]  | 111 [40–100] | 18 [40–100] |
| SV           |                     |                 |                 |                 |                |              |              |             |
| Fixed Angle* | $97.4 \pm 1.4$      | -               | $97.5 \pm 1.3$  | $98.7 \pm 1.0$  | 17 [11–20]     | -            | 16 [10–20]   | 60 [10–20]  |
| Fixed Angle† | $92.0 \pm 3.6$      | -               | $96.3 \pm 1.6$  | $98.0 \pm 1.0$  | 17 [11–20]     | -            | 16 [10–20]   | 60 [10–20]  |
| Fourier*     | $78.1 \pm 13.1$     | $74.2 \pm 12.3$ | $76.8 \pm 11.3$ | $80.7 \pm 14.0$ | 92 [10–20]     | 20 [12–21]   | 180 [10–20]  | 184 [10–20] |
| Interp.*     | $96.5 \pm 1.1$      | $94.7 \pm 1.0$  | $95.4 \pm 1.9$  | $92.1 \pm 6.8$  | 92 [10–20]     | 20 [12–21]   | 180 [10–20]  | 184 [10–20] |
| Linear Ramp  | $95.0 \pm 1.4$      | $94.6 \pm 0.9$  | $93.5 \pm 1.9$  | $96.2 \pm 1.7$  | 92 [10–20]     | 20 [12–21]   | 180 [10–20]  | 184 [10–20] |
| Linear Ramp* | $96.7 \pm 1.1$      | $95.3 \pm 0.8$  | $95.5 \pm 1.6$  | $97.8 \pm 1.2$  | 92 [10–20]     | 20 [12–21]   | 180 [10–20]  | 184 [10–20] |
| Recursive TS | $95.0 \pm 2.0$      | $92.8 \pm 2.4$  | $92.5 \pm 2.2$  | $93.8 \pm 3.1$  | 92 [10–20]     | 20 [12–21]   | 180 [10–20]  | 184 [10–20] |
| TQA          | $91.4 \pm 3.7$      | $89.8 \pm 3.3$  | $84.1 \pm 8.2$  | $91.9 \pm 4.6$  | 92 [10–20]     | 20 [12–21]   | 180 [10–20]  | 184 [10–20] |
| TQA*         | $95.3 \pm 1.3$      | $92.8 \pm 1.9$  | $93.2 \pm 2.5$  | $96.7 \pm 1.5$  | 92 [10–20]     | 20 [12–21]   | 180 [10–20]  | 184 [10–20] |

Table 30: **Avg. Approx. Ratio and Number of Instances for Various Evaluation Methods for MAX-CUT for  $P = 10$ .** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Evaluation methods are shown in different colors for approximation ratios. Darker colors indicating better approximation ratios or more instances. Approximation ratios are in percentage, with  $\pm$  their standard deviation. The range of graph sizes are shown in brackets, next to the number of instances.

# Summary Tables of MIS Data

## 11 Tables for depth $P = 1$

|              | ER      | HH       | LB       | RR       |
|--------------|---------|----------|----------|----------|
| MPS (Aer)    |         |          |          |          |
| Fourier*     | 10 [70] | 10 [144] | 10 [100] | 10 [100] |
| Interp.*     | 10 [70] | 10 [144] | 10 [100] | 10 [100] |
| TQA          | -       | -        | -        | -        |
| TQA*         | -       | -        | -        | -        |
| MPS (Quimb)  |         |          |          |          |
| Fourier*     | -       | -        | -        | -        |
| Interp.*     | -       | -        | -        | -        |
| TQA          | -       | -        | -        | -        |
| TQA*         | -       | -        | -        | -        |
| PP           |         |          |          |          |
| Fourier*     | 10 [70] | 10 [144] | 10 [100] | 10 [100] |
| Interp.*     | 10 [70] | 10 [144] | 10 [100] | 10 [100] |
| TQA          | -       | -        | -        | -        |
| TQA*         | -       | -        | -        | -        |
| SV           |         |          |          |          |
| Fourier*     | 10 [20] | 10 [21]  | 10 [20]  | 7 [20]   |
| Interp.*     | 10 [20] | 10 [21]  | 10 [20]  | 7 [20]   |
| Linear Ramp  | -       | -        | -        | -        |
| Linear Ramp* | -       | -        | -        | -        |
| Recursive TS | 10 [20] | 10 [21]  | 10 [20]  | 7 [20]   |
| TQA          | -       | -        | -        | 1 [20]   |
| TQA*         | -       | -        | -        | 1 [20]   |

Table 31: **Number of Instances (num. nodes in brackets) for  $P = 1$ .** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Cells are colored based on the number of instances for the given configuration, with darker colors indicating more instances.

|              | ER                   | HH               | LB                  | RR                  |
|--------------|----------------------|------------------|---------------------|---------------------|
| MPS (Aer)    |                      |                  |                     |                     |
| Fourier*     | 56682.2 $\pm$ 2535.8 | 7484.9 $\pm$ 0.0 | 15317.6 $\pm$ 0.0   | 9229.9 $\pm$ 7.4    |
| Interp.*     | 56598.1 $\pm$ 2542.4 | 7484.9 $\pm$ 0.0 | 15317.6 $\pm$ 0.0   | 9229.9 $\pm$ 7.4    |
| TQA          | -                    | -                | -                   | -                   |
| TQA*         | -                    | -                | -                   | -                   |
| MPS (Quimb)  |                      |                  |                     |                     |
| Fourier*     | -                    | -                | -                   | -                   |
| Interp.*     | -                    | -                | -                   | -                   |
| TQA          | -                    | -                | -                   | -                   |
| TQA*         | -                    | -                | -                   | -                   |
| PP           |                      |                  |                     |                     |
| Fourier*     | 56682.2 $\pm$ 2535.8 | 7484.9 $\pm$ 0.0 | 15317.6 $\pm$ 0.0   | 9229.9 $\pm$ 7.4    |
| Interp.*     | 56598.1 $\pm$ 2542.4 | 7484.9 $\pm$ 0.0 | 15317.6 $\pm$ 0.0   | 9229.9 $\pm$ 7.4    |
| TQA          | -                    | -                | -                   | -                   |
| TQA*         | -                    | -                | -                   | -                   |
| SV           |                      |                  |                     |                     |
| Fourier*     | 2839.4 $\pm$ 631.4   | 940.5 $\pm$ 0.0  | 7569.7 $\pm$ 4127.4 | 5998.2 $\pm$ 3052.2 |
| Interp.*     | 2839.4 $\pm$ 631.4   | 940.5 $\pm$ 0.0  | 7569.7 $\pm$ 4127.3 | 5998.2 $\pm$ 3052.2 |
| Linear Ramp  | -                    | -                | -                   | -                   |
| Linear Ramp* | -                    | -                | -                   | -                   |
| Recursive TS | 2839.4 $\pm$ 631.4   | 940.5 $\pm$ 0.0  | 7569.7 $\pm$ 4127.3 | 5998.2 $\pm$ 3052.2 |
| TQA          | -                    | -                | -                   | 1806.3              |
| TQA*         | -                    | -                | -                   | 1874.4              |

Table 32: **Avg. Energy  $\pm$  standard deviation for MIS for  $P = 1$ .** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Evaluation methods are shown in different colors, with darker colors indicating better energies. The minimum and maximum energies defining the range of colors are shared between MPS and PP. Note that energies include invalid solutions, which are penalized by the Hamiltonian.

|              | Energy           |              |                 |                 | Num. Instances |          |          |          |
|--------------|------------------|--------------|-----------------|-----------------|----------------|----------|----------|----------|
|              | ER               | HH           | LB              | RR              | ER             | HH       | LB       |          |
| MPS (Aer)    |                  |              |                 |                 |                |          |          |          |
| Fourier*     | 56682.2 ± 2535.8 | 7484.9 ± 0.0 | 15317.6 ± 0.0   | 9229.9 ± 7.4    | 10 [70]        | 10 [144] | 10 [100] | 10 [100] |
| Interp.*     | 56598.1 ± 2542.4 | 7484.9 ± 0.0 | 15317.6 ± 0.0   | 9229.9 ± 7.4    | 10 [70]        | 10 [144] | 10 [100] | 10 [100] |
| TQA          | -                | -            | -               | -               | -              | -        | -        | -        |
| TQA*         | -                | -            | -               | -               | -              | -        | -        | -        |
| MPS (Quimb)  |                  |              |                 |                 |                |          |          |          |
| Fourier*     | -                | -            | -               | -               | -              | -        | -        | -        |
| Interp.*     | -                | -            | -               | -               | -              | -        | -        | -        |
| TQA          | -                | -            | -               | -               | -              | -        | -        | -        |
| TQA*         | -                | -            | -               | -               | -              | -        | -        | -        |
| PP           |                  |              |                 |                 |                |          |          |          |
| Fourier*     | 56682.2 ± 2535.8 | 7484.9 ± 0.0 | 15317.6 ± 0.0   | 9229.9 ± 7.4    | 10 [70]        | 10 [144] | 10 [100] | 10 [100] |
| Interp.*     | 56598.1 ± 2542.4 | 7484.9 ± 0.0 | 15317.6 ± 0.0   | 9229.9 ± 7.4    | 10 [70]        | 10 [144] | 10 [100] | 10 [100] |
| TQA          | -                | -            | -               | -               | -              | -        | -        | -        |
| TQA*         | -                | -            | -               | -               | -              | -        | -        | -        |
| SV           |                  |              |                 |                 |                |          |          |          |
| Fourier*     | 2839.4 ± 631.4   | 940.5 ± 0.0  | 7569.7 ± 4127.4 | 5998.2 ± 3052.2 | 10 [20]        | 10 [21]  | 10 [20]  | 7 [20]   |
| Interp.*     | 2839.4 ± 631.4   | 940.5 ± 0.0  | 7569.7 ± 4127.3 | 5998.2 ± 3052.2 | 10 [20]        | 10 [21]  | 10 [20]  | 7 [20]   |
| Linear Ramp  | -                | -            | -               | -               | -              | -        | -        | -        |
| Linear Ramp* | -                | -            | -               | -               | -              | -        | -        | -        |
| Recursive TS | 2839.4 ± 631.4   | 940.5 ± 0.0  | 7569.7 ± 4127.3 | 5998.2 ± 3052.2 | 10 [20]        | 10 [21]  | 10 [20]  | 7 [20]   |
| TQA          | -                | -            | -               | 1806.3          | -              | -        | -        | 1 [20]   |
| TQA*         | -                | -            | -               | 1874.4          | -              | -        | -        | 1 [20]   |

Table 33: **Avg. Energy and Number of Instances for Various Evaluation Methods for MIS for  $P = 1$ .** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Evaluation methods are shown in different colors for energies. Darker colors indicating better energies or more instances. The minimum and maximum energies defining the range of colors are shared between MPS and PP. Average energies are shown, with  $\pm$  their standard deviation. The range of graph sizes are shown in brackets, next to the number of instances.

## 12 Tables for depth $P = 2$

|              | ER      | HH       | LB       | RR       |
|--------------|---------|----------|----------|----------|
| MPS (Aer)    |         |          |          |          |
| Fourier*     | 10 [70] | 10 [144] | 10 [100] | 10 [100] |
| Interp.*     | 10 [70] | 10 [144] | 10 [100] | 10 [100] |
| TQA          | 10 [70] | 10 [144] | 10 [100] | 10 [100] |
| TQA*         | 10 [70] | 10 [144] | 10 [100] | 10 [100] |
| MPS (Quimb)  |         |          |          |          |
| Fourier*     | -       | -        | -        | -        |
| Interp.*     | -       | -        | -        | -        |
| TQA          | -       | 10 [144] | -        | -        |
| TQA*         | -       | 10 [144] | -        | -        |
| PP           |         |          |          |          |
| Fourier*     | 10 [70] | 10 [144] | 10 [100] | 10 [100] |
| Interp.*     | 10 [70] | 10 [144] | 10 [100] | 10 [100] |
| TQA          | 10 [70] | 10 [144] | 10 [100] | 10 [100] |
| TQA*         | 10 [70] | 10 [144] | 10 [100] | 10 [100] |
| SV           |         |          |          |          |
| Fourier*     | 10 [20] | 10 [21]  | 10 [20]  | 7 [20]   |
| Interp.*     | 10 [20] | 10 [21]  | 10 [20]  | 7 [20]   |
| Linear Ramp  | -       | -        | -        | -        |
| Linear Ramp* | -       | -        | -        | -        |
| Recursive TS | 10 [20] | 10 [21]  | 10 [20]  | 7 [20]   |
| TQA          | -       | -        | -        | 1 [20]   |
| TQA*         | -       | -        | -        | 1 [20]   |

Table 34: **Number of Instances (num. nodes in brackets) for  $P = 2$ .** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Cells are colored based on the number of instances for the given configuration, with darker colors indicating more instances.

|              | ER                   | HH                  | LB                  | RR                  |
|--------------|----------------------|---------------------|---------------------|---------------------|
| MPS (Aer)    |                      |                     |                     |                     |
| Fourier*     | 61936.6 $\pm$ 2524.1 | 15087.0 $\pm$ 19.3  | 23286.0 $\pm$ 577.7 | 16822.9 $\pm$ 133.1 |
| Interp.*     | 53447.9 $\pm$ 6357.5 | 15047.8 $\pm$ 53.3  | 19285.1 $\pm$ 523.1 | 16950.8 $\pm$ 64.1  |
| TQA          | 26859.1 $\pm$ 3717.5 | 11388.1 $\pm$ 8.8   | 21309.1 $\pm$ 17.0  | 15015.9 $\pm$ 134.0 |
| TQA*         | 59220.7 $\pm$ 3629.8 | 15043.9 $\pm$ 77.7  | 23709.2 $\pm$ 162.9 | 16962.7 $\pm$ 77.7  |
| MPS (Quimb)  |                      |                     |                     |                     |
| Fourier*     | -                    | -                   | -                   | -                   |
| Interp.*     | -                    | -                   | -                   | -                   |
| TQA          | -                    | 13385.3 $\pm$ 0.0   | -                   | -                   |
| TQA*         | -                    | 15348.6 $\pm$ 272.1 | -                   | -                   |
| PP           |                      |                     |                     |                     |
| Fourier*     | 66023.6 $\pm$ 2531.7 | 14667.9 $\pm$ 0.0   | 23962.1 $\pm$ 0.0   | 16649.5 $\pm$ 7.4   |
| Interp.*     | 53045.1 $\pm$ 6171.8 | 14668.8 $\pm$ 0.0   | 20205.4 $\pm$ 0.0   | 16648.6 $\pm$ 7.9   |
| TQA          | 23622.4 $\pm$ 2109.4 | 11591.9 $\pm$ 0.0   | 21688.0 $\pm$ 0.0   | 14559.1 $\pm$ 5.9   |
| TQA*         | 64848.8 $\pm$ 5913.4 | 14668.5 $\pm$ 0.0   | 23961.1 $\pm$ 0.0   | 16645.6 $\pm$ 8.6   |
| SV           |                      |                     |                     |                     |
| Fourier*     | 3945.4 $\pm$ 756.2   | 1853.2 $\pm$ 0.0    | 8305.3 $\pm$ 3734.0 | 7126.2 $\pm$ 2801.9 |
| Interp.*     | 3859.0 $\pm$ 719.9   | 934.7 $\pm$ 0.0     | 7320.4 $\pm$ 2999.7 | 6426.5 $\pm$ 2497.8 |
| Linear Ramp  | -                    | -                   | -                   | -                   |
| Linear Ramp* | -                    | -                   | -                   | -                   |
| Recursive TS | 3922.0 $\pm$ 695.1   | 1379.1 $\pm$ 0.0    | 8429.1 $\pm$ 3760.8 | 6989.7 $\pm$ 2874.7 |
| TQA          | -                    | -                   | -                   | 2918.4              |
| TQA*         | -                    | -                   | -                   | 3308.9              |

Table 35: **Avg. Energy  $\pm$  standard deviation for MIS for  $P = 2$ .** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Evaluation methods are shown in different colors, with darker colors indicating better energies. The minimum and maximum energies defining the range of colors are shared between MPS and PP. Note that energies include invalid solutions, which are penalized by the Hamiltonian.

|              | Energy               |                     |                     |                     | Num. Instances |          |          |
|--------------|----------------------|---------------------|---------------------|---------------------|----------------|----------|----------|
|              | ER                   | HH                  | LB                  | RR                  | ER             | HH       | LB       |
| MPS (Aer)    |                      |                     |                     |                     |                |          |          |
| Fourier*     | 61936.6 $\pm$ 2524.1 | 15087.0 $\pm$ 19.3  | 23286.0 $\pm$ 577.7 | 16822.9 $\pm$ 133.1 | 10 [70]        | 10 [144] | 10 [100] |
| Interp.*     | 53447.9 $\pm$ 6357.5 | 15047.8 $\pm$ 53.3  | 19285.1 $\pm$ 523.1 | 16950.8 $\pm$ 64.1  | 10 [70]        | 10 [144] | 10 [100] |
| TQA          | 26859.1 $\pm$ 3717.5 | 11388.1 $\pm$ 8.8   | 21309.1 $\pm$ 17.0  | 15015.9 $\pm$ 134.0 | 10 [70]        | 10 [144] | 10 [100] |
| TQA*         | 59220.7 $\pm$ 3629.8 | 15043.9 $\pm$ 77.7  | 23709.2 $\pm$ 162.9 | 16962.7 $\pm$ 77.7  | 10 [70]        | 10 [144] | 10 [100] |
| MPS (Quimb)  |                      |                     |                     |                     |                |          |          |
| Fourier*     | -                    | -                   | -                   | -                   | -              | -        | -        |
| Interp.*     | -                    | -                   | -                   | -                   | -              | -        | -        |
| TQA          | -                    | 13385.3 $\pm$ 0.0   | -                   | -                   | -              | 10 [144] | -        |
| TQA*         | -                    | 15348.6 $\pm$ 272.1 | -                   | -                   | -              | 10 [144] | -        |
| PP           |                      |                     |                     |                     |                |          |          |
| Fourier*     | 66023.6 $\pm$ 2531.7 | 14667.9 $\pm$ 0.0   | 23962.1 $\pm$ 0.0   | 16649.5 $\pm$ 7.4   | 10 [70]        | 10 [144] | 10 [100] |
| Interp.*     | 53045.1 $\pm$ 6171.8 | 14668.8 $\pm$ 0.0   | 20205.4 $\pm$ 0.0   | 16648.6 $\pm$ 7.9   | 10 [70]        | 10 [144] | 10 [100] |
| TQA          | 23622.4 $\pm$ 2109.4 | 11591.9 $\pm$ 0.0   | 21688.0 $\pm$ 0.0   | 14559.1 $\pm$ 5.9   | 10 [70]        | 10 [144] | 10 [100] |
| TQA*         | 64848.8 $\pm$ 5913.4 | 14668.5 $\pm$ 0.0   | 23961.1 $\pm$ 0.0   | 16645.6 $\pm$ 8.6   | 10 [70]        | 10 [144] | 10 [100] |
| SV           |                      |                     |                     |                     |                |          |          |
| Fourier*     | 3945.4 $\pm$ 756.2   | 1853.2 $\pm$ 0.0    | 8305.3 $\pm$ 3734.0 | 7126.2 $\pm$ 2801.9 | 10 [20]        | 10 [21]  | 10 [20]  |
| Interp.*     | 3859.0 $\pm$ 719.9   | 934.7 $\pm$ 0.0     | 7320.4 $\pm$ 2999.7 | 6426.5 $\pm$ 2497.8 | 10 [20]        | 10 [21]  | 10 [20]  |
| Linear Ramp  | -                    | -                   | -                   | -                   | -              | -        | -        |
| Linear Ramp* | -                    | -                   | -                   | -                   | -              | -        | -        |
| Recursive TS | 3922.0 $\pm$ 695.1   | 1379.1 $\pm$ 0.0    | 8429.1 $\pm$ 3760.8 | 6989.7 $\pm$ 2874.7 | 10 [20]        | 10 [21]  | 10 [20]  |
| TQA          | -                    | -                   | -                   | 2918.4              | -              | -        | -        |
| TQA*         | -                    | -                   | -                   | 3308.9              | -              | -        | -        |

Table 36: **Avg. Energy and Number of Instances for Various Evaluation Methods for MIS for  $P = 2$ .** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Evaluation methods are shown in different colors for energies. Darker colors indicating better energies or more instances. The minimum and maximum energies defining the range of colors are shared between MPS and PP. Average energies are shown, with  $\pm$  their standard deviation. The range of graph sizes are shown in brackets, next to the number of instances.

### 13 Tables for depth $P = 3$

|              | ER      | HH       | LB       | RR       |
|--------------|---------|----------|----------|----------|
| MPS (Aer)    |         |          |          |          |
| Fourier*     | 10 [70] | 10 [144] | 10 [100] | 10 [100] |
| Interp.*     | 10 [70] | 10 [144] | 10 [100] | 10 [100] |
| TQA          | -       | -        | -        | -        |
| TQA*         | -       | -        | -        | -        |
| MPS (Quimb)  |         |          |          |          |
| Fourier*     | -       | -        | -        | -        |
| Interp.*     | -       | -        | -        | -        |
| TQA          | -       | -        | -        | -        |
| TQA*         | -       | -        | -        | -        |
| PP           |         |          |          |          |
| Fourier*     | 10 [70] | 10 [144] | 10 [100] | 10 [100] |
| Interp.*     | 10 [70] | 10 [144] | 10 [100] | 10 [100] |
| TQA          | -       | -        | -        | -        |
| TQA*         | -       | -        | -        | -        |
| SV           |         |          |          |          |
| Fourier*     | 10 [20] | 10 [21]  | 10 [20]  | 7 [20]   |
| Interp.*     | 10 [20] | 10 [21]  | 10 [20]  | 7 [20]   |
| Linear Ramp  | -       | -        | -        | -        |
| Linear Ramp* | -       | -        | -        | -        |
| Recursive TS | 10 [20] | 10 [21]  | 10 [20]  | 7 [20]   |
| TQA          | -       | -        | -        | 1 [20]   |
| TQA*         | -       | -        | -        | 1 [20]   |

Table 37: **Number of Instances (num. nodes in brackets) for  $P = 3$ .** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Cells are colored based on the number of instances for the given configuration, with darker colors indicating more instances.

|              | ER                   | HH                  | LB                  | RR                  |
|--------------|----------------------|---------------------|---------------------|---------------------|
| MPS (Aer)    |                      |                     |                     |                     |
| Fourier*     | 63673.4 $\pm$ 2329.8 | 16820.8 $\pm$ 51.2  | 23195.5 $\pm$ 513.9 | 17145.6 $\pm$ 166.6 |
| Interp.*     | 30291.4 $\pm$ 8828.4 | 15646.5 $\pm$ 437.4 | 20601.2 $\pm$ 388.8 | 17142.0 $\pm$ 124.0 |
| TQA          | -                    | -                   | -                   | -                   |
| TQA*         | -                    | -                   | -                   | -                   |
| MPS (Quimb)  |                      |                     |                     |                     |
| Fourier*     | -                    | -                   | -                   | -                   |
| Interp.*     | -                    | -                   | -                   | -                   |
| TQA          | -                    | -                   | -                   | -                   |
| TQA*         | -                    | -                   | -                   | -                   |
| PP           |                      |                     |                     |                     |
| Fourier*     | 70899.1 $\pm$ 2921.4 | 16638.6 $\pm$ 0.0   | 25141.6 $\pm$ 0.0   | 17514.9 $\pm$ 4.0   |
| Interp.*     | 15399.2 $\pm$ 5117.8 | 16615.9 $\pm$ 0.0   | 23006.9 $\pm$ 0.0   | 17490.7 $\pm$ 13.7  |
| TQA          | -                    | -                   | -                   | -                   |
| TQA*         | -                    | -                   | -                   | -                   |
| SV           |                      |                     |                     |                     |
| Fourier*     | 4260.1 $\pm$ 742.6   | 2069.8 $\pm$ 0.0    | 8957.7 $\pm$ 4034.6 | 7576.3 $\pm$ 3083.8 |
| Interp.*     | 3916.1 $\pm$ 687.1   | 1579.4 $\pm$ 0.0    | 5958.9 $\pm$ 3763.0 | 4536.5 $\pm$ 1679.9 |
| Linear Ramp  | -                    | -                   | -                   | -                   |
| Linear Ramp* | -                    | -                   | -                   | -                   |
| Recursive TS | 4335.6 $\pm$ 684.4   | 1946.3 $\pm$ 0.0    | 8692.1 $\pm$ 3718.1 | 7331.9 $\pm$ 2682.5 |
| TQA          | -                    | -                   | -                   | 3244.3              |
| TQA*         | -                    | -                   | -                   | 3483.0              |

Table 38: **Avg. Energy  $\pm$  standard deviation for MIS for  $P = 3$ .** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Evaluation methods are shown in different colors, with darker colors indicating better energies. The minimum and maximum energies defining the range of colors are shared between MPS and PP. Note that energies include invalid solutions, which are penalized by the Hamiltonian.

|              | Energy               |                     |                     |                     | Num. Instances |          |          |
|--------------|----------------------|---------------------|---------------------|---------------------|----------------|----------|----------|
|              | ER                   | HH                  | LB                  | RR                  | ER             | HH       | LB       |
| MPS (Aer)    |                      |                     |                     |                     |                |          |          |
| Fourier*     | 63673.4 $\pm$ 2329.8 | 16820.8 $\pm$ 51.2  | 23195.5 $\pm$ 513.9 | 17145.6 $\pm$ 166.6 | 10 [70]        | 10 [144] | 10 [100] |
| Interp.*     | 30291.4 $\pm$ 8828.4 | 15646.5 $\pm$ 437.4 | 20601.2 $\pm$ 388.8 | 17142.0 $\pm$ 124.0 | 10 [70]        | 10 [144] | 10 [100] |
| TQA          | -                    | -                   | -                   | -                   | -              | -        | -        |
| TQA*         | -                    | -                   | -                   | -                   | -              | -        | -        |
| MPS (Quimb)  |                      |                     |                     |                     |                |          |          |
| Fourier*     | -                    | -                   | -                   | -                   | -              | -        | -        |
| Interp.*     | -                    | -                   | -                   | -                   | -              | -        | -        |
| TQA          | -                    | -                   | -                   | -                   | -              | -        | -        |
| TQA*         | -                    | -                   | -                   | -                   | -              | -        | -        |
| PP           |                      |                     |                     |                     |                |          |          |
| Fourier*     | 70899.1 $\pm$ 2921.4 | 16638.6 $\pm$ 0.0   | 25141.6 $\pm$ 0.0   | 17514.9 $\pm$ 4.0   | 10 [70]        | 10 [144] | 10 [100] |
| Interp.*     | 15399.2 $\pm$ 5117.8 | 16615.9 $\pm$ 0.0   | 23006.9 $\pm$ 0.0   | 17490.7 $\pm$ 13.7  | 10 [70]        | 10 [144] | 10 [100] |
| TQA          | -                    | -                   | -                   | -                   | -              | -        | -        |
| TQA*         | -                    | -                   | -                   | -                   | -              | -        | -        |
| SV           |                      |                     |                     |                     |                |          |          |
| Fourier*     | 4260.1 $\pm$ 742.6   | 2069.8 $\pm$ 0.0    | 8957.7 $\pm$ 4034.6 | 7576.3 $\pm$ 3083.8 | 10 [20]        | 10 [21]  | 10 [20]  |
| Interp.*     | 3916.1 $\pm$ 687.1   | 1579.4 $\pm$ 0.0    | 5958.9 $\pm$ 3763.0 | 4536.5 $\pm$ 1679.9 | 10 [20]        | 10 [21]  | 10 [20]  |
| Linear Ramp  | -                    | -                   | -                   | -                   | -              | -        | -        |
| Linear Ramp* | -                    | -                   | -                   | -                   | -              | -        | -        |
| Recursive TS | 4335.6 $\pm$ 684.4   | 1946.3 $\pm$ 0.0    | 8692.1 $\pm$ 3718.1 | 7331.9 $\pm$ 2682.5 | 10 [20]        | 10 [21]  | 10 [20]  |
| TQA          | -                    | -                   | -                   | 3244.3              | -              | -        | -        |
| TQA*         | -                    | -                   | -                   | 3483.0              | -              | -        | -        |

Table 39: **Avg. Energy and Number of Instances for Various Evaluation Methods for MIS for  $P = 3$ .** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Evaluation methods are shown in different colors for energies. Darker colors indicating better energies or more instances. The minimum and maximum energies defining the range of colors are shared between MPS and PP. Average energies are shown, with  $\pm$  their standard deviation. The range of graph sizes are shown in brackets, next to the number of instances.

## 14 Tables for depth $P = 4$

|              | ER      | HH       | LB       | RR       |
|--------------|---------|----------|----------|----------|
| MPS (Aer)    |         |          |          |          |
| Fourier*     | 10 [70] | 10 [144] | 10 [100] | 10 [100] |
| Interp.*     | 10 [70] | 10 [144] | 10 [100] | 10 [100] |
| TQA          | 10 [70] | 10 [144] | 10 [100] | 10 [100] |
| TQA*         | 10 [70] | 10 [144] | 10 [100] | 10 [100] |
| MPS (Quimb)  |         |          |          |          |
| Fourier*     | -       | -        | -        | -        |
| Interp.*     | -       | -        | -        | -        |
| TQA          | -       | 10 [144] | -        | -        |
| TQA*         | -       | 10 [144] | -        | -        |
| PP           |         |          |          |          |
| Fourier*     | 10 [70] | 10 [144] | 10 [100] | 10 [100] |
| Interp.*     | 10 [70] | 10 [144] | 10 [100] | 10 [100] |
| TQA          | 10 [70] | 10 [144] | 10 [100] | 10 [100] |
| TQA*         | 10 [70] | 10 [144] | 10 [100] | 10 [100] |
| SV           |         |          |          |          |
| Fourier*     | 10 [20] | 10 [21]  | 10 [20]  | 7 [20]   |
| Interp.*     | 10 [20] | 10 [21]  | 10 [20]  | 7 [20]   |
| Linear Ramp  | -       | -        | -        | -        |
| Linear Ramp* | -       | -        | -        | -        |
| Recursive TS | 10 [20] | 10 [21]  | 10 [20]  | 7 [20]   |
| TQA          | -       | -        | -        | 1 [20]   |
| TQA*         | -       | -        | -        | 1 [20]   |

Table 40: **Number of Instances (num. nodes in brackets) for  $P = 4$ .** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Cells are colored based on the number of instances for the given configuration, with darker colors indicating more instances.

|              | ER                    | HH                  | LB                  | RR                  |
|--------------|-----------------------|---------------------|---------------------|---------------------|
| MPS (Aer)    |                       |                     |                     |                     |
| Fourier*     | 65073.4 $\pm$ 2065.2  | 17025.1 $\pm$ 65.9  | 23889.9 $\pm$ 331.7 | 17227.5 $\pm$ 109.2 |
| Interp.*     | 30289.6 $\pm$ 18260.2 | 16319.8 $\pm$ 349.4 | 22168.3 $\pm$ 982.7 | 17355.5 $\pm$ 48.4  |
| TQA          | 45337.4 $\pm$ 4065.1  | 15942.9 $\pm$ 6.2   | 21463.2 $\pm$ 18.7  | 16587.7 $\pm$ 14.7  |
| TQA*         | 52894.8 $\pm$ 6299.2  | 16825.6 $\pm$ 75.2  | 23803.7 $\pm$ 314.5 | 17309.4 $\pm$ 25.9  |
| MPS (Quimb)  |                       |                     |                     |                     |
| Fourier*     | -                     | -                   | -                   | -                   |
| Interp.*     | -                     | -                   | -                   | -                   |
| TQA          | -                     | 15660.4 $\pm$ 8.7   | -                   | -                   |
| TQA*         | -                     | 16122.0 $\pm$ 113.1 | -                   | -                   |
| PP           |                       |                     |                     |                     |
| Fourier*     | 73002.8 $\pm$ 3089.2  | 17057.3 $\pm$ 0.0   | 25757.2 $\pm$ 0.0   | 17619.5 $\pm$ 15.3  |
| Interp.*     | 27879.7 $\pm$ 27264.2 | 16074.5 $\pm$ 0.0   | 23790.6 $\pm$ 0.0   | 17580.0 $\pm$ 17.3  |
| TQA          | 18316.7 $\pm$ 5870.9  | 16094.1 $\pm$ 0.0   | 21509.8 $\pm$ 0.0   | 16273.7 $\pm$ 4.3   |
| TQA*         | 57385.8 $\pm$ 12377.5 | 16902.2 $\pm$ 0.0   | 24749.9 $\pm$ 0.0   | 17512.7 $\pm$ 19.7  |
| SV           |                       |                     |                     |                     |
| Fourier*     | 4517.7 $\pm$ 742.2    | 2204.0 $\pm$ 0.0    | 9144.0 $\pm$ 4065.7 | 7776.5 $\pm$ 3129.3 |
| Interp.*     | 4042.6 $\pm$ 642.8    | 1280.9 $\pm$ 0.0    | 5135.8 $\pm$ 3812.9 | 4016.3 $\pm$ 2128.0 |
| Linear Ramp  | -                     | -                   | -                   | -                   |
| Linear Ramp* | -                     | -                   | -                   | -                   |
| Recursive TS | 4469.5 $\pm$ 698.7    | 1982.8 $\pm$ 0.0    | 8986.2 $\pm$ 3812.9 | 7659.6 $\pm$ 3026.5 |
| TQA          | -                     | -                   | -                   | 3259.2              |
| TQA*         | -                     | -                   | -                   | 3479.1              |

Table 41: **Avg. Energy  $\pm$  standard deviation for MIS for  $P = 4$ .** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Evaluation methods are shown in different colors, with darker colors indicating better energies. The minimum and maximum energies defining the range of colors are shared between MPS and PP. Note that energies include invalid solutions, which are penalized by the Hamiltonian.

|              | Energy                |                     |                     |                     | Num. Instances |          |          |
|--------------|-----------------------|---------------------|---------------------|---------------------|----------------|----------|----------|
|              | ER                    | HH                  | LB                  | RR                  | ER             | HH       | LB       |
| MPS (Aer)    |                       |                     |                     |                     |                |          |          |
| Fourier*     | 65073.4 $\pm$ 2065.2  | 17025.1 $\pm$ 65.9  | 23889.9 $\pm$ 331.7 | 17227.5 $\pm$ 109.2 | 10 [70]        | 10 [144] | 10 [100] |
| Interp.*     | 30289.6 $\pm$ 18260.2 | 16319.8 $\pm$ 349.4 | 22168.3 $\pm$ 982.7 | 17355.5 $\pm$ 48.4  | 10 [70]        | 10 [144] | 10 [100] |
| TQA          | 45337.4 $\pm$ 4065.1  | 15942.9 $\pm$ 6.2   | 21463.2 $\pm$ 18.7  | 16587.7 $\pm$ 14.7  | 10 [70]        | 10 [144] | 10 [100] |
| TQA*         | 52894.8 $\pm$ 6299.2  | 16825.6 $\pm$ 75.2  | 23803.7 $\pm$ 314.5 | 17309.4 $\pm$ 25.9  | 10 [70]        | 10 [144] | 10 [100] |
| MPS (Quimb)  |                       |                     |                     |                     |                |          |          |
| Fourier*     | -                     | -                   | -                   | -                   | -              | -        | -        |
| Interp.*     | -                     | -                   | -                   | -                   | -              | -        | -        |
| TQA          | -                     | 15660.4 $\pm$ 8.7   | -                   | -                   | -              | 10 [144] | -        |
| TQA*         | -                     | 16122.0 $\pm$ 113.1 | -                   | -                   | -              | 10 [144] | -        |
| PP           |                       |                     |                     |                     |                |          |          |
| Fourier*     | 73002.8 $\pm$ 3089.2  | 17057.3 $\pm$ 0.0   | 25757.2 $\pm$ 0.0   | 17619.5 $\pm$ 15.3  | 10 [70]        | 10 [144] | 10 [100] |
| Interp.*     | 27879.7 $\pm$ 27264.2 | 16074.5 $\pm$ 0.0   | 23790.6 $\pm$ 0.0   | 17580.0 $\pm$ 17.3  | 10 [70]        | 10 [144] | 10 [100] |
| TQA          | 18316.7 $\pm$ 5870.9  | 16094.1 $\pm$ 0.0   | 21509.8 $\pm$ 0.0   | 16273.7 $\pm$ 4.3   | 10 [70]        | 10 [144] | 10 [100] |
| TQA*         | 57385.8 $\pm$ 12377.5 | 16902.2 $\pm$ 0.0   | 24749.9 $\pm$ 0.0   | 17512.7 $\pm$ 19.7  | 10 [70]        | 10 [144] | 10 [100] |
| SV           |                       |                     |                     |                     |                |          |          |
| Fourier*     | 4517.7 $\pm$ 742.2    | 2204.0 $\pm$ 0.0    | 9144.0 $\pm$ 4065.7 | 7776.5 $\pm$ 3129.3 | 10 [20]        | 10 [21]  | 10 [20]  |
| Interp.*     | 4042.6 $\pm$ 642.8    | 1280.9 $\pm$ 0.0    | 5135.8 $\pm$ 3812.9 | 4016.3 $\pm$ 2128.0 | 10 [20]        | 10 [21]  | 10 [20]  |
| Linear Ramp  | -                     | -                   | -                   | -                   | -              | -        | -        |
| Linear Ramp* | -                     | -                   | -                   | -                   | -              | -        | -        |
| Recursive TS | 4469.5 $\pm$ 698.7    | 1982.8 $\pm$ 0.0    | 8986.2 $\pm$ 3812.9 | 7659.6 $\pm$ 3026.5 | 10 [20]        | 10 [21]  | 10 [20]  |
| TQA          | -                     | -                   | -                   | 3259.2              | -              | -        | -        |
| TQA*         | -                     | -                   | -                   | 3479.1              | -              | -        | -        |

Table 42: **Avg. Energy and Number of Instances for Various Evaluation Methods for MIS for  $P = 4$ .** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Evaluation methods are shown in different colors for energies. Darker colors indicating better energies or more instances. The minimum and maximum energies defining the range of colors are shared between MPS and PP. Average energies are shown, with  $\pm$  their standard deviation. The range of graph sizes are shown in brackets, next to the number of instances.

## 15 Tables for depth $P = 5$

|              | ER      | HH       | LB       | RR       |
|--------------|---------|----------|----------|----------|
| MPS (Aer)    |         |          |          |          |
| Fourier*     | 10 [70] | 10 [144] | 10 [100] | 10 [100] |
| Interp.*     | 10 [70] | 10 [144] | 10 [100] | 10 [100] |
| TQA          | -       | -        | -        | -        |
| TQA*         | -       | -        | -        | -        |
| MPS (Quimb)  |         |          |          |          |
| Fourier*     | -       | -        | -        | -        |
| Interp.*     | -       | -        | -        | -        |
| TQA          | -       | -        | -        | -        |
| TQA*         | -       | -        | -        | -        |
| PP           |         |          |          |          |
| Fourier*     | 10 [70] | 10 [144] | 10 [100] | 10 [100] |
| Interp.*     | 10 [70] | 10 [144] | 10 [100] | 10 [100] |
| TQA          | -       | -        | -        | -        |
| TQA*         | -       | -        | -        | -        |
| SV           |         |          |          |          |
| Fourier*     | 10 [20] | 10 [21]  | 10 [20]  | 7 [20]   |
| Interp.*     | 10 [20] | 10 [21]  | 10 [20]  | 7 [20]   |
| Linear Ramp  | -       | -        | -        | -        |
| Linear Ramp* | -       | -        | -        | -        |
| Recursive TS | 10 [20] | 10 [21]  | 10 [20]  | 7 [20]   |
| TQA          | -       | -        | -        | 1 [20]   |
| TQA*         | -       | -        | -        | 1 [20]   |

Table 43: **Number of Instances (num. nodes in brackets) for  $P = 5$ .** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Cells are colored based on the number of instances for the given configuration, with darker colors indicating more instances.

|              | ER                    | HH                  | LB                  | RR                  |
|--------------|-----------------------|---------------------|---------------------|---------------------|
| MPS (Aer)    |                       |                     |                     |                     |
| Fourier*     | 65502.3 $\pm$ 2332.4  | 17150.3 $\pm$ 42.1  | 24101.5 $\pm$ 259.9 | 17355.1 $\pm$ 70.2  |
| Interp.*     | 27765.9 $\pm$ 13747.4 | 16526.5 $\pm$ 257.7 | 22601.5 $\pm$ 712.6 | 17424.0 $\pm$ 38.1  |
| TQA          | -                     | -                   | -                   | -                   |
| TQA*         | -                     | -                   | -                   | -                   |
| MPS (Quimb)  |                       |                     |                     |                     |
| Fourier*     | -                     | -                   | -                   | -                   |
| Interp.*     | -                     | -                   | -                   | -                   |
| TQA          | -                     | -                   | -                   | -                   |
| TQA*         | -                     | -                   | -                   | -                   |
| PP           |                       |                     |                     |                     |
| Fourier*     | 79758.6 $\pm$ 2424.4  | 17143.2 $\pm$ 0.0   | 26076.0 $\pm$ 0.0   | 17773.9 $\pm$ 20.7  |
| Interp.*     | 35592.9 $\pm$ 25747.1 | 16105.4 $\pm$ 0.0   | 19595.6 $\pm$ 0.0   | 17756.7 $\pm$ 27.4  |
| TQA          | -                     | -                   | -                   | -                   |
| TQA*         | -                     | -                   | -                   | -                   |
| SV           |                       |                     |                     |                     |
| Fourier*     | 4652.9 $\pm$ 746.8    | 2217.9 $\pm$ 0.0    | 9295.9 $\pm$ 4091.1 | 7870.5 $\pm$ 3186.7 |
| Interp.*     | 4078.6 $\pm$ 624.5    | 1440.4 $\pm$ 0.0    | 4298.4 $\pm$ 2632.0 | 4072.0 $\pm$ 2156.8 |
| Linear Ramp  | -                     | -                   | -                   | -                   |
| Linear Ramp* | -                     | -                   | -                   | -                   |
| Recursive TS | 4533.5 $\pm$ 714.4    | 1991.6 $\pm$ 0.0    | 9066.1 $\pm$ 3849.1 | 7772.1 $\pm$ 3092.2 |
| TQA          | -                     | -                   | -                   | 3308.0              |
| TQA*         | -                     | -                   | -                   | 3476.4              |

Table 44: **Avg. Energy  $\pm$  standard deviation for MIS for  $P = 5$ .** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Evaluation methods are shown in different colors, with darker colors indicating better energies. The minimum and maximum energies defining the range of colors are shared between MPS and PP. Note that energies include invalid solutions, which are penalized by the Hamiltonian.

|              | Energy                |                     |                     |                     | Num. Instances |          |          |
|--------------|-----------------------|---------------------|---------------------|---------------------|----------------|----------|----------|
|              | ER                    | HH                  | LB                  | RR                  | ER             | HH       | LB       |
| MPS (Aer)    |                       |                     |                     |                     |                |          |          |
| Fourier*     | 65502.3 $\pm$ 2332.4  | 17150.3 $\pm$ 42.1  | 24101.5 $\pm$ 259.9 | 17355.1 $\pm$ 70.2  | 10 [70]        | 10 [144] | 10 [100] |
| Interp.*     | 27765.9 $\pm$ 13747.4 | 16526.5 $\pm$ 257.7 | 22601.5 $\pm$ 712.6 | 17424.0 $\pm$ 38.1  | 10 [70]        | 10 [144] | 10 [100] |
| TQA          | -                     | -                   | -                   | -                   | -              | -        | -        |
| TQA*         | -                     | -                   | -                   | -                   | -              | -        | -        |
| MPS (Quimb)  |                       |                     |                     |                     |                |          |          |
| Fourier*     | -                     | -                   | -                   | -                   | -              | -        | -        |
| Interp.*     | -                     | -                   | -                   | -                   | -              | -        | -        |
| TQA          | -                     | -                   | -                   | -                   | -              | -        | -        |
| TQA*         | -                     | -                   | -                   | -                   | -              | -        | -        |
| PP           |                       |                     |                     |                     |                |          |          |
| Fourier*     | 79758.6 $\pm$ 2424.4  | 17143.2 $\pm$ 0.0   | 26076.0 $\pm$ 0.0   | 17773.9 $\pm$ 20.7  | 10 [70]        | 10 [144] | 10 [100] |
| Interp.*     | 35592.9 $\pm$ 25747.1 | 16105.4 $\pm$ 0.0   | 19595.6 $\pm$ 0.0   | 17756.7 $\pm$ 27.4  | 10 [70]        | 10 [144] | 10 [100] |
| TQA          | -                     | -                   | -                   | -                   | -              | -        | -        |
| TQA*         | -                     | -                   | -                   | -                   | -              | -        | -        |
| SV           |                       |                     |                     |                     |                |          |          |
| Fourier*     | 4652.9 $\pm$ 746.8    | 2217.9 $\pm$ 0.0    | 9295.9 $\pm$ 4091.1 | 7870.5 $\pm$ 3186.7 | 10 [20]        | 10 [21]  | 10 [20]  |
| Interp.*     | 4078.6 $\pm$ 624.5    | 1440.4 $\pm$ 0.0    | 4298.4 $\pm$ 2632.0 | 4072.0 $\pm$ 2156.8 | 10 [20]        | 10 [21]  | 10 [20]  |
| Linear Ramp  | -                     | -                   | -                   | -                   | -              | -        | -        |
| Linear Ramp* | -                     | -                   | -                   | -                   | -              | -        | -        |
| Recursive TS | 4533.5 $\pm$ 714.4    | 1991.6 $\pm$ 0.0    | 9066.1 $\pm$ 3849.1 | 7772.1 $\pm$ 3092.2 | 10 [20]        | 10 [21]  | 10 [20]  |
| TQA          | -                     | -                   | -                   | 3308.0              | -              | -        | -        |
| TQA*         | -                     | -                   | -                   | 3476.4              | -              | -        | -        |

Table 45: **Avg. Energy and Number of Instances for Various Evaluation Methods for MIS for  $P = 5$ .** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Evaluation methods are shown in different colors for energies. Darker colors indicating better energies or more instances. The minimum and maximum energies defining the range of colors are shared between MPS and PP. Average energies are shown, with  $\pm$  their standard deviation. The range of graph sizes are shown in brackets, next to the number of instances.

## 16 Tables for depth $P = 6$

|              | ER      | HH       | LB       | RR       |
|--------------|---------|----------|----------|----------|
| MPS (Aer)    |         |          |          |          |
| Fourier*     | 10 [70] | 10 [144] | 10 [100] | 10 [100] |
| Interp.*     | 10 [70] | 10 [144] | 10 [100] | 10 [100] |
| TQA          | 10 [70] | 10 [144] | 10 [100] | 10 [100] |
| TQA*         | 10 [70] | 10 [144] | 10 [100] | 10 [100] |
| MPS (Quimb)  |         |          |          |          |
| Fourier*     | -       | -        | -        | -        |
| Interp.*     | -       | -        | -        | -        |
| TQA          | -       | 10 [144] | -        | -        |
| TQA*         | -       | 10 [144] | -        | -        |
| PP           |         |          |          |          |
| Fourier*     | 10 [70] | 10 [144] | 10 [100] | 10 [100] |
| Interp.*     | 10 [70] | 10 [144] | 10 [100] | 10 [100] |
| TQA          | 10 [70] | 10 [144] | 10 [100] | 10 [100] |
| TQA*         | 10 [70] | 10 [144] | 10 [100] | 10 [100] |
| SV           |         |          |          |          |
| Fourier*     | 10 [20] | 10 [21]  | 10 [20]  | 7 [20]   |
| Interp.*     | 10 [20] | 10 [21]  | 10 [20]  | 7 [20]   |
| Linear Ramp  | 10 [20] | 10 [21]  | 10 [20]  | 7 [20]   |
| Linear Ramp* | 10 [20] | 10 [21]  | 10 [20]  | 7 [20]   |
| Recursive TS | 10 [20] | 10 [21]  | 10 [20]  | 7 [20]   |
| TQA          | 10 [20] | 10 [21]  | 10 [20]  | 7 [20]   |
| TQA*         | 10 [20] | 10 [21]  | 10 [20]  | 7 [20]   |

Table 46: **Number of Instances (num. nodes in brackets) for  $P = 6$ .** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Cells are colored based on the number of instances for the given configuration, with darker colors indicating more instances.

|              | ER                    | HH                  | LB                  | RR                  |
|--------------|-----------------------|---------------------|---------------------|---------------------|
| MPS (Aer)    |                       |                     |                     |                     |
| Fourier*     | 65882.6 $\pm$ 2605.2  | 17238.8 $\pm$ 41.9  | 24117.0 $\pm$ 532.0 | 17398.2 $\pm$ 62.9  |
| Interp.*     | 41477.0 $\pm$ 10764.6 | 16407.2 $\pm$ 251.6 | 22849.9 $\pm$ 266.4 | 17427.2 $\pm$ 68.3  |
| TQA          | 12617.7 $\pm$ 2917.7  | 16331.0 $\pm$ 4.1   | 21160.4 $\pm$ 21.2  | 16771.7 $\pm$ 14.6  |
| TQA*         | 32061.9 $\pm$ 11764.8 | 17136.6 $\pm$ 34.3  | 24234.3 $\pm$ 142.4 | 17309.3 $\pm$ 40.8  |
| MPS (Quimb)  |                       |                     |                     |                     |
| Fourier*     | -                     | -                   | -                   | -                   |
| Interp.*     | -                     | -                   | -                   | -                   |
| TQA          | -                     | 16300.9 $\pm$ 0.7   | -                   | -                   |
| TQA*         | -                     | 17190.4 $\pm$ 15.8  | -                   | -                   |
| PP           |                       |                     |                     |                     |
| Fourier*     | 78103.2 $\pm$ 7976.1  | 17172.0 $\pm$ 0.0   | 25853.4 $\pm$ 0.0   | 17940.8 $\pm$ 30.8  |
| Interp.*     | 55315.5 $\pm$ 32067.7 | 16369.2 $\pm$ 0.0   | 25263.7 $\pm$ 0.0   | 17786.4 $\pm$ 18.8  |
| TQA          | 3285.8 $\pm$ 753.9    | 15990.3 $\pm$ 0.0   | 18897.2 $\pm$ 0.0   | 16844.9 $\pm$ 5.5   |
| TQA*         | 19827.1 $\pm$ 3918.3  | 17075.8 $\pm$ 0.0   | 25357.7 $\pm$ 0.0   | 17627.0 $\pm$ 81.7  |
| SV           |                       |                     |                     |                     |
| Fourier*     | 4716.6 $\pm$ 742.4    | 2212.8 $\pm$ 0.0    | 9290.9 $\pm$ 4162.0 | 7664.8 $\pm$ 3014.5 |
| Interp.*     | 4089.5 $\pm$ 607.9    | 1711.2 $\pm$ 0.0    | 4905.3 $\pm$ 3834.9 | 3880.6 $\pm$ 2462.0 |
| Linear Ramp  | 4777.5 $\pm$ 779.0    | 2107.4 $\pm$ 0.0    | 6402.1 $\pm$ 2074.9 | 6448.6 $\pm$ 2419.1 |
| Linear Ramp* | 4829.3 $\pm$ 792.9    | 2231.4 $\pm$ 0.0    | 9282.9 $\pm$ 3910.1 | 7918.1 $\pm$ 3162.0 |
| Recursive TS | 4578.2 $\pm$ 720.8    | 2094.1 $\pm$ 0.0    | 9099.4 $\pm$ 3870.4 | 7803.1 $\pm$ 3117.3 |
| TQA          | 2644.2 $\pm$ 630.6    | 2126.2 $\pm$ 0.0    | 2784.5 $\pm$ 831.0  | 4028.6 $\pm$ 1065.1 |
| TQA*         | 4239.2 $\pm$ 561.7    | 2228.6 $\pm$ 0.0    | 6051.2 $\pm$ 1720.6 | 6584.1 $\pm$ 2329.4 |

Table 47: **Avg. Energy  $\pm$  standard deviation for MIS for  $P = 6$ .** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Evaluation methods are shown in different colors, with darker colors indicating better energies. The minimum and maximum energies defining the range of colors are shared between MPS and PP. Note that energies include invalid solutions, which are penalized by the Hamiltonian.

|              | Energy                |                     |                     |                     | Num. Instances |          |          |
|--------------|-----------------------|---------------------|---------------------|---------------------|----------------|----------|----------|
|              | ER                    | HH                  | LB                  | RR                  | ER             | HH       | LB       |
| MPS (Aer)    |                       |                     |                     |                     |                |          |          |
| Fourier*     | 65882.6 $\pm$ 2605.2  | 17238.8 $\pm$ 41.9  | 24117.0 $\pm$ 532.0 | 17398.2 $\pm$ 62.9  | 10 [70]        | 10 [144] | 10 [100] |
| Interp.*     | 41477.0 $\pm$ 10764.6 | 16407.2 $\pm$ 251.6 | 22849.9 $\pm$ 266.4 | 17427.2 $\pm$ 68.3  | 10 [70]        | 10 [144] | 10 [100] |
| TQA          | 12617.7 $\pm$ 2917.7  | 16331.0 $\pm$ 4.1   | 21160.4 $\pm$ 21.2  | 16771.7 $\pm$ 14.6  | 10 [70]        | 10 [144] | 10 [100] |
| TQA*         | 32061.9 $\pm$ 11764.8 | 17136.6 $\pm$ 34.3  | 24234.3 $\pm$ 142.4 | 17309.3 $\pm$ 40.8  | 10 [70]        | 10 [144] | 10 [100] |
| MPS (Quimb)  |                       |                     |                     |                     |                |          |          |
| Fourier*     | -                     | -                   | -                   | -                   | -              | -        | -        |
| Interp.*     | -                     | -                   | -                   | -                   | -              | -        | -        |
| TQA          | -                     | 16300.9 $\pm$ 0.7   | -                   | -                   | -              | 10 [144] | -        |
| TQA*         | -                     | 17190.4 $\pm$ 15.8  | -                   | -                   | -              | 10 [144] | -        |
| PP           |                       |                     |                     |                     |                |          |          |
| Fourier*     | 78103.2 $\pm$ 7976.1  | 17172.0 $\pm$ 0.0   | 25853.4 $\pm$ 0.0   | 17940.8 $\pm$ 30.8  | 10 [70]        | 10 [144] | 10 [100] |
| Interp.*     | 55315.5 $\pm$ 32067.7 | 16369.2 $\pm$ 0.0   | 25263.7 $\pm$ 0.0   | 17786.4 $\pm$ 18.8  | 10 [70]        | 10 [144] | 10 [100] |
| TQA          | 3285.8 $\pm$ 753.9    | 15990.3 $\pm$ 0.0   | 18897.2 $\pm$ 0.0   | 16844.9 $\pm$ 5.5   | 10 [70]        | 10 [144] | 10 [100] |
| TQA*         | 19827.1 $\pm$ 3918.3  | 17075.8 $\pm$ 0.0   | 25357.7 $\pm$ 0.0   | 17627.0 $\pm$ 81.7  | 10 [70]        | 10 [144] | 10 [100] |
| SV           |                       |                     |                     |                     |                |          |          |
| Fourier*     | 4716.6 $\pm$ 742.4    | 2212.8 $\pm$ 0.0    | 9290.9 $\pm$ 4162.0 | 7664.8 $\pm$ 3014.5 | 10 [20]        | 10 [21]  | 10 [20]  |
| Interp.*     | 4089.5 $\pm$ 607.9    | 1711.2 $\pm$ 0.0    | 4905.3 $\pm$ 3834.9 | 3880.6 $\pm$ 2462.0 | 10 [20]        | 10 [21]  | 10 [20]  |
| Linear Ramp  | 4777.5 $\pm$ 779.0    | 2107.4 $\pm$ 0.0    | 6402.1 $\pm$ 2074.9 | 6448.6 $\pm$ 2419.1 | 10 [20]        | 10 [21]  | 10 [20]  |
| Linear Ramp* | 4829.3 $\pm$ 792.9    | 2231.4 $\pm$ 0.0    | 9282.9 $\pm$ 3910.1 | 7918.1 $\pm$ 3162.0 | 10 [20]        | 10 [21]  | 10 [20]  |
| Recursive TS | 4578.2 $\pm$ 720.8    | 2094.1 $\pm$ 0.0    | 9099.4 $\pm$ 3870.4 | 7803.1 $\pm$ 3117.3 | 10 [20]        | 10 [21]  | 10 [20]  |
| TQA          | 2644.2 $\pm$ 630.6    | 2126.2 $\pm$ 0.0    | 2784.5 $\pm$ 831.0  | 4028.6 $\pm$ 1065.1 | 10 [20]        | 10 [21]  | 10 [20]  |
| TQA*         | 4239.2 $\pm$ 561.7    | 2228.6 $\pm$ 0.0    | 6051.2 $\pm$ 1720.6 | 6584.1 $\pm$ 2329.4 | 10 [20]        | 10 [21]  | 10 [20]  |

Table 48: **Avg. Energy and Number of Instances for Various Evaluation Methods for MIS for  $P = 6$ .** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Evaluation methods are shown in different colors for energies. Darker colors indicating better energies or more instances. The minimum and maximum energies defining the range of colors are shared between MPS and PP. Average energies are shown, with  $\pm$  their standard deviation. The range of graph sizes are shown in brackets, next to the number of instances.

## 17 Tables for depth $P = 7$

|              | ER      | HH       | LB      | RR     |
|--------------|---------|----------|---------|--------|
| MPS (Aer)    |         |          |         |        |
| Fourier*     | -       | 10 [144] | -       | -      |
| Interp.*     | -       | 10 [144] | -       | -      |
| TQA          | -       | -        | -       | -      |
| TQA*         | -       | -        | -       | -      |
| MPS (Quimb)  |         |          |         |        |
| Fourier*     | -       | -        | -       | -      |
| Interp.*     | -       | -        | -       | -      |
| TQA          | -       | -        | -       | -      |
| TQA*         | -       | -        | -       | -      |
| PP           |         |          |         |        |
| Fourier*     | -       | 10 [144] | -       | -      |
| Interp.*     | -       | 10 [144] | -       | -      |
| TQA          | -       | -        | -       | -      |
| TQA*         | -       | -        | -       | -      |
| SV           |         |          |         |        |
| Fourier*     | 10 [20] | 10 [21]  | 10 [20] | 7 [20] |
| Interp.*     | 10 [20] | 10 [21]  | 10 [20] | 7 [20] |
| Linear Ramp  | -       | -        | -       | -      |
| Linear Ramp* | -       | -        | -       | -      |
| Recursive TS | 10 [20] | 10 [21]  | 10 [20] | 7 [20] |
| TQA          | -       | -        | -       | 1 [20] |
| TQA*         | -       | -        | -       | 1 [20] |

Table 49: **Number of Instances (num. nodes in brackets) for  $P = 7$ .** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Cells are colored based on the number of instances for the given configuration, with darker colors indicating more instances.

|              | ER                 | HH                  | LB                  | RR                  |
|--------------|--------------------|---------------------|---------------------|---------------------|
| MPS (Aer)    |                    |                     |                     |                     |
| Fourier*     | -                  | 17289.5 $\pm$ 51.9  | -                   | -                   |
| Interp.*     | -                  | 16398.9 $\pm$ 190.0 | -                   | -                   |
| TQA          | -                  | -                   | -                   | -                   |
| TQA*         | -                  | -                   | -                   | -                   |
| MPS (Quimb)  |                    |                     |                     |                     |
| Fourier*     | -                  | -                   | -                   | -                   |
| Interp.*     | -                  | -                   | -                   | -                   |
| TQA          | -                  | -                   | -                   | -                   |
| TQA*         | -                  | -                   | -                   | -                   |
| PP           |                    |                     |                     |                     |
| Fourier*     | -                  | 17176.5 $\pm$ 0.0   | -                   | -                   |
| Interp.*     | -                  | 16528.1 $\pm$ 0.0   | -                   | -                   |
| TQA          | -                  | -                   | -                   | -                   |
| TQA*         | -                  | -                   | -                   | -                   |
| SV           |                    |                     |                     |                     |
| Fourier*     | 4747.8 $\pm$ 745.9 | 2208.8 $\pm$ 0.0    | 9028.3 $\pm$ 3854.4 | 7571.2 $\pm$ 2958.1 |
| Interp.*     | 4100.8 $\pm$ 606.4 | 1640.5 $\pm$ 0.0    | 4115.2 $\pm$ 3116.2 | 3306.0 $\pm$ 2667.3 |
| Linear Ramp  | -                  | -                   | -                   | -                   |
| Linear Ramp* | -                  | -                   | -                   | -                   |
| Recursive TS | 4605.1 $\pm$ 707.9 | 2114.6 $\pm$ 0.0    | 9138.6 $\pm$ 3920.3 | 7845.0 $\pm$ 3152.0 |
| TQA          | -                  | -                   | -                   | 3258.0              |
| TQA*         | -                  | -                   | -                   | 3490.7              |

Table 50: **Avg. Energy  $\pm$  standard deviation for MIS for  $P = 7$ .** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Evaluation methods are shown in different colors, with darker colors indicating better energies. The minimum and maximum energies defining the range of colors are shared between MPS and PP. Note that energies include invalid solutions, which are penalized by the Hamiltonian.

|              | Energy             |                     |                     |                     | Num. Instances |          |         |        |
|--------------|--------------------|---------------------|---------------------|---------------------|----------------|----------|---------|--------|
|              | ER                 | HH                  | LB                  | RR                  | ER             | HH       | LB      | RR     |
| MPS (Aer)    |                    |                     |                     |                     |                |          |         |        |
| Fourier*     | -                  | 17289.5 $\pm$ 51.9  | -                   | -                   | -              | 10 [144] | -       | -      |
| Interp.*     | -                  | 16398.9 $\pm$ 190.0 | -                   | -                   | -              | 10 [144] | -       | -      |
| TQA          | -                  | -                   | -                   | -                   | -              | -        | -       | -      |
| TQA*         | -                  | -                   | -                   | -                   | -              | -        | -       | -      |
| MPS (Quimb)  |                    |                     |                     |                     |                |          |         |        |
| Fourier*     | -                  | -                   | -                   | -                   | -              | -        | -       | -      |
| Interp.*     | -                  | -                   | -                   | -                   | -              | -        | -       | -      |
| TQA          | -                  | -                   | -                   | -                   | -              | -        | -       | -      |
| TQA*         | -                  | -                   | -                   | -                   | -              | -        | -       | -      |
| PP           |                    |                     |                     |                     |                |          |         |        |
| Fourier*     | -                  | 17176.5 $\pm$ 0.0   | -                   | -                   | -              | 10 [144] | -       | -      |
| Interp.*     | -                  | 16528.1 $\pm$ 0.0   | -                   | -                   | -              | 10 [144] | -       | -      |
| TQA          | -                  | -                   | -                   | -                   | -              | -        | -       | -      |
| TQA*         | -                  | -                   | -                   | -                   | -              | -        | -       | -      |
| SV           |                    |                     |                     |                     |                |          |         |        |
| Fourier*     | 4747.8 $\pm$ 745.9 | 2208.8 $\pm$ 0.0    | 9028.3 $\pm$ 3854.4 | 7571.2 $\pm$ 2958.1 | 10 [20]        | 10 [21]  | 10 [20] | 7 [20] |
| Interp.*     | 4100.8 $\pm$ 606.4 | 1640.5 $\pm$ 0.0    | 4115.2 $\pm$ 3116.2 | 3306.0 $\pm$ 2667.3 | 10 [20]        | 10 [21]  | 10 [20] | 7 [20] |
| Linear Ramp  | -                  | -                   | -                   | -                   | -              | -        | -       | -      |
| Linear Ramp* | -                  | -                   | -                   | -                   | -              | -        | -       | -      |
| Recursive TS | 4605.1 $\pm$ 707.9 | 2114.6 $\pm$ 0.0    | 9138.6 $\pm$ 3920.3 | 7845.0 $\pm$ 3152.0 | 10 [20]        | 10 [21]  | 10 [20] | 7 [20] |
| TQA          | -                  | -                   | -                   | 3258.0              | -              | -        | -       | 1 [20] |
| TQA*         | -                  | -                   | -                   | 3490.7              | -              | -        | -       | 1 [20] |

Table 51: **Avg. Energy and Number of Instances for Various Evaluation Methods for MIS for  $P = 7$ .** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Evaluation methods are shown in different colors for energies. Darker colors indicating better energies or more instances. The minimum and maximum energies defining the range of colors are shared between MPS and PP. Average energies are shown, with  $\pm$  their standard deviation. The range of graph sizes are shown in brackets, next to the number of instances.

## 18 Tables for depth $P = 8$

|              | ER      | HH       | LB      | RR     |
|--------------|---------|----------|---------|--------|
| MPS (Aer)    |         |          |         |        |
| Fourier*     | -       | 10 [144] | -       | -      |
| Interp.*     | -       | 10 [144] | -       | -      |
| TQA          | -       | 10 [144] | -       | -      |
| TQA*         | -       | 10 [144] | -       | -      |
| MPS (Quimb)  |         |          |         |        |
| Fourier*     | -       | -        | -       | -      |
| Interp.*     | -       | -        | -       | -      |
| TQA          | -       | 10 [144] | -       | -      |
| TQA*         | -       | 10 [144] | -       | -      |
| PP           |         |          |         |        |
| Fourier*     | -       | 10 [144] | -       | -      |
| Interp.*     | -       | 10 [144] | -       | -      |
| TQA          | -       | 10 [144] | -       | -      |
| TQA*         | -       | 10 [144] | -       | -      |
| SV           |         |          |         |        |
| Fourier*     | 10 [20] | 10 [21]  | 10 [20] | 7 [20] |
| Interp.*     | 10 [20] | 10 [21]  | 10 [20] | 7 [20] |
| Linear Ramp  | -       | -        | -       | -      |
| Linear Ramp* | -       | -        | -       | -      |
| Recursive TS | 10 [20] | 10 [21]  | 10 [20] | 7 [20] |
| TQA          | -       | -        | -       | 1 [20] |
| TQA*         | -       | -        | -       | 1 [20] |

Table 52: **Number of Instances (num. nodes in brackets) for  $P = 8$ .** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Cells are colored based on the number of instances for the given configuration, with darker colors indicating more instances.

|              | ER                 | HH                  | LB                  | RR                  |
|--------------|--------------------|---------------------|---------------------|---------------------|
| MPS (Aer)    |                    |                     |                     |                     |
| Fourier*     | -                  | 17315.2 $\pm$ 34.1  | -                   | -                   |
| Interp.*     | -                  | 16279.0 $\pm$ 420.1 | -                   | -                   |
| TQA          | -                  | 17132.3 $\pm$ 3.1   | -                   | -                   |
| TQA*         | -                  | 17252.6 $\pm$ 20.1  | -                   | -                   |
| MPS (Quimb)  |                    |                     |                     |                     |
| Fourier*     | -                  | -                   | -                   | -                   |
| Interp.*     | -                  | -                   | -                   | -                   |
| TQA          | -                  | 16609.5 $\pm$ 2.2   | -                   | -                   |
| TQA*         | -                  | 17070.9 $\pm$ 43.3  | -                   | -                   |
| PP           |                    |                     |                     |                     |
| Fourier*     | -                  | 17161.2 $\pm$ 0.0   | -                   | -                   |
| Interp.*     | -                  | 16582.6 $\pm$ 0.0   | -                   | -                   |
| TQA          | -                  | 16788.3 $\pm$ 0.0   | -                   | -                   |
| TQA*         | -                  | 17137.5 $\pm$ 0.1   | -                   | -                   |
| SV           |                    |                     |                     |                     |
| Fourier*     | 4758.6 $\pm$ 746.4 | 2168.6 $\pm$ 0.0    | 9217.4 $\pm$ 3993.7 | 7590.1 $\pm$ 3001.8 |
| Interp.*     | 4094.5 $\pm$ 620.2 | 1600.4 $\pm$ 0.0    | 4043.8 $\pm$ 3198.8 | 3025.0 $\pm$ 2890.9 |
| Linear Ramp  | -                  | -                   | -                   | -                   |
| Linear Ramp* | -                  | -                   | -                   | -                   |
| Recursive TS | 4616.3 $\pm$ 710.9 | 2120.1 $\pm$ 0.0    | 9206.3 $\pm$ 4027.3 | 7866.3 $\pm$ 3170.9 |
| TQA          | -                  | -                   | -                   | 3327.0              |
| TQA*         | -                  | -                   | -                   | 3451.4              |

Table 53: **Avg. Energy  $\pm$  standard deviation for MIS for  $P = 8$ .** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Evaluation methods are shown in different colors, with darker colors indicating better energies. The minimum and maximum energies defining the range of colors are shared between MPS and PP. Note that energies include invalid solutions, which are penalized by the Hamiltonian.

|              | Energy             |                     |                     |                     | Num. Instances |          |         |        |
|--------------|--------------------|---------------------|---------------------|---------------------|----------------|----------|---------|--------|
|              | ER                 | HH                  | LB                  | RR                  | ER             | HH       | LB      | RR     |
| MPS (Aer)    |                    |                     |                     |                     |                |          |         |        |
| Fourier*     | -                  | 17315.2 $\pm$ 34.1  | -                   | -                   | -              | 10 [144] | -       | -      |
| Interp.*     | -                  | 16279.0 $\pm$ 420.1 | -                   | -                   | -              | 10 [144] | -       | -      |
| TQA          | -                  | 17132.3 $\pm$ 3.1   | -                   | -                   | -              | 10 [144] | -       | -      |
| TQA*         | -                  | 17252.6 $\pm$ 20.1  | -                   | -                   | -              | 10 [144] | -       | -      |
| MPS (Quimb)  |                    |                     |                     |                     |                |          |         |        |
| Fourier*     | -                  | -                   | -                   | -                   | -              | -        | -       | -      |
| Interp.*     | -                  | -                   | -                   | -                   | -              | -        | -       | -      |
| TQA          | -                  | 16609.5 $\pm$ 2.2   | -                   | -                   | -              | 10 [144] | -       | -      |
| TQA*         | -                  | 17070.9 $\pm$ 43.3  | -                   | -                   | -              | 10 [144] | -       | -      |
| PP           |                    |                     |                     |                     |                |          |         |        |
| Fourier*     | -                  | 17161.2 $\pm$ 0.0   | -                   | -                   | -              | 10 [144] | -       | -      |
| Interp.*     | -                  | 16582.6 $\pm$ 0.0   | -                   | -                   | -              | 10 [144] | -       | -      |
| TQA          | -                  | 16788.3 $\pm$ 0.0   | -                   | -                   | -              | 10 [144] | -       | -      |
| TQA*         | -                  | 17137.5 $\pm$ 0.1   | -                   | -                   | -              | 10 [144] | -       | -      |
| SV           |                    |                     |                     |                     |                |          |         |        |
| Fourier*     | 4758.6 $\pm$ 746.4 | 2168.6 $\pm$ 0.0    | 9217.4 $\pm$ 3993.7 | 7590.1 $\pm$ 3001.8 | 10 [20]        | 10 [21]  | 10 [20] | 7 [21] |
| Interp.*     | 4094.5 $\pm$ 620.2 | 1600.4 $\pm$ 0.0    | 4043.8 $\pm$ 3198.8 | 3025.0 $\pm$ 2890.9 | 10 [20]        | 10 [21]  | 10 [20] | 7 [21] |
| Linear Ramp  | -                  | -                   | -                   | -                   | -              | -        | -       | -      |
| Linear Ramp* | -                  | -                   | -                   | -                   | -              | -        | -       | -      |
| Recursive TS | 4616.3 $\pm$ 710.9 | 2120.1 $\pm$ 0.0    | 9206.3 $\pm$ 4027.3 | 7866.3 $\pm$ 3170.9 | 10 [20]        | 10 [21]  | 10 [20] | 7 [21] |
| TQA          | -                  | -                   | -                   | 3327.0              | -              | -        | -       | 1 [21] |
| TQA*         | -                  | -                   | -                   | 3451.4              | -              | -        | -       | 1 [21] |

Table 54: **Avg. Energy and Number of Instances for Various Evaluation Methods for MIS for  $P = 8$ .** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Evaluation methods are shown in different colors for energies. Darker colors indicating better energies or more instances. The minimum and maximum energies defining the range of colors are shared between MPS and PP. Average energies are shown, with  $\pm$  their standard deviation. The range of graph sizes are shown in brackets, next to the number of instances.

## 19 Tables for depth $P = 9$

|              | ER      | HH       | LB      | RR     |
|--------------|---------|----------|---------|--------|
| MPS (Aer)    |         |          |         |        |
| Fourier*     | -       | 10 [144] | -       | -      |
| Interp.*     | -       | 10 [144] | -       | -      |
| TQA          | -       | -        | -       | -      |
| TQA*         | -       | -        | -       | -      |
| MPS (Quimb)  |         |          |         |        |
| Fourier*     | -       | -        | -       | -      |
| Interp.*     | -       | -        | -       | -      |
| TQA          | -       | -        | -       | -      |
| TQA*         | -       | -        | -       | -      |
| PP           |         |          |         |        |
| Fourier*     | -       | 10 [144] | -       | -      |
| Interp.*     | -       | 10 [144] | -       | -      |
| TQA          | -       | -        | -       | -      |
| TQA*         | -       | -        | -       | -      |
| SV           |         |          |         |        |
| Fourier*     | 10 [20] | 10 [21]  | 10 [20] | 7 [20] |
| Interp.*     | 10 [20] | 10 [21]  | 10 [20] | 7 [20] |
| Linear Ramp  | -       | -        | -       | -      |
| Linear Ramp* | -       | -        | -       | -      |
| Recursive TS | 10 [20] | 10 [21]  | 10 [20] | 7 [20] |
| TQA          | -       | -        | -       | 1 [20] |
| TQA*         | -       | -        | -       | 1 [20] |

Table 55: **Number of Instances (num. nodes in brackets) for  $P = 9$ .** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Cells are colored based on the number of instances for the given configuration, with darker colors indicating more instances.

|              | ER                 | HH                  | LB                  | RR                  |
|--------------|--------------------|---------------------|---------------------|---------------------|
| MPS (Aer)    |                    |                     |                     |                     |
| Fourier*     | -                  | 17318.8 $\pm$ 27.4  | -                   | -                   |
| Interp.*     | -                  | 16162.6 $\pm$ 379.1 | -                   | -                   |
| TQA          | -                  | -                   | -                   | -                   |
| TQA*         | -                  | -                   | -                   | -                   |
| MPS (Quimb)  |                    |                     |                     |                     |
| Fourier*     | -                  | -                   | -                   | -                   |
| Interp.*     | -                  | -                   | -                   | -                   |
| TQA          | -                  | -                   | -                   | -                   |
| TQA*         | -                  | -                   | -                   | -                   |
| PP           |                    |                     |                     |                     |
| Fourier*     | -                  | 17132.5 $\pm$ 0.0   | -                   | -                   |
| Interp.*     | -                  | 16628.0 $\pm$ 0.0   | -                   | -                   |
| TQA          | -                  | -                   | -                   | -                   |
| TQA*         | -                  | -                   | -                   | -                   |
| SV           |                    |                     |                     |                     |
| Fourier*     | 4779.6 $\pm$ 762.1 | 2185.7 $\pm$ 0.0    | 9151.8 $\pm$ 3924.7 | 7659.2 $\pm$ 3043.9 |
| Interp.*     | 4055.4 $\pm$ 575.4 | 1553.8 $\pm$ 0.0    | 3306.2 $\pm$ 2844.6 | 3000.7 $\pm$ 2768.1 |
| Linear Ramp  | -                  | -                   | -                   | -                   |
| Linear Ramp* | -                  | -                   | -                   | -                   |
| Recursive TS | 4616.4 $\pm$ 710.9 | 2123.9 $\pm$ 0.0    | 9212.5 $\pm$ 4035.7 | 7871.9 $\pm$ 3175.0 |
| TQA          | -                  | -                   | -                   | 3299.8              |
| TQA*         | -                  | -                   | -                   | 3448.1              |

Table 56: **Avg. Energy  $\pm$  standard deviation for MIS for  $P = 9$ .** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Evaluation methods are shown in different colors, with darker colors indicating better energies. The minimum and maximum energies defining the range of colors are shared between MPS and PP. Note that energies include invalid solutions, which are penalized by the Hamiltonian.

|              | Energy         |                 |                 |                 | Num. Instances |          |         |        |
|--------------|----------------|-----------------|-----------------|-----------------|----------------|----------|---------|--------|
|              | ER             | HH              | LB              | RR              | ER             | HH       | LB      | RR     |
| MPS (Aer)    |                |                 |                 |                 |                |          |         |        |
| Fourier*     | -              | 17318.8 ± 27.4  | -               | -               | -              | 10 [144] | -       | -      |
| Interp.*     | -              | 16162.6 ± 379.1 | -               | -               | -              | 10 [144] | -       | -      |
| TQA          | -              | -               | -               | -               | -              | -        | -       | -      |
| TQA*         | -              | -               | -               | -               | -              | -        | -       | -      |
| MPS (Quimb)  |                |                 |                 |                 |                |          |         |        |
| Fourier*     | -              | -               | -               | -               | -              | -        | -       | -      |
| Interp.*     | -              | -               | -               | -               | -              | -        | -       | -      |
| TQA          | -              | -               | -               | -               | -              | -        | -       | -      |
| TQA*         | -              | -               | -               | -               | -              | -        | -       | -      |
| PP           |                |                 |                 |                 |                |          |         |        |
| Fourier*     | -              | 17132.5 ± 0.0   | -               | -               | -              | 10 [144] | -       | -      |
| Interp.*     | -              | 16628.0 ± 0.0   | -               | -               | -              | 10 [144] | -       | -      |
| TQA          | -              | -               | -               | -               | -              | -        | -       | -      |
| TQA*         | -              | -               | -               | -               | -              | -        | -       | -      |
| SV           |                |                 |                 |                 |                |          |         |        |
| Fourier*     | 4779.6 ± 762.1 | 2185.7 ± 0.0    | 9151.8 ± 3924.7 | 7659.2 ± 3043.9 | 10 [20]        | 10 [21]  | 10 [20] | 7 [20] |
| Interp.*     | 4055.4 ± 575.4 | 1553.8 ± 0.0    | 3306.2 ± 2844.6 | 3000.7 ± 2768.1 | 10 [20]        | 10 [21]  | 10 [20] | 7 [20] |
| Linear Ramp  | -              | -               | -               | -               | -              | -        | -       | -      |
| Linear Ramp* | -              | -               | -               | -               | -              | -        | -       | -      |
| Recursive TS | 4616.4 ± 710.9 | 2123.9 ± 0.0    | 9212.5 ± 4035.7 | 7871.9 ± 3175.0 | 10 [20]        | 10 [21]  | 10 [20] | 7 [20] |
| TQA          | -              | -               | -               | 3299.8          | -              | -        | -       | 1 [20] |
| TQA*         | -              | -               | -               | 3448.1          | -              | -        | -       | 1 [20] |

Table 57: **Avg. Energy and Number of Instances for Various Evaluation Methods for MIS for  $P = 9$ .** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Evaluation methods are shown in different colors for energies. Darker colors indicating better energies or more instances. The minimum and maximum energies defining the range of colors are shared between MPS and PP. Average energies are shown, with  $\pm$  their standard deviation. The range of graph sizes are shown in brackets, next to the number of instances.

## 20 Tables for depth $P = 10$

|              | ER      | HH       | LB      | RR     |
|--------------|---------|----------|---------|--------|
| MPS (Aer)    |         |          |         |        |
| Fourier*     | -       | 10 [144] | -       | -      |
| Interp.*     | -       | 10 [144] | -       | -      |
| TQA          | -       | 10 [144] | -       | -      |
| TQA*         | -       | 10 [144] | -       | -      |
| MPS (Quimb)  |         |          |         |        |
| Fourier*     | -       | -        | -       | -      |
| Interp.*     | -       | -        | -       | -      |
| TQA          | -       | 10 [144] | -       | -      |
| TQA*         | -       | 10 [144] | -       | -      |
| PP           |         |          |         |        |
| Fourier*     | -       | 10 [144] | -       | -      |
| Interp.*     | -       | 10 [144] | -       | -      |
| TQA          | -       | 10 [144] | -       | -      |
| TQA*         | -       | 10 [144] | -       | -      |
| SV           |         |          |         |        |
| Fourier*     | 10 [20] | 10 [21]  | 10 [20] | 7 [20] |
| Interp.*     | 10 [20] | 10 [21]  | 10 [20] | 7 [20] |
| Linear Ramp  | -       | -        | -       | -      |
| Linear Ramp* | -       | -        | -       | -      |
| Recursive TS | 10 [20] | 10 [21]  | 10 [20] | 7 [20] |
| TQA          | -       | -        | -       | 1 [20] |
| TQA*         | -       | -        | -       | 1 [20] |

Table 58: **Number of Instances (num. nodes in brackets) for  $P = 10$ .** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Cells are colored based on the number of instances for the given configuration, with darker colors indicating more instances.

|              | ER                 | HH                  | LB                  | RR                  |
|--------------|--------------------|---------------------|---------------------|---------------------|
| MPS (Aer)    |                    |                     |                     |                     |
| Fourier*     | -                  | 17326.1 $\pm$ 29.2  | -                   | -                   |
| Interp.*     | -                  | 16072.7 $\pm$ 324.4 | -                   | -                   |
| TQA          | -                  | 17167.3 $\pm$ 2.4   | -                   | -                   |
| TQA*         | -                  | 17298.2 $\pm$ 14.8  | -                   | -                   |
| MPS (Quimb)  |                    |                     |                     |                     |
| Fourier*     | -                  | -                   | -                   | -                   |
| Interp.*     | -                  | -                   | -                   | -                   |
| TQA          | -                  | 16800.2 $\pm$ 1.5   | -                   | -                   |
| TQA*         | -                  | 17178.2 $\pm$ 11.7  | -                   | -                   |
| PP           |                    |                     |                     |                     |
| Fourier*     | -                  | 17128.7 $\pm$ 0.0   | -                   | -                   |
| Interp.*     | -                  | 16575.6 $\pm$ 0.0   | -                   | -                   |
| TQA          | -                  | 16835.4 $\pm$ 0.0   | -                   | -                   |
| TQA*         | -                  | 17142.7 $\pm$ 0.0   | -                   | -                   |
| SV           |                    |                     |                     |                     |
| Fourier*     | 4800.7 $\pm$ 779.0 | 2182.8 $\pm$ 0.0    | 9305.2 $\pm$ 4020.2 | 7689.7 $\pm$ 3060.3 |
| Interp.*     | 4051.0 $\pm$ 615.0 | 1607.1 $\pm$ 0.0    | 3504.3 $\pm$ 2465.6 | 2826.1 $\pm$ 2759.5 |
| Linear Ramp  | -                  | -                   | -                   | -                   |
| Linear Ramp* | -                  | -                   | -                   | -                   |
| Recursive TS | 4616.4 $\pm$ 710.9 | 2124.1 $\pm$ 0.0    | 9223.0 $\pm$ 4053.3 | 7873.1 $\pm$ 3175.7 |
| TQA          | -                  | -                   | -                   | 3276.3              |
| TQA*         | -                  | -                   | -                   | 3470.8              |

Table 59: **Avg. Energy  $\pm$  standard deviation for MIS for  $P = 10$ .** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Evaluation methods are shown in different colors, with darker colors indicating better energies. The minimum and maximum energies defining the range of colors are shared between MPS and PP. Note that energies include invalid solutions, which are penalized by the Hamiltonian.

|              | Energy             |                     |                     |                     | Num. Instances |          |         |        |
|--------------|--------------------|---------------------|---------------------|---------------------|----------------|----------|---------|--------|
|              | ER                 | HH                  | LB                  | RR                  | ER             | HH       | LB      | RR     |
| MPS (Aer)    |                    |                     |                     |                     |                |          |         |        |
| Fourier*     | -                  | 17326.1 $\pm$ 29.2  | -                   | -                   | -              | 10 [144] | -       | -      |
| Interp.*     | -                  | 16072.7 $\pm$ 324.4 | -                   | -                   | -              | 10 [144] | -       | -      |
| TQA          | -                  | 17167.3 $\pm$ 2.4   | -                   | -                   | -              | 10 [144] | -       | -      |
| TQA*         | -                  | 17298.2 $\pm$ 14.8  | -                   | -                   | -              | 10 [144] | -       | -      |
| MPS (Quimb)  |                    |                     |                     |                     |                |          |         |        |
| Fourier*     | -                  | -                   | -                   | -                   | -              | -        | -       | -      |
| Interp.*     | -                  | -                   | -                   | -                   | -              | -        | -       | -      |
| TQA          | -                  | 16800.2 $\pm$ 1.5   | -                   | -                   | -              | 10 [144] | -       | -      |
| TQA*         | -                  | 17178.2 $\pm$ 11.7  | -                   | -                   | -              | 10 [144] | -       | -      |
| PP           |                    |                     |                     |                     |                |          |         |        |
| Fourier*     | -                  | 17128.7 $\pm$ 0.0   | -                   | -                   | -              | 10 [144] | -       | -      |
| Interp.*     | -                  | 16575.6 $\pm$ 0.0   | -                   | -                   | -              | 10 [144] | -       | -      |
| TQA          | -                  | 16835.4 $\pm$ 0.0   | -                   | -                   | -              | 10 [144] | -       | -      |
| TQA*         | -                  | 17142.7 $\pm$ 0.0   | -                   | -                   | -              | 10 [144] | -       | -      |
| SV           |                    |                     |                     |                     |                |          |         |        |
| Fourier*     | 4800.7 $\pm$ 779.0 | 2182.8 $\pm$ 0.0    | 9305.2 $\pm$ 4020.2 | 7689.7 $\pm$ 3060.3 | 10 [20]        | 10 [21]  | 10 [20] | 7 [20] |
| Interp.*     | 4051.0 $\pm$ 615.0 | 1607.1 $\pm$ 0.0    | 3504.3 $\pm$ 2465.6 | 2826.1 $\pm$ 2759.5 | 10 [20]        | 10 [21]  | 10 [20] | 7 [20] |
| Linear Ramp  | -                  | -                   | -                   | -                   | -              | -        | -       | -      |
| Linear Ramp* | -                  | -                   | -                   | -                   | -              | -        | -       | -      |
| Recursive TS | 4616.4 $\pm$ 710.9 | 2124.1 $\pm$ 0.0    | 9223.0 $\pm$ 4053.3 | 7873.1 $\pm$ 3175.7 | 10 [20]        | 10 [21]  | 10 [20] | 7 [20] |
| TQA          | -                  | -                   | -                   | 3276.3              | -              | -        | -       | 1 [20] |
| TQA*         | -                  | -                   | -                   | 3470.8              | -              | -        | -       | 1 [20] |

Table 60: **Avg. Energy and Number of Instances for Various Evaluation Methods for MIS for  $P = 10$ .** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Evaluation methods are shown in different colors for energies. Darker colors indicating better energies or more instances. The minimum and maximum energies defining the range of colors are shared between MPS and PP. Average energies are shown, with  $\pm$  their standard deviation. The range of graph sizes are shown in brackets, next to the number of instances.